

SCHOOL OF SCIENCES

MATHEMATICS DEPARTMENT

Prepared for School of ICT (Computer Engineering and Computer Science)

Module Title: MATHEMATICS FOR ENGINEERS I

COURSE CODE: AM80166

Level : 1, Semester: 1 Credits: 10

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1. Module Title: MATHEMATICS FOR ENGINEERS I (AM80166).
2. Level: 1 Semester: 1 Credits: 10.
3. Pre-requisite or co-requisite modules, excluded combinations. Pre-requisite: Pass A level Mathematics

4. **Brief description of aims and content**

The aim is to provide a link between mathematics that required for Engineering. A review unit is followed by units on calculus on functions of one real variable, numerical and functions series, introduction to matrices theory, and element of vector's spaces.

LEARNING OUTCOMES

A. Knowledge and Understanding

Having successfully completed the module, students should be able to demonstrate knowledge and understanding of:

- A.1. The basics of calculus on functions of one variable, functions series, vectors and matrices.
- A.2. The implications of the basic mathematical theories.

B. Cognitive/Intellectual skills/Applications of Knowledge

Having successfully completed the module, students should be able to:

- B.1. Present simple arguments and conclusion using the mathematical theories.
- B.2. Analyze and evaluate problems in mathematical and engineering.

C. Communication /ICT/Numeracy/Analytic Techniques/Practical Skills.

Having successfully completed the module, students should be able to:

- C.1. Apply the mathematical knowledge to solve problems in range of engineering situations.
- C.2. Carry out mathematical and numerical manipulation with confidence and accuracy.

D. General transferable skills

Having successfully completed the module, students should be able to:

- D.1. Assimilate Abstract Ideas;
- D.2. Communicate information having a mathematical content.

INDICATIVE CONTENT

Introduction: Recall of some background tools: sets of real numbers, interval, numerical function and their classical property.

CHAPTER I: Calculus and functions of one real variable

- I.1. Definitions of limits and continuity, techniques of finding limits.
- I.2. Techniques of differentiation, tangent lines and rates of change.
- I.3. Applications of derivatives: Indeterminate form and L' hospital's rule and Extreme values
- I.4. Techniques of indefinite integration.
- I.5. Definite integrals, properties of definite integrals.
- I.6. Applications of definite integrals.
- I.7. Improper integrals.

Chapter II: Numerical and functional series.

- II.1. Convergence and divergence of numerical sequences and series,
- II.2. Techniques for testing the convergence of numerical sequences and series.
- II.3. Power series, Maclaurin series, Taylor series and their applications.

Chapter III. Introduction to matrices theory.

- III.1. Introduction to matrices, Algebra of matrices, Special matrices.Determinants and their properties.
- III.2. System of linear equations, solution by Elimination Methods (Gauss-Elimination, Gauss-Jordan algorithms)

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Chapter IV. Element of Vector's spaces

- IV.1. Vectors, operation by vectors, scalar, vector and triple products.
- IV.2. Linear independence, bases, Vector space.
- IV.3. Linear operators, Eigenvalues, Eigenvectors; Orthogonal matrices.

$$6. \text{ ASSESSMENT STRATEGY: } \left\{ \begin{array}{l} \text{Quizzes: } 10 \\ \text{Assignments: } 10 \\ \text{CAT (2) : } 30 \\ \text{Final Exam: } 50 \end{array} \right.$$

7. REFERENCES:

- 1. Linear Algebra with applications; sixth Edition, Steven J.Leon; 2002 by Printice-Hall, Inc.
- 2. Mathematics for science, S.M. UPPAL,1996, New Age International(P),Limited, publishers.
- 3. Calculus with analytic geometry Howard Anton, second Edition, 1984, by Anton Textbooks, Inc.
- 4. Recueil d'exercices et de problèmes d'analyse mathématique.

GENERAL INTRODUCTION

0.1 The Set $\mathbb{R}$ of Real Numbers

a) $(\mathbb{R}, +, \cdot)$ is a **field** because:

- (1) $(\mathbb{R}, +)$ is an abelian group with identity element equal 0;
- (2) the multiplication is distributive with respect to the addition in $\mathbb{R}$;
- (3) $(\mathbb{R}/\{0\}, \cdot)$ is an abelian group with identity element equal 1.

b) $(\mathbb{R}, \leq)$ is a **completely ordered field** because: the relation $\dots \leq, \dots$ has three following properties:

- (1) $\forall x \in \mathbb{R} : x \leq x$ (**reflexivity**)
- (2) $\forall x, y \in \mathbb{R}$: if $x \leq y$ and if $y \leq x$ then $x = y$ (**antisymmetry**)
- (3) $\forall x, y, z \in \mathbb{R}$: if $x \leq y$ and $y \leq z$ then $x \leq z$ (**transitivity**)

c) The absolute value of a real number **a** is a positive number defined by

$$|a| = \begin{cases} a, & \text{if } a > 0 \\ 0, & \text{if } a = 0 \\ -a, & \text{if } a < 0 \end{cases}$$

and it verifies the following properties:

- (1) $\forall x \in \mathbb{R} : |x| = |-x|$
- (2) $\forall x \in \mathbb{R} : |x| = 0 \Leftrightarrow x = 0$
- (3) $\forall x, y \in \mathbb{R} : |x| = |y| \Leftrightarrow x = y \text{ or } x = -y$
- (4) $\forall x, y \in \mathbb{R} : |x \cdot y| = |x| \cdot |y|$
- (5) $\forall x, y \in \mathbb{R} : \left| \frac{x}{y} \right| = \frac{|x|}{|y|}$; for $y \neq 0$.
- (6) $\forall x, y \in \mathbb{R} : |x + y| \leq |x| + |y|$
- (7) $\forall x, y \in \mathbb{R} : |x - y| \geq |x| - |y|$

d) **The identities are verified in:** $\forall x, y, z \in \mathbb{R}; \forall n, p \in \mathbb{N}$:

- (1) (i) $(x \cdot y \cdot z)^n = x^n \cdot y^n \cdot z^n$; and (ii) $\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$; $y \neq 0$
- (2) (i) $(x^n)^p = x^{n \cdot p}$; and (ii) $x^n x^p = x^{n+p}$; and (iii) $\frac{x^n}{x^p} = x^{n-p}$; $x \neq 0$
- (3) (i) $\sqrt[n]{x \cdot y \cdot z} = \sqrt[n]{x} \cdot \sqrt[n]{y} \cdot \sqrt[n]{z}$; and (ii) $\sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}}$, $y \neq 0$
- (4) (i) $(\sqrt{x})^p = \sqrt[p]{x^p}$; and (ii) $\sqrt[p]{x^p} = x^{\frac{p}{p}}$
- (5) (i) $(x + y)^2 = x^2 + 2xy + y^2$; and (ii) $(x - y)^2 = x^2 - 2xy + y^2$
- (iii) $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$; and (iv) $(x - y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$
- (v) $(x + y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$; and (vi) $(x - y)^4 = x^4 - 4x^3y + 6x^2y^2 - 4xy^3 + y^4$
- (6) (i) $(x - y)(x + y) = x^2 - y^2$; and (ii) $(x - y)(x^2 + xy + y^2) = x^3 - y^3$
- (iii) $(x + y)((x^2 - xy + y^2) = x^3 + y^3$

e) **The interval in $\mathbb{R}$:**

Let $a, b \in \mathbb{R} : a < b$:

- The set $]a, b[= \{x \in \mathbb{R} : a < x < b\}$ is called the **open interval** of bounds a and b .
- The set $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$ is the **closed interval** of bounds a and b .
- The set $[a, b[= \{x \in \mathbb{R} : a \leq x < b\}$ and the set $]a, b] = \{x \in \mathbb{R} : a < x \leq b\}$ are called the **semi-open interval** of bounds a and b .

In the cases which are above; the centre of the interval is $x_0 = \frac{a+b}{2}$, and the radius of the interval is $r = |\frac{a-b}{2}|$. We denote $I =]x_0 - r; x_0 + r[$ to express an open interval I of centre x_0 and radius r .

- Let $+\infty$ and $-\infty$ be two symbols used to indicate a number such that: $\forall x \in \mathbb{R} : x < +\infty$ and $x > -\infty$.

We call the extended real number line, the set $\mathbb{R} \cup \{-\infty, +\infty\}$ denoted by $\mathbb{R}'$ and $\mathbb{R} =]-\infty, +\infty[$.

Rules:

$$(1) \quad \forall x \in \mathbb{R} : x.(+\infty) = \begin{cases} +\infty; & \text{if } x > 0 \\ -\infty; & \text{if } x < 0 \end{cases}; \quad \text{and} \quad x.(-\infty) = \begin{cases} +\infty; & \text{if } x < 0 \\ -\infty; & \text{if } x > 0 \end{cases}$$

$$(2) \quad \forall x \in \mathbb{R} : x + (+\infty) = (+\infty); \quad \text{and} \quad x + (-\infty) = (-\infty)$$

$$(3) \quad (+\infty)(+\infty) = (+\infty); \quad \text{and} \quad (+\infty) + (+\infty) = (+\infty);$$

$$(4) \quad (-\infty)(-\infty) = (+\infty); \quad \text{and} \quad (-\infty) + (-\infty) = (-\infty); \quad \text{and} \quad (+\infty)(-\infty) = (-\infty)$$

Remark: The expressions $0.\infty$; $(+\infty) + (-\infty)$; $\frac{\infty}{\infty}$; $\frac{0}{0}$; $1^\infty, 0^0$ are called **indeterminate forms and can not indicate a real quality**.

- Let $a \in \mathbb{R}$, the neighbourhood of a is very part V of $\mathbb{R}$ such that there exists an open interval containing a and included in V ; i.e., $V \subset \mathbb{R}$ and $a \in \mathbb{R}$, V is a neighbourhood of $a \Leftrightarrow \exists r > 0 :]a - r, a + r[\subset V$

Interval	Set notation	Geometric picture	Namely
$[a, b]$	$\{x : a \leq x \leq b\}$		Closed interval
(a, b)	$\{x : a < x < b\}$		Open interval
$(a, b]$	$\{x : a < x \leq b\}$		Half closed
$[a, b)$	$\{x : a \leq x < b\}$		Half open
$(-\infty, b]$	$\{x : x \leq b\}$		Closed above
$(-\infty, b)$	$\{x : x < b\}$		Half open
$[a, +\infty)$	$\{x : x \geq a\}$		Half closed
$(a, +\infty)$	$\{x : x > a\}$		
$(-\infty, +\infty)$	$\{x : x \text{ is a real number}\}$		Open

EXERCISES 0.1

1. For each pair of sets, find $A \cap B$ and $A \cup B$.

a) $A = \{2, 4, 6, 8\}$, $B = \{1, 3, 4, 6, 7\}$

b) $A = \{5, 10, 15, \dots\}$, $B = \{10, 20, 30, 40, \dots\}$

c) $A = \{1, 3, 5, 7, \dots\}$, $B = \{2, 4, 6, 8, \dots\}$

2. Given the set $U = \{2, 3, 4, 5, 6, 7, 8\}$ $A = \{3, 5, 7, 8\}$, $B = \{2, 4, 5, 6, 7\}$.

Find a) A' , b) B' , c) $A' \cup B'$, d) $A' \cap B'$, e) $A \cap B'$, f) $A \cup B'$, g) $A' \cap B$

3. If $U = \{0, 1, 2, 3, \dots\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7\}$, and $C = \{3, 5, 6, 7, 8\}$.

Find a) $A' \cap B'$ b) $(A \cap B) \cap C$ c) $(A \cup B) \cup C$ d) $(A \cap B)' \cup C$ e) $A \cup (B \cap C)$

f) $A' \cap (B' \cap C')$.

4. a) If $A = \{5, 10\}$, $B = \{x, y, z\}$, then find: $B \times A$ and $A \times B$

b) If $C = \{4, 6, 8\}$ $D = \{3, 5, 7\}$, then find: $C \times D$, $D \times C$ and $\text{card}(C \times D)$

c) If $A = \{1, 2, 3\}$, $B = \{3, 4, 5, 6\}$ and $C = \{4, 5, 6, 7\}$ find each of the following:

(i) $(A \cap C) \times B$ (ii) $(B \cap C) \times A$ (iii) $\text{card}(B \times C)$ (iv) $(B \cup C) \times A$ (v) $(A \cap B) \times (B \cap C)$.

5. Calculate in real number:

a) $[-3, 2] \cup [1, 4]$ b) $[4, 6] \cup [8, 11]$ c) $(-4, \cup(-5, 1)$ d) $[2, 4) \cap (0, 5]$ e) $[1.2, 3) \cup (1.4, \sqrt{2})$

f) $(-\infty, -1) \cup (-3, +\infty)$ g) $(-\infty, 5) \cap [0, \infty)$

6. If $U = \mathbb{R}$ and $A =]-1, 1[$, $B = [0, 1]$, $C = [1, 2[$, $D = \{0\}$, $E =]-\infty, 0]$,

$F =]-\infty, 1]$, $G = [1, +\infty[$.

Find: a) $A', B', C', D', E', F', G'$

b) $A \cap B$, $A \cup B$, $(A \cup B)'$, $(A \cap B)'$, $A' \cup A'$, $A' \cap B'$

c) $A - C$, $(B - D) \cup F$, $(E \cup F)', F - E$, $(A \cap B) \cup C$

7. Give us all subst of the following sets: $U = \{2, 3, 4, 5, 6, 7, 8\}$, $A = \{3, 5, 7, 8\}$

1 Chapter 1: Real Valued Function

1.1 Some important definitions

A **real-valued function** f defined on an interval I of real number is a rule which assigns to each number x in exactly one real number $f(x)$. We have $f : I \rightarrow \mathbb{R}$, $x \rightarrow f(x) \in \mathbb{R}$.

The set I of all those numbers x for which $f(x)$ defined is called the **domain or domain of definition** of the function f and it is denoted by $dom f$.

The number $f(x)$, read “ f of x ”, is called the **value of the function f at the point x** .

The set of all values $y = f(x)$ is called the **range** of f . The range of f is denoted by $Im f$. Thus $Im f = \{y : y = f(x), x \in dom f\}$. Where we describe the function f by writing a formula $y = f(x)$, we call x **independent variable** and y **dependent variable** because the value of y depends through f upon the choice of x .

The graph of f is denoted by G_f and defined as follows.

$$G_f = \{(x, f(x)) : (x \in dom f) \wedge (f(x) \in Im f)\}$$

1.1.1 Monotonic - Increasing and Decreasing Functions

The function f is **increasing** on the interval I , if $\forall x_1, x_2 \in I : x_1 < x_2 \implies f(x_1) < f(x_2)$.

The function f is **decreasing** on the interval I , if $\forall x_1, x_2 \in I : x_1 < x_2 \implies f(x_1) > f(x_2)$.

The function f is said **monotonic**, if it is either increasing or decreasing or constant.

1.1.2 Bounded Function

The function f is **bounded above** by number M , if $\forall x \in dom f$ implies $f(x) \leq M$.

The function f is **bounded below** by a number m , if $\forall x \in dom f$ implies $f(x) \geq m$.

The function f is said **bounded**, if it is bounded above and below simultaneously i.e., $\forall x \in dom f; \exists m, M : m \leq f(x) \leq M$.

The numbers M and m are referred to as an **upper bound** and **lower bound**, respectively.

The **least upper bound** for f is denoted by $sup f$ and the **greatest lower bound** by $inf f(x)$.

1.1.3 Odd and Even Function

The function $f(x)$ is an **even function** on interval I , if $\forall x, -x \in I : f(-x) = f(x)$.

The function $f(x)$ is an **odd function** on interval I , if $\forall x, -x \in I : f(-x) = -f(x)$.

An even function is symmetric with respect to y-axis. An odd function is symmetric with respect to the origin of coordinate axes.

1.1.4 Periodic Function

The function f is called **periodic function** with period T , if $f(x + T) = f(x)$, where T is the smallest positive number which satisfies this property for all values of x .

1.1.5 Composite function or Function of function

The **composite** of functions f and g is a new function denoted by $f \circ g$ and defined as follows: $(f \circ g)(x) = f(g(x))$. The $f \circ g$ is called a **composite function**.

1.1.6 Inverse function

Two functions $f : I_1 \rightarrow J_1$, $g : I_2 \rightarrow J_2$ are said to be **inverse functions**, or **inverse** of each other if and only if:

- i. The domain of each is the range of the other i.e. $I_1 = J_2$ and $J_1 = I_2$;
- ii. $(f(g(x))) = x$, $\forall x \in I_2 \wedge g(f(x)) = x$

Remark 1.1. (i) Implies that the range of values of f is the domain of g and vice-versa.

Is equivalent to $\forall x \in \text{dom } f, \forall y \in \text{dom } g : f(x) = y$ if and only if $g(y) = x$.

That is the graphs of two equations $y = f(x)$ and $x = f(y)$ is identical. On the other hand, the graphs of two equations $x = g(y)$ and $y = g(x)$ are reflections of one another across the line $y = x$. The inverse of f is denoted by f^{-1} .

Remark 1.2. The function f has inverse if and only if the function f is a one-to-one function.

1.1.7 Operations over functions

Many important functions are built of simpler ones. Suppose that f and g are functions, and c a fixed real number.

The scalar multiple cf , the sum $f + g$, the difference $f - g$, the product $f.g$ and the quotient $\frac{f}{g}$ are new functions determined by the formula:

$$(cf)(x) = cf(x); \quad (f \pm g)(x) = f(x) \pm g(x); \quad (f.g)(x) = f(x).g(x); \quad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}.$$

Remark 1.3:

$$f^{-1} \neq \frac{1}{f}; \quad \text{where} \quad \frac{1}{f}(x) = \frac{1}{f(x)}.$$

1.2 A brief catalog of elementary functions

1.2.1 Polynomials

A **polynomial** of degree n is a function of the form

$$p_n(x) = a_0 + a_1x + \dots + a_{n-1}x^{n-1} + a_nx^n,$$

where the coefficients $a_0, a_1, \dots, a_n$ are fixed real numbers. Thus a n^{th} degree polynomial is a sum of constant multiples of the **power functions** $1, x, x^2, \dots, x^n$.

1.2.2 Rational functions

A **rational function** is a quotient

$$f(x) = \frac{P_n(x)}{Q_n(x)} \quad \text{of two polynomials } P_n(x) \text{ and } Q_n(x).$$

Graphs of rational functions and polynomials have several features in common.

For instance, a rational function has only a finite number of zeros, because $f(x)$ can be zero only when $P_n(x)$ is zero. The graph of a rational function has only a finite number of bends. The denominator $Q_n(x)$ may be zero at point $x = a$, where the numerator is non zero. In this case the value of $f(x)$ will be extremely large when x is very close to a .

1.2.3 Trigonometric functions

There exists trigonometric functions: **sine, cosine, tangent, cosecant, secant and cotangent** denoted respectively by $\sin, \cos, \tan, \cos ec, \cot$. The sine and cosine functions describe oscillatory systems such as wave motion, electromagnetic radiation or simple harmonic motion.

The tangent function is undefined for $x = n\frac{\pi}{2}$, where n is an odd integer. Cosecant, secant and cotangent are simply the reciprocals of the sine, cosine and tangent.

$$\cos ec x = \frac{1}{\sin x}; \quad \sec x = \frac{1}{\cos x}; \quad \cot x = \frac{1}{\tan x}.$$

These functions are periodic functions.

$$\begin{aligned} \sin(x + 2\pi) &= \sin x; \quad \cos(x + 2\pi) = \cos x; \quad \cos ec(x + 2\pi) = \cos ec x; \\ \sec(x + 2\pi) &= \sec x; \quad \tan(x + \pi) = \tan x; \quad \cot(x + \pi) = \cot x. \end{aligned}$$

Cosine and secant are even functions and the remaining function are odd functions. Sine and cosine are bounded functions: $-1 \leq \sin x; \cos x \leq 1$.

1.2.4 Inverse trigonometric functions

The notation $\arcsin x$ is used to denote any angle whose sin is x . $\text{Arc} \sin x$ is usually used to denote the principal value. Similar notation is used for another inverse trigonometric functions. The domains and ranges of principal values of the inverse of trigonometric functions are defined as follows.

$$\arcsin x = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]; \quad \arccos x : [0, \pi] \rightarrow [-1, 1]; \quad \arctan x = \left]-\frac{\pi}{2}, \frac{\pi}{2}\right[\rightarrow]-\infty, +\infty[.$$

1.2.5 Exponential and Logarithmic functions

Definition of Exponential Function

The exponential function f with base a is denoted by $f(x) = a^x$, where $a \neq 1$, ($a \in \mathbb{R}_{0+}/\{1\}$) and x is any real number. The function value will be positive because a positive base raised to any power is positive. This means that the graph of the exponential function $f(x) = a^x$ will be located in quadrants I and II .

For example, if the base is 2 and $x = 4$, the function value $f(4)$ will equal to 16. A corresponding point on the graph of $f(x) = 2^x$ would be $(4, 16)$.

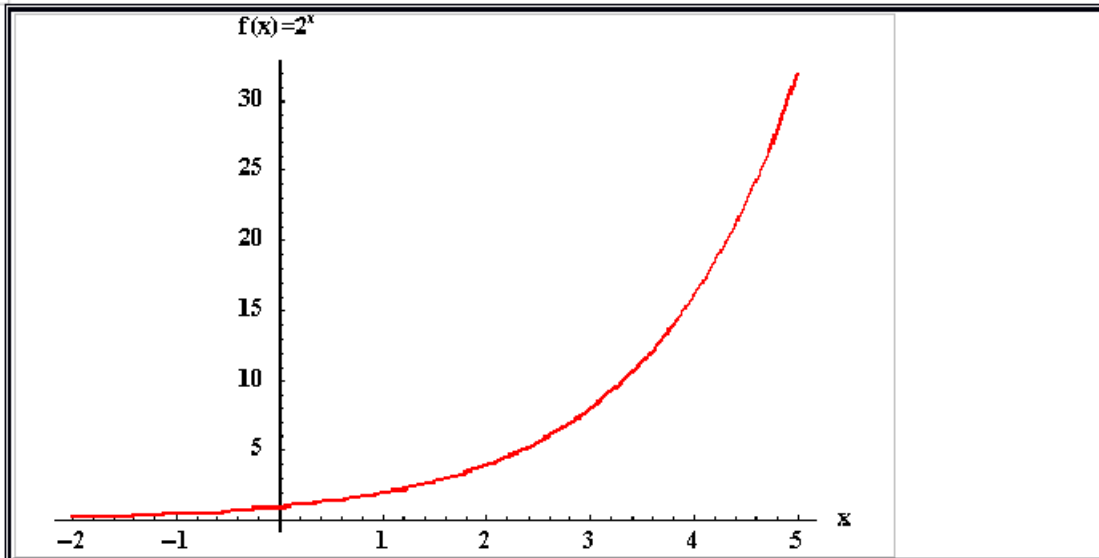


Figure 1: A graph of the exponential function $f(x) = 2^x$.

Definition of Logarithmic function

For $x > 0$, $a > 0$, and $a \neq 1$, we have

$$f(x) = \log_a(x) \text{ if and only if } a^{f(x)} = x.$$

Since $x > 0$, the graph of the above function will be in quadrants *I* and *IV*.

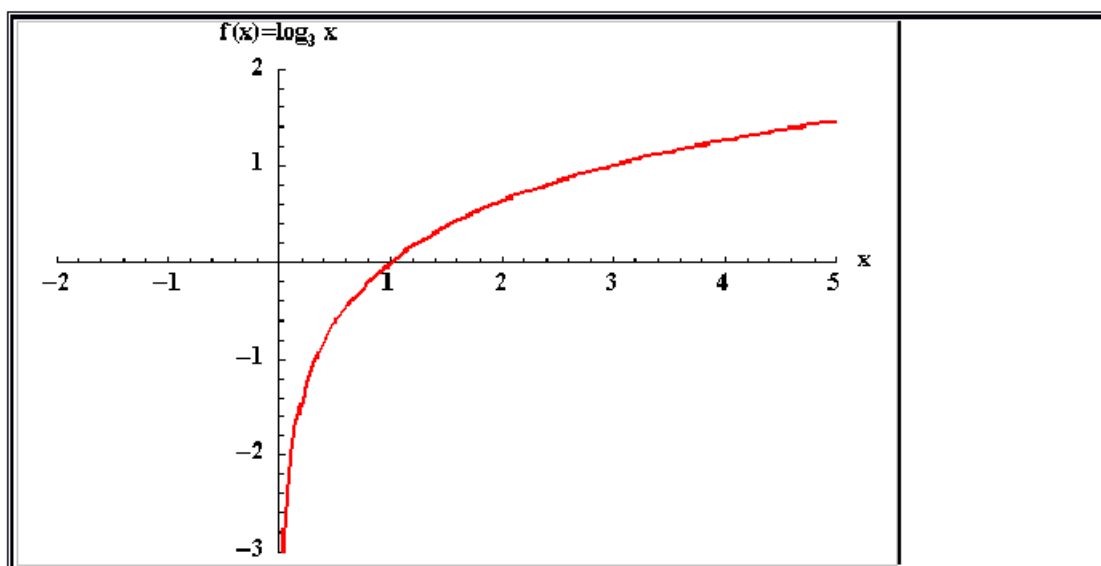


Figure 2: A graph of the logarithmic function $f(x) = \log_3 x$.

Comments on Logarithmic functions

- The exponential equation $4^3 = 64$ could be written in terms of a logarithmic equation as $\log_4(64) = 3$.

- The exponential equation $5^{-2} = \frac{1}{25}$ can be written as the logarithmic equation as $\log_5 \frac{1}{25} = -2$.
- Since logarithms are nothing more than exponents, you can use the rules of exponents with logarithms.
- Logarithmic functions are the inverse of exponential functions. For example if $(4, 16)$ is a point on the graph of an exponential function, then $(16, 4)$ would be the corresponding point on the graph of the inverse logarithmic function.

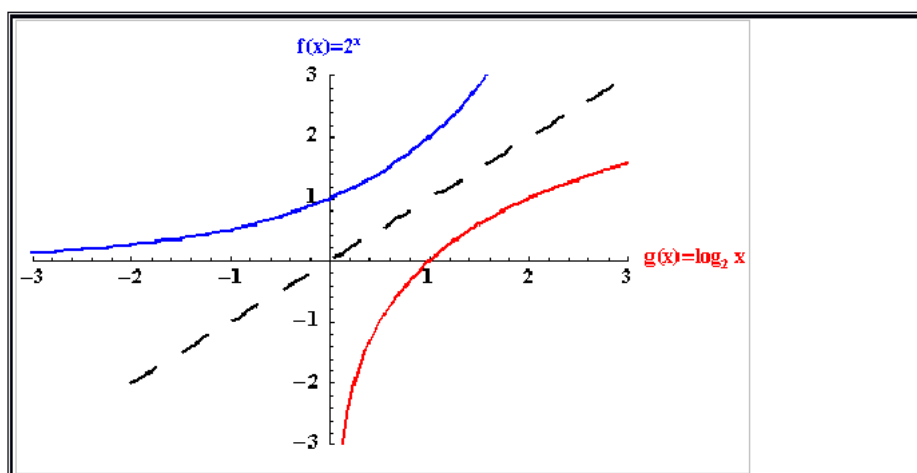


Figure 3: A graph of the exponential function and logarithmic function.

Properties

- $e^0 = 1 \iff \ln 1 = 0$
- $e^1 = e \iff \ln e = 1$
- $e^{\ln(x)} = x$ if $x > 0$
- $\ln(e^x) = x$.
- $\ln(x) < \ln(y)$ for $0 < x < y$.
- Like all logarithms, the natural logarithm maps multiplication into addition:

$$\ln(xy) = \ln x + \ln y$$

Thus, the logarithm function is an **isomorphism** from the **group** of positive real numbers under multiplication to the group of real numbers under addition, represented as a **functor**:

Logarithmic rules

As logarithms are exponents, mathematical operations involving them follow the same rules as those for exponents.

Common Logarithm	Natural Logarithm
$\log xy = \log x + \log y$	$\ln xy = \ln x + \ln y$
$\log \frac{x}{y} = \log x - \log y$	$\ln \frac{x}{y} = \ln x - \ln y$
$\log x^y = y \log x$	$\ln x^y = y \ln x$
$\log \sqrt[y]{x} = \log x^{\frac{1}{y}} = \left(\frac{1}{y}\right) \log x$	$\ln \sqrt[y]{x} = \ln x^{\frac{1}{y}} = \left(\frac{1}{y}\right) \ln x$

Change of base

While there are several useful identities, the most important for calculator use lets one find logarithms with bases other than those built into the calculator (usually $\log_e$ and $\log_{10}$). To find a logarithm with base r , using any other base b .

$$\log_r(x) = \frac{\log_b(x)}{\log_b(r)} \quad \text{for any base } b \text{ or simply} \quad \log_r(x) = \frac{\log(x)}{\log(r)}$$

This is because the definition of logarithm says that

$$x = b^a \iff \log_b x = a.$$

But we can also get a by using the base k logarithm and then get

$$\log_k x = \log_k b^a = a \log_k b \iff \frac{\log_k x}{\log_k b} = a,$$

with $b \neq 1$, because $\log_k 1 = 0$. Any number to the power 0 is equal to 1.

Moreover, this result implies that all logariyhm functions (whatever the base) are **similar** to each other.

Formulae

$$\begin{aligned} \log_k x &= \log_k l \cdot \log_l x \\ \log_k x &= \frac{1}{\log_x k} \\ \log_k x &= \frac{\log_k l}{\log_x l} \\ &= \frac{\log_l x}{\log_l k} \end{aligned}$$

EXERCISES 0.2

Problem 1: Solve for x in the equation

$$8 + 5^{2x+2} = 12$$

Problem 2: Solve for x in the equation

$$\frac{4000}{2 + 7^{3x}} = 5.$$

Problem 3: Solve for x in the equation

$$10e^{2x} - 31e^x + 15 = 0.$$

Problem 4: Solve for x in the equation

$$\left(1 + \frac{10}{12}\right)^{12x} = 2.$$

Problem 5: Solve for x in the equation

$$400 = 5000 \left(1 - \frac{4}{4 + e^{-0.002x}}\right).$$

Problem 6: Solve for x in the equation

$$5(8e^{2x} - 3)^3 = 625.$$

EXERCISES II

1. Solve for x in the equation: $4 + \log_3(7x) = 10$.
2. Solve for x in the equation: $\ln(2x - 1) = 3$.
3. Solve for x in the equation: $\ln(x + 1)^2 = 2$.
4. Solve for x in the equation: $\log_4 x - \log_4(x - 1) = \frac{1}{2}$.
5. Solve for x in the equation: $\log_2(x + 5) + \log_2(x + 2) = \log_2(x + 6)$.
6. Solve for x in the equation: $\log 8x - \log(1 + \sqrt{x}) = 2$.

1.3 Calculus on functions of one real variable

Limits and Continuity

1.3.1 Concept of a limit (existence, unicity) and properties

a) Concept of a limit at a point

Let $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 2x + 2$ and $g : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto \frac{2x^2-2}{x-1}$ two functions. Find the domain of definition, the graph and the limit at 1.

For $f(x)$, $\text{dom} f = \mathbb{R}$ and the graph is a straight line which passes through $(0, 2)$; $(-1, 0)$ and $(1, 4)$.

For $g(x)$, $\text{dom} g = \mathbb{R}/\{1\}$ and the graph is straight line which passes through $(0, 2)$; $(-1, 0)$ but doesn't pass through $(1, 4)$ because 1 is not element of the domain of g ; but when x is a neighbourhood of 1 we see that $g(x)$ is 4. We say that 4 is a limit of f and of g at 1 and we denote it by

$$\begin{aligned} f(x) &= \lim_{x \rightarrow 1} f(x) = 4 \\ &= \lim_{x \rightarrow 1} g(x). \end{aligned}$$

We say that

$$\lim_{x \rightarrow a} f(x) = b \Leftrightarrow \forall \epsilon > 0, \exists \delta > 0 : 0 < |x - a| < \delta \implies |f(x) - b| < \epsilon.$$

Example 1. *Prove that*

$$(1) \quad \lim_{x \rightarrow 2} (2x + 5) = 9; \quad (2) \quad \lim_{x \rightarrow -2} (2x^3 - 10x) = 4$$

. Solutions:

$$(1) \quad \lim_{x \rightarrow 2} (2x + 5) = 9 \Leftrightarrow \forall \epsilon > 0, \exists \delta > 0 : 0 < |x - 2| < \delta \implies |f(x) - 9| < \epsilon$$

Or

$$|f(x) - 9| = |(2x + 5) - 9| = |2x - 4| = 2|x - 2| < 2\delta$$

; then we can get $2\delta \leq \epsilon \implies \delta \leq \frac{\epsilon}{2}$. Consequently, we get $\delta(\epsilon) \leq \frac{\epsilon}{2}$ because if $|x - 2| < \delta$ with $\delta \leq \frac{\epsilon}{2}$ we are sure that $|(2x + 5) - 9| < \epsilon$.

(2)

$$\lim_{x \rightarrow -2} (2x^3 - 10x) = 4 \Leftrightarrow \forall \epsilon > 0, \exists \delta > 0 : 0 < |x + 2| < \delta \implies |f(x) - 4| < \epsilon$$

Or

$$\begin{aligned} |f(x) - 4| &= |(2x^3 - 10x) - 4| = |2x^3 - 10x - 4| = 2|x^3 - 5x - 2| \\ &= 2|(x + 2)^3 - 5x - 2 - (6x^2 + 12x + 8)| = 2|(x + 2)^3 - 6x^2 - 17x - 10| < \epsilon \\ &= 2|(x + 2)^3 - 6(x + 2)^2 + 7x + 14| = 2|(x + 2)^3 - 6(x + 2)^2 + 7(x + 2)| \\ &\leq 2|(x + 2)^3| + 12|(x + 2)^2| + 14|(x + 2)| \\ &< 2|x + 2|^2 + 12|x + 2|^2 + 12|x + 2| \\ &< 2\delta^3 + 12\delta^2 + 14\delta \quad \text{because} \quad \delta^3 < \delta^2 \quad \text{for the very small numbers.} \end{aligned}$$

Then we can get $28\delta \leq \epsilon \implies \delta \leq \frac{\epsilon}{28}$.

Limit on the right and limit on the left:

Let $a \in \mathbb{R}$, the variable x can approach a from the right or from the left side. We distinguish these 2 limits by $x \rightarrow a^+$ and $x \rightarrow a^-$. Then $\lim_{x \rightarrow a^+} f(x) = b$ indicates that b is the limit of f on the right side at a . And $\lim_{x \rightarrow a^-} f(x) = c$ indicates that c is the limit of f on the left side at a .

Theorem 1.1. *If the limit of a function f at a point a exists, this limit is unique.*

Consequence: If $\lim_{x \rightarrow a^+} f(x) \neq \lim_{x \rightarrow a^-} f(x)$, then the limit at a does not exist.

Example 2. *Let*

$$f(x) = \frac{|x - 3|}{2x - 6}$$

, find the limit at 3.

$$\begin{aligned} |x - 3| &= \begin{cases} x - 3, & \text{if } x > 3 \\ -x + 3, & \text{if } x < 3 \end{cases} \quad ; \quad \text{then} \\ \lim_{x \rightarrow 3^+} f(x) &= \lim_{x \rightarrow 3^+} \frac{x - 3}{2(x - 3)} = \frac{1}{2}; \quad \text{and} \\ \lim_{x \rightarrow 3^-} f(x) &= \lim_{x \rightarrow 3^-} \frac{-(x - 3)}{2(x - 3)} = -\frac{1}{2}; \\ &\text{then the limit at 3 does not exist.} \end{aligned}$$

Properties of a limit at a point

If the functions $f(x)$ and $g(x)$ admit a limit at the point a then:

$$\begin{aligned} (1) \quad \lim_{x \rightarrow a} (f(x) \pm g(x)) &= \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x). & (2) \quad \lim_{x \rightarrow a} f(x) \cdot g(x) &= \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x). \\ (3) \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}. & (4) \quad \lim_{x \rightarrow a} f(x)^{g(x)} &= \left(\lim_{x \rightarrow a} f(x) \right)^{\lim_{x \rightarrow a} g(x)}. \end{aligned}$$

Examples: Let $f(x) = 2x + 3$ and $g(x) = x^2 - 5$. We calculate that

$$\begin{aligned} (1) \quad \lim_{x \rightarrow -1} (f(x) - g(x)) &= 5 & (2) \quad \lim_{x \rightarrow 2} f(x) \cdot g(x) &= -7 \\ (3) \quad \lim_{x \rightarrow 3} \frac{f(x)}{g(x)} &= \frac{9}{4} & (4) \quad \lim_{x \rightarrow 1} f(x)^{g(x)} &= \frac{1}{625} \end{aligned}$$

Extension of the limit at infinite

The concept of the limit $\lim_{x \rightarrow a} f(x) = b$ can be defined for a or b which are infinite:

(1) If a is infinite and b is finite:

$$\begin{aligned} \lim_{x \rightarrow -\infty} f(x) = b &\Leftrightarrow \forall \epsilon > 0, \exists A > 0 : x < -A \implies |f(x) - b| < \epsilon \\ \lim_{x \rightarrow +\infty} f(x) = b &\Leftrightarrow \forall \epsilon > 0, \exists A > 0 : x > A \implies |f(x) - b| < \epsilon \end{aligned}$$

(2) If a is finite and b is infinite:

$$\lim_{x \rightarrow a} f(x) = -\infty \Leftrightarrow \forall B > 0, \exists \delta > 0 : |x - a| < \delta \Rightarrow f(x) < -B.$$

$$\lim_{x \rightarrow a} f(x) = +\infty \Leftrightarrow \forall B > 0, \exists \delta > 0 : |x - a| < \delta \Rightarrow f(x) > B.$$

(3) If a is infinite and b is infinite:

$$\lim_{x \rightarrow -\infty} f(x) = -\infty \Leftrightarrow \forall B > 0, \exists A > 0 : x < -A \Rightarrow f(x) < -B.$$

$$\lim_{x \rightarrow -\infty} f(x) = +\infty \Leftrightarrow \forall B > 0, \exists A > 0 : x < -A \Rightarrow f(x) > B.$$

$$\lim_{x \rightarrow +\infty} f(x) = -\infty \Leftrightarrow \forall B > 0, \exists A > 0 : x > A \Rightarrow f(x) < -B.$$

$$\lim_{x \rightarrow +\infty} f(x) = +\infty \Leftrightarrow \forall B > 0, \exists A > 0 : x > A \Rightarrow f(x) > B.$$

Remark: The symbols 0^+ and 0^- are called the infinitely smalls.

Examples:

$$(1) \quad \lim_{x \rightarrow 3^-} \frac{2x}{3-x} = \frac{6}{0^+} = +\infty; \quad (2) \quad \lim_{x \rightarrow 3^+} \frac{2x}{3-x} = \frac{6}{0^-} = -\infty; \quad \lim_{x \rightarrow +\infty} 2x \cdot (3-x) = +\infty \cdot (-\infty) = -\infty$$

Indeterminate forms: ($0 \cdot \infty$; $\infty - \infty$; $\frac{\infty}{\infty}$; $\frac{0}{0}$; 1^∞ , 0^0 , ∞^0 , 0^∞).

The symbols ($0 \cdot \infty$; $\infty - \infty$; $\frac{\infty}{\infty}$; $\frac{0}{0}$; 1^∞ , 0^0 , ∞^0 , 0^∞). don't have a signification quality, and are called indeterminate forms in calculation of limits. To give a precision to the forms above we use some theorems in this operation.

Theorem 1.2. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$

Theorem 1.3. If $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ and if $g(x) = b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0$, then

$$(1) \quad \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} a_n x^n; \quad \text{and} \quad \lim_{x \rightarrow \infty} g(x) = \lim_{x \rightarrow \infty} b_m x^m$$

$$(2) \quad \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \begin{cases} 0, & \text{if } n < m \\ \frac{a_n}{b_m}, & \text{if } n = m \\ \infty, & \text{if } n > m \end{cases}$$

Theorem 1.4. Let $e = 2.71828\dots$, then number of the Neper; then

$$\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e \quad \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e.$$

Suppose that we need to find the limit of a quantity $y = [f(x)]^{g(x)}$, where the limits of f and g as $x \rightarrow a$ are such that one of the indeterminate forms: 0^0 , ∞^0 or 1^∞ is produced.

We have the following four steps for finding the limit of $[f(x)]^{g(x)}$ as $x \rightarrow a$:

- i. Let $y = [f(x)]^{g(x)}$
- ii. Simplify $\ln y = g(x) \ln f(x)$
- iii. Evaluate $\lim_{x \rightarrow a} \ln y$
- iv. $\lim_{x \rightarrow a} [f(x)]^{g(x)} = e^{\lim_{x \rightarrow a} \ln y}$

Indeterminate form of type $\frac{\infty}{\infty}$

$$\lim_{x \rightarrow +\infty} \frac{\sum_{k=0}^n a_k x^k}{\sum_{i=0}^m b_i x^i} = \begin{cases} \frac{a_n}{b_m}, & n = m \\ 0, & n < m \\ \infty, & n > m \end{cases}$$

Indeterminate form of type $\frac{0}{0}$.

$$a) \quad \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \left(\frac{0}{0} \right) = \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x-1} = \lim_{x \rightarrow 1} x + 1 = 2.$$

$$b) \quad \lim_{x \rightarrow 0} \frac{\sin 4x}{\sin 2x} = \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{4x \frac{\sin 4x}{4x}}{2x \frac{\sin 2x}{2x}} = \lim_{x \rightarrow 0} \frac{4x}{2x} \cdot \frac{\lim_{x \rightarrow 0} \frac{\sin 4x}{4x}}{\lim_{x \rightarrow 0} \frac{\sin 2x}{2x}} = 2$$

Exercises

1. Prove the 3 theorems above.
2. Calculate:

$$(1) \quad \lim_{x \rightarrow 0} \frac{2x + \sin 4x}{x - \sin 3x};$$

$$(2) \quad \lim_{x \rightarrow \infty} \left(\frac{-3x^2 + x}{-x^2 - x + 2} \right);$$

$$(3) \quad \lim_{x \rightarrow 2} \frac{x - 2}{\sqrt[3]{3x + 2} - 2};$$

$$(4) \quad \lim_{x \rightarrow \infty} \left(\frac{x + 4}{x + 2} \right)^{3x};$$

$$(5) \quad \lim_{x \rightarrow 1} \left(\frac{-x^2 + x}{-x^2 - x + 2} \right)$$

Solutions

(1) is an indeterminate form $\frac{0}{0}$. Then:

$$\lim_{x \rightarrow 0} \frac{2x + \sin 4x}{x - \sin 3x} = \frac{\lim_{x \rightarrow 0} x \left(2 + 4 \frac{\sin 4x}{4x} \right)}{x \left(1 - \frac{\sin 3x}{3x} \right)} = \frac{2 + 4 \lim_{x \rightarrow 0} \frac{\sin 4x}{4x}}{1 - 3 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}} = -3$$

(2) is an indeterminate form $\frac{\infty}{\infty}$. Then we apply theorem 2 and

$$(2) \quad \lim_{x \rightarrow \infty} \left(\frac{-3x^2 + x}{-x^2 - x + 2} \right) = 3$$

(3) is an indeterminate form $\frac{0}{0}$.

$$\lim_{x \rightarrow 2} \frac{x-2}{\sqrt[3]{3x+2}-2} = \lim_{x \rightarrow 2} \frac{(x-2) \left(\sqrt[3]{(3x+2)^2} + 2\sqrt[3]{3x+2} + 4 \right)}{(\sqrt[3]{3x+2}-2) \left(\sqrt[3]{(3x+2)^2} + 2\sqrt[3]{3x+2} + 4 \right)} = \frac{12}{-6} = -3$$

(4) is an indeterminate form 1^∞ ; or

$$\lim_{x \rightarrow \infty} \left(\frac{x+4}{x+2} \right)^{3x} = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{\frac{x+2}{2}} \right)^{3x}.$$

If $t = \frac{x+2}{2}$ the limit become $\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t} \right)^{12t-6} = e^{12}$

(5) is an indeterminate form $\frac{0}{0}$. Then

$$\lim_{x \rightarrow 1} \left(\frac{x(1-x)}{(1-x)(x+2)} \right) = \frac{1}{2}$$

Indeterminate form of type 1^∞

$$a) \quad \lim_{x \rightarrow +\infty} \left(1 + \frac{2}{x} \right)^x = (1^\infty). \quad \text{Put } x = 2t, \text{ then } t = \frac{x}{2} \text{ and if } x \rightarrow +\infty \implies t \rightarrow +\infty$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x} \right)^x = \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t} \right)^{2t} = \left[\lim_{t \rightarrow +\infty} \left(1 + \frac{1}{t} \right)^t \right]^2 = e^2$$

$$b) \quad \lim_{x \rightarrow +\infty} \left(1 + \frac{k}{x} \right)^x = (1)^\infty. \quad \text{Put } x = kt, \text{ then } t = \frac{x}{k} \text{ and if } x \rightarrow +\infty \implies t \rightarrow +\infty$$

$$\lim_{x \rightarrow +\infty} \left(1 + \frac{k}{x} \right)^x = \lim_{t \rightarrow +\infty} \left(1 + \frac{1}{t} \right)^{kt} = \left[\lim_{t \rightarrow +\infty} \left(1 + \frac{1}{t} \right)^t \right]^k = e^k$$

$$c) \quad \lim_{x \rightarrow +\infty} \left(\frac{x+2}{x-1} \right)^x = (1^\infty); \quad \frac{x+2}{x-1} = 1 + \frac{3}{x-1}; \quad \lim_{x \rightarrow +\infty} \left(\frac{x+2}{x-1} \right)^x = \lim_{x \rightarrow +\infty} \left(1 + \frac{3}{x-1} \right)^x.$$

Put $t = \frac{1}{3}(x-1) \implies x = 3t+1$. If $x \rightarrow +\infty \implies t \rightarrow +\infty$

$$\lim_{x \rightarrow +\infty} \left(1 + \frac{3}{x-1} \right)^x = \lim_{t \rightarrow +\infty} \left(1 + \frac{1}{t} \right)^{3t+1} = \left[\lim_{t \rightarrow +\infty} \left(1 + \frac{1}{t} \right)^t \right]^3 \times \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t} \right) = e^3 \cdot e = e^4$$

1.3.2 Continuity of a function at a point.

Definitions and Examples

A function f is continuous at the point a . This is equivalent to the following three conditions that are verified as:

- (i) $f(a)$ exists (i.e., $a \in \text{dom} f$).
- (ii) $\lim_{x \rightarrow a} f(x)$ exists (i.e., $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$).
- (iii) $\lim_{x \rightarrow a} f(x) = f(a)$.

If one of the 3 conditions is not satisfied, we say that the function f is discontinuous at the point a .

The definition above is equivalent to the following definition: A function f is continuous at the point $a \in \text{dom} f$

$$\Leftrightarrow \forall \epsilon > 0, \exists \delta > 0; \forall x : |x - a| < \delta \implies |f(x) - f(a)| < \epsilon.$$

Examples:

$$(1) \quad f(x) = \begin{cases} \frac{\sin 2x}{x}, & \text{if } x \neq 0 \\ 2, & \text{if } x = 0 \end{cases} \quad \text{is discontinuous at the point } x = 0.$$

$$(2) \quad f(x) = \begin{cases} 1, & \text{if } x > 0 \\ 10, & \text{if } x = 0 \\ -2, & \text{if } x < 0 \end{cases} \quad \text{is discontinuous at the point } x = 0.$$

$$(3) \quad f(x) = \frac{x+2}{(x+1)(x-2)} \quad \text{is discontinuous at the point } x = 2 \text{ and at the point } x = -1.$$

$$(4) \quad f(x) = \frac{x^2-1}{x-1} \quad \text{is discontinuous at the point } x = 1.$$

Exercise: Study the continuity of the functions above and verify the conclusion given.

Classification of the discontinuity points of a function.

A) Eliminate or apparent discontinuity.

If $f(x)$ is not defined at a , but $\lim_{x \rightarrow a} f(x)$ exists, we say that is a point of an apparent or eliminable discontinuity. For example, $f(x) = \frac{x^2-1}{x-1}$ has an apparent discontinuity at $x = 1$. But if we

take $f(1) = 2$ the function is prolonged by continuity and in this case $f(x) = \begin{cases} \frac{x^2-1}{x-1}, & \text{if } x \neq 1 \\ 2, & \text{if } x = 1 \end{cases}$

is continuous at the point $x = 1$.

B) Simple discontinuity or discontinuity of the first type.

If $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ exist but are not equal, we say that $f(x)$ presents a simple discontinuity at $x = a$ (or a is a point of discontinuity of the first type).

For example, $f(x) = \begin{cases} 1, & \text{if } x > 0 \\ 10, & \text{if } x = 0 \\ -2, & \text{if } x < 0 \end{cases}$ has at $x = 0$ a point of discontinuity of the first type.

C) Discontinuity of the second type.

We say that a function f has a discontinuity of second type at a if one at least of its limits on the left and on the right does not or is infinite.

For example, $f(x) = \frac{x^2+1}{x-1}$ has a discontinuity of the second type at $x = 1$ because

$$\lim_{x \rightarrow 1^+} f(x) = +\infty \quad \text{and} \quad \lim_{x \rightarrow 1^-} f(x) = -\infty$$

.

Continuity of function on an interval.

A function f is continuous on an interval I of $\mathbb{R}$ if it is continuous at each point of I .

For example, $f(x) = \frac{1}{x}$ is continuous on $I = [1, 3]$ but is discontinuous on $I = [-1, 2]$

Properties:

(1) The functions $y = \sin x$, $y = \cos x$, $y = e^x$, polynomial functions are continuous on $\mathbb{R}$.

(2) If $f(x)$ and $g(x)$ are continuous on $\mathbb{R}$ then the function $\frac{f(x)}{g(x)}$ is continuous on $\mathbb{R}/\{x : g(x) = 0\}$.

(3) Theorem of Bolzano - Cauchy:

(Bolzano (1781 - 1848) is a Tcheque mathematician; Cauchy(1789 - 1857) is a French mathematician.)

Theorem 1.5. If the function $f(x)$ is continuous on the interval $[a, b]$ and if $f(a)$ and $f(b)$ take the value of different signs, then exists at least a point $c \in]a, b[$ such that $f(c) = 0$.

Application: Let $f(x) = x^3 + x^2 - 2x - 3$, verify if the graph of f passes through x -axis in the interval $[1, 2]$.

Verification: As a polynomial function $f(x)$ is continuous on $\mathbb{R}$ and in a very subset of $\mathbb{R}$ including $[1, 2]$. Secondly $f(1) = -3$ and $f(2) = 5$ and have different signs. Consequently there is $c \in]1, 2[$ such that $f(c) = 0$. Then c is a zero of $f(x)$ and the graph of f passes through x -axis at $(c, 0)$.

EXERCISES 1.1

1. Prove the statement using the ϵ and δ definition of limit.

$$a) \quad \lim_{x \rightarrow 3} \frac{x}{5} = \frac{3}{5} \quad b) \quad \lim_{x \rightarrow 6} \left(\frac{x}{4} + 3 \right) = \frac{9}{2} \quad c) \quad \lim_{x \rightarrow 3} (x^2 + x - 4) = 8$$

2. In the following exercises we told that $\lim_{x \rightarrow a} f(x) = l$, find a number δ such that $|f(x) - l| < \epsilon$ whenever $0 < |x - a| < \delta$:

$$a) \quad \lim_{x \rightarrow 3} (5x - 2) = 13, \quad \epsilon = 0.01 \quad b) \quad \lim_{x \rightarrow 6} \frac{x^2 - 1}{x + 1} = -2, \quad \epsilon = 0.05$$

3. Calculate the following limits:

$$a) \quad \lim_{x \rightarrow 8} \left(\frac{x - 8}{\sqrt[3]{x - 2}} \right) \quad b) \quad \lim_{h \rightarrow 0} \frac{\sqrt{x + h} - \sqrt{x}}{h}$$
$$c) \quad \lim_{x \rightarrow 0} \frac{\sin 5x}{\sin 2x} \quad d) \quad \lim_{x \rightarrow 0} \frac{\cos x - \cos a}{x - a} \quad e) \quad \lim_{n \rightarrow \infty} \left(n \sin \frac{\pi}{n} \right)$$

4. Find the following limits:

$$a) \quad \lim_{x \rightarrow 2^-} \left(\frac{x}{x - 2} \right) \quad b) \quad \lim_{x \rightarrow 2^+} \left(\frac{x}{x - 2} \right)$$
$$c) \quad \lim_{x \rightarrow 1^-} \left(\frac{x - 1}{|x - 1|} \right) \quad d) \quad \lim_{x \rightarrow 0} \left(\frac{\cosh x - 1}{x^2} \right)$$

5. Study the continuity of:

$$a) \quad y = \frac{x^2}{x - 2} \quad b) \quad y = \frac{1 + x^3}{1 + x}$$
$$c) \quad y = \sin \frac{\pi}{x} \quad d) \quad y = e^{\frac{1}{x+1}}$$

1.4 DIFFERENTIATION

1. Definition

Increment: If a variable x is given an increment Δx from $x = x$, an arbitrary but fixed value of x in its range, a function $y = f(x)$ will be given in turn an increment,

$$\Delta y = \Delta f(x) = f(x + \Delta x) - f(x).$$

The **average rate of change** of a function $f(x)$ with respect to x is given by $\frac{\Delta f(x)}{\Delta x}$.

The derivative of a function $f(x)$ with respect to x is denoted by $f'(x)$ and defined as

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}, \quad \text{provided the limit exists.}$$

Example: If $f(x) = x^2 + x$, calculate $f'(2)$ by definition.

The formula give

$$f'(2) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0} \frac{(2+h)^2 + (2+h) - (4+2)}{h} = \lim_{h \rightarrow 0} \frac{h(5+h)}{h} = 5.$$

Remark: It should be understood that the existence of a derivative is a local property of a function, that is, it is a property for a given value of x and not for the entire function. The process of finding the derivative f' is called **differentiation** of f .

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta f(x)}{\Delta x} = \tan \theta;$$

where θ : angle between x -axis and the tangenr line of $y = f(x)$ at point x . $f'(x)$ is the slope of tangent line of $y = f(x)$ at $(x, f(x))$.

Geometrical interpretation

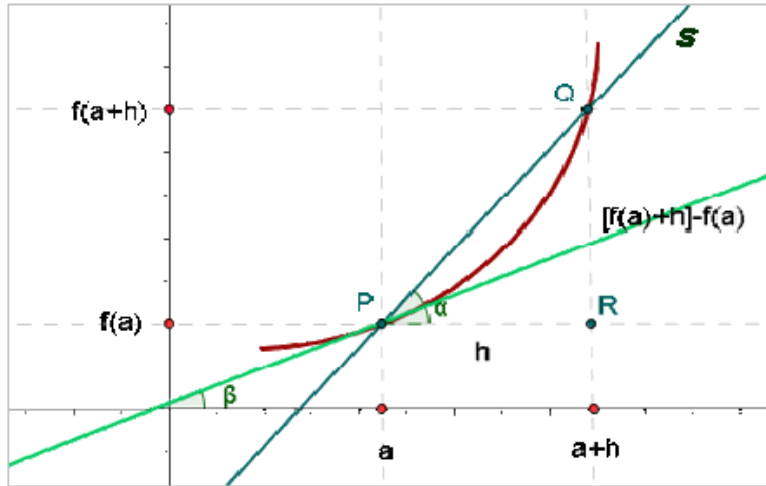


Figure 4: When h approaches zero, the point Q approaches point P . At this time, the secant line begins to resemble the tangent to the function $f(x)$ at Point P , and thus the angle α tends to be β .

$$\tan \beta = \lim_{h \rightarrow 0} \frac{\Delta y}{h} = f'(a)$$

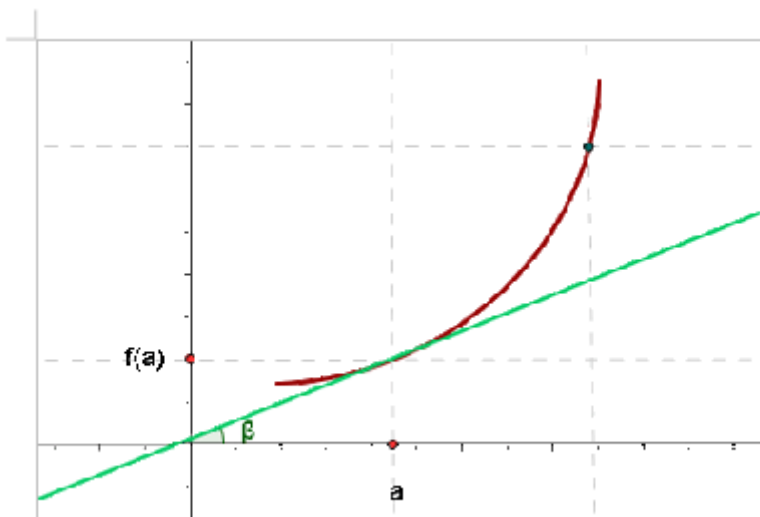


Figure 5: The gradient or slope of the tangent to the curve at a point is equal to the derivative of the function at that point.

$$m_t = f'(a)$$

Example 3. *Given the parabola $f(x) = x^2$, find the points where the tangent line is parallel to the bisector of the first quadrant.*

The bisector of the first quadrant has the equation $y = x$, so its slope is $m = 1$. Since the two lines are parallel they have the same slope, so: $f'(a) = 1$. Since the slope of the tangent to the curve equals the derivative at $x = a$.

$$f'(a) = \lim_{h \rightarrow 0} \frac{(a+h)^2 - a^2}{h} = \lim_{h \rightarrow 0} \frac{a^2 + 2ah + h^2 - a^2}{h} = \lim_{h \rightarrow 0} \frac{2ah + h^2}{h} = \lim_{h \rightarrow 0} (2a + h) = 2a$$

$$2a = 1 \quad a = \frac{1}{2} \quad P\left(\frac{1}{2}, \frac{1}{4}\right).$$

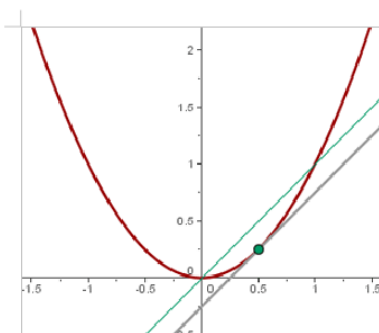


Figure 6: We prove that $f'(a)$ is the slope of the tangent T of the representative curve of $f(x)$ at the point a . Then $T \equiv y - f(a) = f'(a)(x - a)$ and if N is the normal of $f(x)$ at the point a then the equation of N is $N \equiv y - f(a) = \frac{-1}{f'(a)}(x - a)$.

For example, if $f(x) = x^2 + x$ then the tangent t and the normal N of $f(x)$ at $x = 2$, are

$$T \equiv y - 6 = 5(x - 2) \quad \text{and} \quad N \equiv y - 6 = \frac{-1}{5}(x - 2).$$

The tangent line problem

What do we mean when we say that a line is tangent to curve at a point? For a circle, we can characterize the tangent line at a point p as the line that is perpendicular to the radial line at point p . However, for a general curve the problem is more difficult.

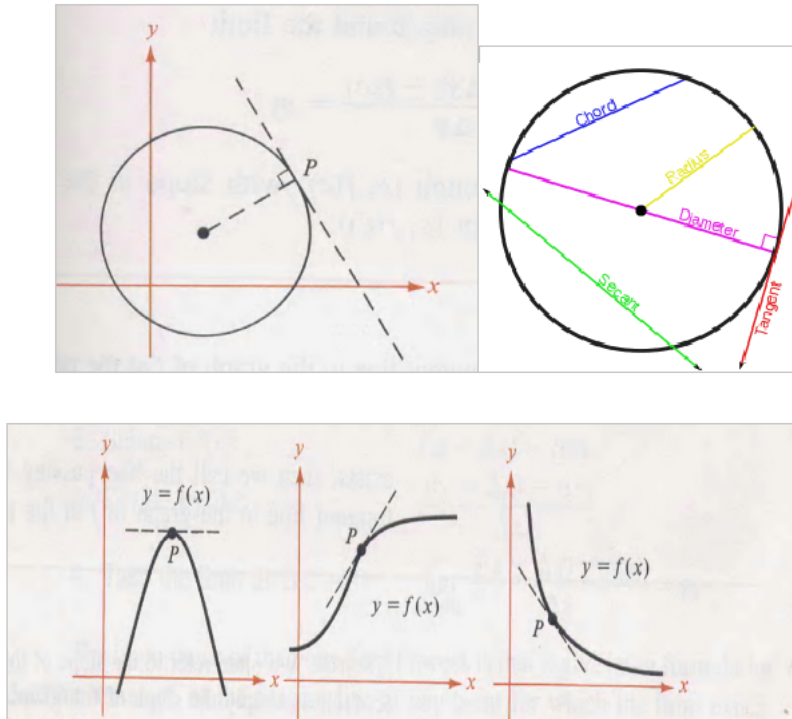
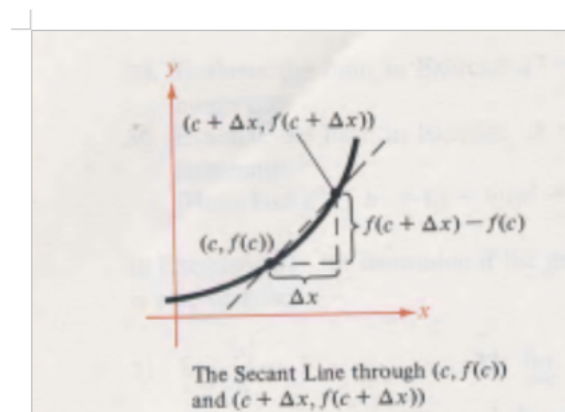


Figure 7: Essentially, the problem of finding the tangent line at a point P boils down to the problem of finding the slope of the tangent line at P . We can approximate this slope by making use of line through the point of tangency and a second point on the curve as shown in the following figure. We call such a line a **secant line**.



If $(c, f(c))$ is the point of tangency and $(c + \Delta x, f(c + \Delta x))$ is a second point on the graph of $f(x)$, then the slope of the secant line through these two points is given by

$$m_{sec} = \frac{f(c + \Delta x) - f(c)}{(c + \Delta x) - c} = \frac{f(x + \Delta x) - f(c)}{\Delta x} \quad \text{slope of secant line.}$$

This formula is called a Difference quotient. The denominator Δx is called the change in x , and the numerator is called the change in y , denoted by $\Delta y = f(c + \Delta x) - f(c)$ change in y .

The beauty of this procedure is that we can obtain better approximations to the slope of the tangent line by choosing the second point closer to the point of tangency.

The equation of the tangent line is: $y - f(x_0) = f'(x_0)(x - x_0)$

The right and the left derivative:

- We call derivative on the right of a function $f(x)$ at a point $x = a$, the limit

$$\lim_{h \rightarrow 0^+} \frac{f(a + h) - f(a)}{h} \quad \text{and denoted } f'(a^+);$$

And the derivative on the left of $f(x)$ at $x = a$ is

$$f'(a^-) = \lim_{h \rightarrow 0^-} \frac{f(a + h) - f(a)}{h}$$

Example 4. Verify if $f(x) = |2x - 6|$ is derivable at $x = 3$.

We define

$$|2x - 6| = \begin{cases} 2x - 6 & \text{if } x > 3 \\ -2x + 6 & \text{if } x \leq 3 \end{cases} \quad \text{then } f'(3^+) = 2 \quad f'(3^-) = -2.$$

Then the two derivative are not equal, so the function is not derivable at $x = 3$. If a function $f(x)$ is differentiable at point a , then it is continuous at that point.

2. Rules for differentiation

1. $C' = 0$ where C is a constant
2. $(f(x) \pm g(x))' = f'(x) \pm g'(x)$
3. $(cf(x))' = cf'(x)$
4. $(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$
5. $\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$
6. The chain rule (differentiation of a function of function) $[f(u(x))]' = f'(u(x)).u'(x)$
7. The differentiation of the inverse function $y'_x = \frac{1}{x'_y}$

3. Some standard derivatives

1. $(x^n)' = nx^{n-1}$
2. $(\sin x)' = \cos x$
3. $(\cos x)' = -\sin x$
4. $(\tan x)' = \sec^2 x$

$$\begin{array}{lll}
5. (\cot x)' = -\cos \sec^2 x & 6. (\arcsin s)' = \frac{1}{\sqrt{1-s^2}} & 7. (\arccos x)' = \frac{-1}{\sqrt{1-x^2}} \\
8. (\arctan x)' = \frac{1}{1+x^2} & 9. (\operatorname{arc cot} x)' = \frac{-1}{1+x^2} & 10. (\ln x)' = \frac{1}{x} \quad 11. (\log_a x)' = \frac{1}{x \ln a} \\
12. (e^x)' = e^x & 13. (a^x)' = a^x \ln a & 14. (shx)' = chx \quad 15. (chx)' = shx
\end{array}$$

4. Successive differentiation. Higher derivatives

If $f(x)$ is a differentiable function of x then $f'(x)$ is called **the first derivative** of $f(x)$. If $f'(x)$ which is also a function of x is differentiable, then its derivative is called the **second derivative** of $f(x)$ and is denoted by $f''(x)$ this $f''(x) = (f'(x))'$. The third and higher derivatives of $f(x)$ are defined similarly. Then n^{th} derivative of $f(x)$ is denoted by $f^{(n)}(x)$.

- Note that for $f^{(n)}(x)$ to exist, $f^{(n-1)}(x)$ must be differentiable.
- $f(x)$ is considered as $f^{(0)}(x)$.
- $f(x), f'(x), f''(x), \dots, f^{(n)}(x)$ are called **successive derivatives** of $f(x)$

Example: i) Let $f(x) = x^6$. Find $f^{(6)}(x)$

Solution:

$$f'(x) = 6x^5; \quad f''(x) = 30x^4; \quad f'''(x) = 120x^3 \quad f^{(4)} = 360x^2; \quad f^{(5)} = 720x; \quad f^{(6)} = 720$$

5. Differentials

Let $f(x)$ be a differentiable function, the differential of the function $f(x)$ is denoted by $df(x)$ and defined as $df(x) = f'(x) \Delta x$, where Δx is the increment of the independent variable x . If $f(x) = x$, then $df(x) = dx = x' \Delta x$. Thus $dx = \Delta x$ and $df(x) = f'(x)dx$.

The differential of the independent variable x is equal to the increment of the variable, but the differential of the function is not equal to the increment of the function.

From the relation $df(x) = f'(x)dx$ we can write $f'(x) = \frac{df(x)}{dx}$. The derivative of a function $f(x)$ with respect to x is equal to quotient $\frac{df}{dx}$ of the differentials.

Properties

$$\begin{array}{ll}
1. dc = 0 & 2. d(f(x) \pm g(x)) = df(x) \pm dg(x) \\
3. d(f(x)g(x)) = g(x)df(x) + f(x)dg(x) & 4. d\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)df(x) - f(x)dg(x)}{g^2(x)}
\end{array}$$

Higher differentials

Higher differentials are defined as similarly ways as higher derivatives. Thus

$$d^2 f(x) = d(df(x)); \quad d^3 f(x) = d(d^2 f(x)); \quad \text{etc.}$$

We can also write: $f''(x) = \frac{d^2 f}{dx^2}$; $f'''(x) = \frac{d^3 f}{dx^3}$; $\dots$; $f^{(n)} = \frac{d^n f(x)}{dx^n}$

Remark: If the function is written as implicit function with together y and x ; we differentiate with respect to x the two sides of the equation and we deduce y'_x .

For example, if $y^6 - x^2 - y + 3x = 0$ then $y' = \frac{2x-3}{6y^5-1}$

Remark: If the function is written as a parametric form: $\begin{cases} x = h(t) \\ y = g(t) \end{cases}$ where $t \in \mathbb{R}$;

Then the derivative function is $\frac{dy}{dx} = \frac{g'(t)dt}{h'(t)dt}$.

For example, if f is defined by $\begin{cases} x = a(\sin t)^3 \\ y = a(\cos t)^3 \end{cases}$ then $\frac{dy}{dx} = -\cot t$.

EXERCISES 1.2

1. Calculate the increment of $y = \sqrt[3]{x}$, if :

a) $x = 0, \Delta x = 0.001$ b) $x = 8, \Delta x = -9$ c) $x = c, \Delta x = h$

2. Calculate the derivative of: a) $y = x^7 e^x$ b) $\frac{x^2}{\ln 3x}$ c) $y = e^x \sin^{-1} x$.

3. Calculate $f'(x) = \begin{cases} 1-x, & \text{for } x \leq 0 \\ e^{-x}, & \text{for } x > 0 \end{cases}$

4. Calculate $f'(0)$, if $f(x) = e^{-x} \cos 3x$.

5. Calculate $f'(0^+)$ and $f'(0^-)$ for the following functions:

a) $f(x) = \sqrt{\sin(x^2)}$ b) $f(x) = \sin^{-1} \left(\frac{a^2 - x^2}{a^2 + x^2} \right)$ c) $f(x) = \frac{x}{1+e^{\frac{1}{x}}}, x \neq 0, f(0) = 0$

d) $f(x) = x^2 \sin \frac{1}{x}, x \neq 0, f(0) = 0$ e) $f(x) = x \sin \frac{1}{x}, x \neq 0, f(0) = 0$

6. Given the function $f(x) = e^{-x}$, calculate the expression: $f(0) + x f'(0)$.

7. Calculate the following derivative:

a) $\begin{cases} x = 2t - 1 \\ y = t^3 \end{cases}$ b) $\begin{cases} x = \sqrt{t} \\ y = \sqrt[3]{t} \end{cases}$ c) $\begin{cases} x = a \cos^3 t \\ y = b \sin^3 t \end{cases}$ d) $\begin{cases} x = e^{-t} \\ y = e^{2t} \end{cases}$

8. Calculate the second derivative of: a) $y = x^7 e^x$ b) $y = \frac{x^2}{\ln 3x}$ c) $y = e^x \sin^{-1} x$.

1.5 APPLICATIONS OF DIFFERENTIAL CALCULUS

1.5.1 Application of derivatives in limits calculation (L'Hospital's rule)

Indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$

L'Hospital's Rule

Let $f(x)$ and $g(x)$ be two functions defined on $[a, b]$ such that f and g are continuous on $[a, b]$ and derivative on $]a, b[$. If $f(x_0) = g(x_0) = 0$ and if $\lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)}$ exists then

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)}.$$

Example: Find the following limits by using the Hospital's rule:

$$(1) \quad \lim_{x \rightarrow 1} \frac{\ln x}{1 - x^2}; \quad (2) \quad \lim_{x \rightarrow \pi} \frac{\sin 3x}{1 - \cos 2x}$$

Solutions: (1) the indeterminate form is $\frac{0}{0}$. The Hospital's rule gives

$$\lim_{x \rightarrow 1} \frac{\ln x}{1 - x^2} = \lim_{x \rightarrow 1} \frac{\frac{1}{x}}{-2x} = \frac{-1}{2}.$$

(2) The indeterminate form is $\frac{0}{0}$. We have

$$\lim_{x \rightarrow \pi} \frac{\sin 3x}{1 - \cos 2x} = \lim_{x \rightarrow \pi} \frac{3 \cos 3x}{2 \sin 2x} = \pm \infty \quad (+ \text{ at the right; } - \text{ at the left}).$$

Limits and Asymptotes

A straight line (d) is said to be an asymptote of an infinite branch of a curve γ if, as a variable point P recedes to infinity along the branch, the perpendicular distance of the P from the line (d) tends to zero.

Vertical asymptote (V.A)

The line $V \equiv x = a$ is a **vertical asymptote** of a curve if either

$$\lim_{x \rightarrow a^-} f(x) = \pm \infty \quad \text{or} \quad \lim_{x \rightarrow a^+} f(x) = \pm \infty \quad \text{or both.}$$

We can also write

$$\lim_{x \rightarrow a} f(x) = \pm \infty.$$

Horizontal asymptote (H.A)

The line $H \equiv y = b$ is a **horizontal asymptote** of a curve

$$\gamma \equiv y = f(x) \quad \text{if either} \quad \lim_{x \rightarrow +\infty} f(x) = b \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = b.$$

Example: $V \equiv x = 0$ and $H \equiv y = 0$ are vertical and horizontal asymptotes of the curve $\gamma = \frac{1}{x}$.

Oblique asymptote (O.A)

Not all asymptotes are horizontal or vertical the non vertical line $O \equiv y = kx + m$ oblique asymptote if

$$\lim_{x \rightarrow \infty} [f(x) - (kx + m)] = 0$$

or both m and k are determined by

$$k = \lim_{x \rightarrow \infty} \frac{f(x)}{x} \quad \text{and} \quad m = \lim_{x \rightarrow \infty} [f(x) - kx]$$

1.5.2 Increasing and decreasing functions. Extreme values

The rate of change of a function $f(x)$ with respect to x is given by $f'(x)$.

If $f'(x) > 0, \forall x \in]a, b[$, then f is an **increasing function** $[a, b]$.

If $f'(x) < 0, \forall x \in]a, b[$, then f is an **decreasing function** $[a, b]$.

Critical values for the function $f(x)$ are values of x for which the **function** is defined and for which $f'(x) = 0$ or becomes infinity.

A function $f(x)$ has a **relative maximum** value for $x = x_0$ if $f(x_0)$ is **greater** than immediately **preceding** and **succeeding** values of the function.

In each case, $f(x)$ changes from an increasing to a decreasing function and $f'(x)$ changes sign from positive to negative as x increases through the corresponding critical value.

A function $f(x)$ has a **relative minimum value** for $x = x_0$ if $f(x_0)$ is **smaller** than immediately **preceding** and **succeeding** values of the function.

In each case, $f(x)$ changes from a decreasing to an increasing function and $f'(x)$ changes sign from negative to positive as x increases through the corresponding critical value.

If x is a point of interval I , then $f(x_0)$ is called **absolute maximum** value of $f(x)$ on I if $f(x_0) \geq f(x), \forall x \in I$.

If x is a point of interval I , then $f(x_0)$ is called **absolute minimum** value of $f(x)$ on I if $f(x_0) \leq f(x), \forall x \in I$.

Tests for Extreme values of the function

a) First Derivative Method

i. Find $f'(x)$ and the critical values

ii. For a critical value $x = x_0$.

- $f(x)$ has a maximum value $[= f(x_0)]$ if $f'(x)$ changes from positive to negative.
- $f(x)$ has a minimum value $[= f(x_0)]$ if $f'(x)$ changes from negative to positive.
- $f(x)$ has neither a maximum nor minimum if $f'(x)$ does not change the sign.

Finding the extreme values of a continuous function f on a closed interval $[a, b]$

Step 1: Find the critical points of F in (a, b) .

Step 2: Evaluate f at the critical points and the endpoints a and b .

Step 3: The largest of the values in step 2 is the maximum and the smallest is the minimum.

Example 5. Find the maximum and minimum values of the function $f(x) = 2x^3 - 15x^2 + 36x$ on the interval $[1, 5]$, and determine where they occur.

Solution: Since polynomials are differentiable everywhere, f is differentiable everywhere on the interval $(1, 5)$. Thus, if the extreme value occurs on the open interval $(1, 5)$, it must occur at a point where the derivative is zero.

Step 1: If $f(x) = 2x^3 - 15x^2 + 36x$, then $f'(x) = 6x^2 - 30x + 36$
 $f'(x) = 0 \implies 6x^2 - 30x + 36 = 0 \Leftrightarrow x^2 - 5x + 6 = 0 \implies x = 2 \vee x = 3$.

The critical points of f in $(1, 5)$ are $x = 2 \vee x = 3$.

Step 2: Evaluating f at these critical points and the end points 1 and 5.

$$f(1) = 23, \quad f(2) = 28, \quad f(3) = 27, \quad f(5) = 55$$

Step 3: Thus, the minimum value is 23 and the maximum value is 55. The minimum occurs at $x = 1$ and the maximum occurs at $x = 5$.

b) Second Derivative Method

i. Find $f'(x)$ and critical values;

ii. Find $f''(x)$;

iii. For a critical value $x = x_0$, $f(x)$ has a maximum value $[= f(x_0)]$ if $f''(x_0) < 0$; $f(x)$ has a minimum value $[= f(x_0)]$ if $f''(x_0) > 0$. The test fails if $f''(x_0) = 0$ or becomes infinite.

Second derivative, Concavity and point of inflection

An arc of a curve $\gamma \equiv y = f(x)$ is **concave up ward or bending up ward** if, at each its points, the arc lies above tangent.

As x increases, $f'(x)$ either is of the same sign and increasing or change sign from negative to positive. In either case, the slope $f'(x)$ of curve is increasing and f'' is positive.

An arc of a curve $\gamma \equiv y = f(x)$ is **concave down ward or bending down ward** if, at each of its points, the arc lies below the tangent.

As x increases, $f'(x)$ either is of the same sign and decreasing or changes sign from positive to negative, In either case, the slope $f'(x)$ of the curve is decreasing and f'' is negative.

A point of inflection (P.I) is a point at which a curve is changing from concave up ward to concave down ward, or vice versa.

Test for concavity and inflection point:

Let $f(x) \in I$.

- If $f''(x) > 0$, then f is concave up ward. We have $f : \cup$.
- If $f''(x) < 0$, then f is concave down ward. We have $f : \cap$.
- The point $x_0 \in I$ is an inflection point of $f(x)$ if: (i) $f(x_0)$ is defined; (ii) $f''(x_0) = 0$ or does not exist; (iii) $f''(x_0) < 0$, on one side of x_0 and $f''(x_0) > 0$ on the other side.

Theorem 1.6. Rolle's theorem: *If the function $f(x)$ is continuous on the segment $[a, b]$, derivable at any point inside the segment (a, b) and if $f(a) = f(b)$; then there is at least one point $c \in (a, b)$ such that $f'(c) = 0$.*

Example 6. *Show that the function $f(x) = \sin x$ satisfies the Rolle's theorem everywhere?*

Solution: $f(x) = \sin x$ is both continuous and differentiable everywhere, so that $f(x) = \sin x$ is continuous on $[0, 2\pi]$ and differentiable on $(0, 2\pi)$.

$$f(0) = \sin 0 = 0 \quad \text{and} \quad f(2\pi) = \sin 2\pi = 0.$$

So that f satisfies the hypotheses of Rolle's theorem on the interval $[0, 2\pi]$.

Thus, we are guaranteed that there is at least one point $c \in (a, b)$ such that $f'(c) = 0$.

$$f'(c) = 0 \longleftrightarrow \cos c = 0 \longrightarrow c = \frac{\pi}{2} \vee c = \frac{3\pi}{2}$$

Remark: Sometimes, however, the function f is sufficiently complicated that it is difficult or impossible to find a value for c .

Theorem 1.7. Mean-Value theorem: *If the function $f(x)$ is continuous on the segment $[a, b]$, differentiable at any point inside the segment (a, b) , then $\exists c \in (a, b)$ such that $\frac{f(b)-f(a)}{b-a} = f'(c)$.*

Example 7. *Show that the function $f(x) = x^3 + 1$ satisfies the mean-value theorem on the interval $[1, 2]$ and find all values of c in this interval whose existence is guaranteed by the theorem?*

Solution: Because polynomials are differentiable everywhere, f is differentiable everywhere on the interval $(1, 2)$ and continuous on $[1, 2]$. If we take $a = 1$ and $b = 2$,

$$f(a) = f(1) = 2, \quad f(b) = f(2) = 9$$

$$f'(x) = 3x^2, \quad f'(c) = 3c^2 \quad f'(c) = \frac{f(b) - f(a)}{b - a} \longrightarrow \frac{9 - 2}{2 - 1} = 3c^2 \longrightarrow 3c^2 = 7 \longrightarrow c = \sqrt{\frac{7}{3}} \wedge c = -\sqrt{\frac{7}{3}}$$

As we are on $[1, 2]$, the accepted value of c is $\sqrt{\frac{7}{3}}$

EXERCISES 1.3

1. Find the equation of the tangent line and normal line at x_0 .

a) $f(x) = x^2 - 6x$; $x_0 = -1$ b)

c) $f(x) = \sqrt{x}$; $x_0 = \frac{1}{4}$. Find the maximum and the minimum values of $f(x)$ on the given closed interval and

a) $f(x) = 4x^2 - 4x + 1$; $[0, 1]$ b) $f(x) = 8x - x^2$; $[0, 6]$

d) $f(x) = \sin x - \cos x$; $[0, \pi]$ e)

3. Find the dimensions of a rectangle with perimeter $100m$ whose area is as large as possible?

4. Find two numbers whose sum is 20 and whose product is: a) maximum b) minimum

4. Verify that the hypotheses of Rolle's theorem are satisfied on the given interval and find all values of c that satisfy the conclusion of the theorem:

a) $f(x) = x - x^3$; $[-1, 0]$ b) $f(x) = x^2 - 6x + 8$; $[2, 4]$ c) $f(x) = \frac{x^2 - 1}{x - 2}$; $[-1, 1]$

5. Verify that the hypotheses of Mean value theorem are satisfied on the given interval and find all values of c that satisfy the conclusion of the theorem:

a) $f(x) = x^2 + x$; $[-4, 6]$ b) $f(x) = x^3 + x - 6$; $[-1, 24]$ c) $f(x) = \sqrt{x + 1}$; $[0, 3]$

6. Calculate the following limits:

a) $\lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{1}{x^2} \right)$ b) $\lim_{x \rightarrow 0} \left(\frac{\sin^2 x}{x} \right)$ c) $\lim_{x \rightarrow 0} \frac{(1 - \cos x) \cos x}{\sin x}$ d) $\lim_{x \rightarrow 1} (1 - x)^{\cos \frac{\pi x}{2}}$

1.6 INDEFINITE INTEGRALS.

1.6.1 Definitions and Properties

Let $f(x)$ a continuous function on an interval $[a, b]$; and $F(x)$ a derivative function on an interval $]a, b[$. $F(x)$ is an anti-derivative of $f(x)$ on an interval $]a, b[$, if $F'(x) = f(x)$ for all $x \in]a, b[$.

Example 8. $g(x) = 3x + 4$ is an anti-derivative of $f(x) = 3$; $h(x) = x^2 + 6$ is an anti-derivative of $f(x) = 2x$.

Theorem 1.8. If $F(x)$ is an anti-derivative of a function $f(x)$ on $]a, b[$; any other anti-derivative of $f(x)$ can be put in the form $F(x) + c$, where c is an arbitrary constant on this interval.

-If a function $F(x)$ is an anti-derivative of a function $f(x)$, the set of the function $F(x) + c$, where $c \in \mathbb{R}$ is called an indefinite integral of $f(x)$ on $]a, b[$ and we write

$$\int f(x) dx = F(x) + c$$

-The symbol $\int$ is the sign of integration or sign sum.

-The function $f(x)$ is called integrating or a function under the sign sum or a function to be integrated. x is the variable of integration and $f(x)dx$ is called the expression under the sign sum.

Example 9.

$$\int (3x^2 + 5) dx = x^3 + 5x + c \quad \text{because} \quad (x^3 + 5x + c)' = 3x^2 + 5.$$

Properties

$$(1) \quad \left(\int f(x) dx \right)' = f(x); \quad \int F'(x) dx = F(x) + c.$$

$$(2) \quad \int k \cdot f(x) dx = k \cdot \int f(x) dx; \quad k \in \mathbb{R}.$$

$$(3) \quad \int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx.$$

1.6.2 The common anti-derivatives.

The differentiation of function give the tabular integrals or elementary integrals which are useful:

$$(1) \quad \int \frac{x^n}{n} dx = \frac{x^{n+1}}{n+1} + c; \quad \text{if } n \neq -1.$$

$$(2) \quad \int \frac{1}{x} dx = \ln |x| + c.$$

$$(3) \quad \int e^x dx = e^x + c.$$

$$(4) \quad \int a^x dx = \frac{a^x}{\ln a} + c.$$

$$(5) \quad \int \cos x dx = \sin x + c.$$

$$(6) \quad \int \sin x dx = -\cos x + c.$$

$$(7) \quad \int (\sec x)^2 dx = \int [1 + (\tan x)^2] dx = \tan x + c.$$

$$(8) \quad \int (\operatorname{cosec} x)^2 dx = \int [1 + (\cot x)^2] dx = -\cot x + c.$$

$$(9) \quad \int \sec x \tan x dx = \sec x + c.$$

$$(10) \quad \int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + c.$$

$$(11) \quad \int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + c.$$

$$(12) \quad \int \frac{dx}{1+x^2} = \arctan x + c.$$

$$(13) \quad \int \frac{dx}{x\sqrt{x^2-1}} = \operatorname{arcsec} x + c.$$

$$(14) \quad \int \frac{u'(x)}{u(x)} dx = \ln |u(x)| + c.$$

$$(15) \quad \int \frac{dx}{x^2+a^2} = \frac{1}{a} \arctan \frac{x}{a} + c.$$

$$(16) \quad \int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin \frac{x}{a} + c.$$

$$(17) \quad \int \frac{dx}{x^2-a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + c.$$

$$(18) \quad \int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a} \operatorname{arcsec} \frac{x}{a} + c.$$

$$(19) \quad \int \tan x dx = \ln |\sec x| + c.$$

$$(20) \quad \int \cot x dx = \ln |\sin x| + c.$$

$$(21) \quad \int \sec x dx = \ln |\sec x + \tan x| + c.$$

$$(22) \quad \int \operatorname{cosec} x dx = \ln |\operatorname{cosec} x - \cot x| + c.$$

$$(23) \quad \int \frac{dx}{x^2+a^2} = \ln (x + \sqrt{x^2+a^2}) + c.$$

$$(24) \quad \int \frac{dx}{\sqrt{x^2-a^2}} = \ln |x + \sqrt{x^2-a^2}| + c.$$

$$(25) \quad \int \ln x dx = x \ln x - x + c.$$

Remark: The formulas (14) to (25) can be deduced by using a method of integration.

Example 10. Find (1) $\int (\tan x)^2 dx$; (2) $\int \frac{2x}{1+x^2} dx$

Solutions: (1) $\int (\tan x)^2 dx = \tan x - x + c$; (2) $\int \frac{2x}{1+x^2} dx = \ln(1+x^2) + c.$

1.6.3 MAIN METHODS OF INTEGRATION

a) Direct Integration

The method apply directly the table of elementary integrals and the properties of indefinite integrals ;like the following examples:

$$(1) \quad \int \left(5 \cos x + 2 - 3x^2 + \frac{1}{x} - \frac{4}{x^2+1} \right) dx = 5 \sin x + 2x - x^3 + \ln |x| - 4 \arctan x + c$$

$$(2) \quad \int \left(-3(\sec x)^2 + \frac{10}{25+x^2} - \frac{24}{x^2-16} \right) dx = -3 \tan x + 2 \arctan \frac{x}{5} - 3 \ln \left| \frac{x-4}{x+4} \right| + c.$$

b) Integration by change of variable

With we introduce a new variable which permits to transform a given integral in a tabular integral like the theorem below:

Theorem 1.9. Let $u = g(x)$ be a define and derivable function on the interval $[a, b]$ and let B be its set of the values; finally let $f(u)$ be defined on B . If $f(u)$ admits an anti-derivative on B then

$$\int f(g(x)).g'(x)dx = \int f(u)du \quad \text{on } [a, b].$$

Example 11. Calculate (1) $A = \int \frac{6(x^2+1)}{\sqrt{x^3+3x}} dx$; (2) $B = \int \frac{\ln x}{x} dx$

Solution:

$$(1) \text{ Let } u = x^3 + 3x \implies du = 3(x^2 + 1)dx \implies A = \int \frac{2du}{\sqrt{u}} = 2 \int u^{-\frac{1}{2}} du = 2 \cdot \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + c =$$

$$4\sqrt{x^3 + 3x} + c.$$

$$(2) \text{ Let } u = \ln x \implies du = \frac{dx}{x} \implies B = \int u du = \frac{u^2}{2} + c = \frac{(\ln x)^2}{2} + c$$

Remark: It is easy to find the integral $I = \int \frac{dx}{ax^2+bx+c}$ because the transformation of ax^2+bx+c give a sum or a difference of two squares which are $u^2 \pm k^2$ where

$$u = x + \frac{b}{2a} \quad \text{and} \quad \pm k^2 = \frac{c}{a} - \frac{b^2}{4a^2} = \frac{-\Delta}{4a^2}$$

Then

$$\int \frac{dx}{ax^2 + bx + c} \quad \text{becomes} \quad \begin{cases} \frac{1}{a} \int \frac{du}{u^2+k^2} = \frac{1}{ak} \arctan \frac{u}{k} + c & \text{if } \Delta < 0. \\ \frac{1}{a} \int \frac{du}{u^2} = \frac{-1}{au} + c & \text{if } \Delta = 0. \\ \frac{1}{a} \int \frac{du}{u^2-k^2} = \frac{1}{2ak} \ln \left| \frac{u-k}{u+k} \right| + c & \text{if } \Delta > 0. \end{cases}$$

Exercise: Verify that

$$(1) A = \int \frac{dx}{3x^2+48x+21} = \frac{1}{6\sqrt{57}} \ln \left| \frac{x+8-\sqrt{57}}{x+8+\sqrt{57}} \right| + c;$$

$$(2) B = \int \frac{dx}{2x^2-x+6} = \frac{2}{\sqrt{47}} \arctan \frac{4x-1}{\sqrt{47}} + c;$$

$$(3) C = \int \frac{dx}{3x^2+6x+3} = \frac{-1}{3(x+1)} + c.$$

c) Integration by parts.

The method is based on the differentiation formula of a product of two functions:

$$d(uv) = u dv + v du. \quad \text{Then} \quad \int d(uv) = \int u dv + \int v du \implies \int u dv = uv - \int v du.$$

Example 12. Find $A = \int (x+1) \cos 2x dx$.

Solution: Let $u = x+1$ and $dv = \cos 2x dx \implies du = dx$ and $v = \frac{\sin 2x}{2}$. Then

$$A = \int (x+1) \cos 2x dx = (x+1) \frac{\sin 2x}{2} - \frac{1}{2} \int \sin 2x dx = (x+1) \frac{\sin 2x}{2} + \frac{1}{4} \cos 2x + c.$$

Example 13. Find $B = \int x^2 e^{3x} dx$.

Solution: Let $u = x^2$ and $dv = e^{3x} dx \implies du = 2x dx$ and $v = \frac{e^{3x}}{3}$; then

$$B = \int x^2 e^{3x} dx = x^2 \frac{e^{3x}}{3} - \frac{2}{3} \int x e^{3x} dx. \quad \text{For the second time suppose } u = x \text{ and } dv = e^{3x} dx$$

$$\implies du = dx \quad \text{and} \quad v = \frac{e^{3x}}{3}; \quad B = \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left(\frac{x e^{3x}}{3} - \frac{1}{3} \int e^{3x} dx \right) \implies B = \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left(\frac{x e^{3x}}{3} - \frac{1}{9} e^{3x} \right) + c$$

Example 14. Find $D = \int e^{2x} \cos 4x dx$.

Solution: Let $u = e^{2x}$ and $dv = \cos 4x dx$, then $du = 2e^{2x} dx$ and $v = \frac{\sin 4x}{4}$. Then

$$D = e^{2x} \frac{\sin 4x}{4} - \frac{1}{2} \int e^{2x} \sin 4x dx.$$

Let again $u = e^{2x}$ and $dv = \sin 4x dx \Rightarrow du = 2e^{2x}$ and $v = -\frac{\cos 4x}{4}$. We find

$$\begin{aligned}
 D &= \frac{e^{2x} \sin 4x}{4} - \frac{1}{2} \left(-e^{2x} \frac{\cos 4x}{4} + \frac{1}{2} \int e^{2x} \cos 4x dx \right) \\
 &= \frac{e^{2x} \sin 4x}{4} + \frac{e^{2x} \cos 4x}{8} - \frac{1}{4} \int e^{2x} \cos 4x dx \\
 \Rightarrow D + \frac{D}{4} &= \frac{2^{2x} \sin 4x}{4} + \frac{e^{2x} \cos 4x}{8} + c; \quad \text{finally we find after calculation} \\
 D &= \frac{e^{2x}}{10} (2 \sin 4x + \cos 4x) + c.
 \end{aligned}$$

d) Integration of rational functions.

-All function which is on the form $f(x) = \frac{N(x)}{D(x)}$ where $N(x)$ and $D(x)$ are polynomials functions, then $f(x)$ is called a rational function.

- If the degree of the numerator is higher or equal to the denominator ($\deg N \geq \deg D$.)

We divide directly and we find $f(x) = \frac{N(x)}{D(x)} = q(x) + \frac{r(x)}{D(x)}$, where $r(x)$ is the **remainder** of the fraction and $q(x)$ is called the **quotient** of the fraction. Then

$$\int \frac{N(x)}{D(x)} dx = \int q(x) dx + \int \frac{r(x)}{D(x)} dx.$$

Example 15. Calculate

$$A = \int \frac{x^4 - 2x^2 + 5x + 2}{x^2 + 4} dx$$

Solutom:

The division gives $\frac{x^4 - 2x^2 + 5x + 2}{x^2 + 4} = x^2 - 6 + \frac{5x + 26}{x^2 + 4}$;

then $A = \int (x^2 - 6) dx + 5 \int \frac{x}{x^2 + 4} dx + 26 \int \frac{dx}{x^2 + 4}$

and after calculation we have $A = \frac{x^3}{3} - 6x + \frac{5}{2} \ln(x^2 + 4) + 13 \arctan \frac{x}{2} + c$.

Exercise: Verify if

$$\int \frac{x^4 - 2x^2 + 5x + 2}{x^2 - 4} dx = \frac{x^3}{3} + 2x + \frac{5}{2} \ln(x^2 - 4) + 2 \ln \frac{|x - 2|}{|x + 2|} + c.$$

- If the degree of the numerator is lower or equal than the degree of the denominator ($\deg N(x) \leq \deg D(x)$:)

We do the decomposition into partial fractions and we find the integral of the results. The decomposition depends on the form of the factors which are the denominator. The table below is a summary of different cases:

No.	Denominator factor form	Partial fraction correspondig to each factor
(1)	$ax + b$	$\frac{A}{ax+b}$
(2)	$(ax + b)^n$	$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \dots + \frac{A_n}{(ax+b)^n}$
(3)	$ax^2 + bx + c; (\Delta < 0)$	$\frac{Ax+B}{ax^2+bx+c}; (\Delta < 0)$
(4)	$(ax^2 + bx + c)^n; (\Delta < 0)$	$\frac{A_1x+B_1}{ax^2+bx+c} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + \dots + \frac{A_nx+B_n}{(ax^2+bx+c)^n}$

Example 16. Calculate

$$I = \int \frac{2x^2 + 3x + 6}{x^3 + x^2 - 2x} dx$$

Solution: We have $x^3 + x^2 - 2x = x(x+2)(x-1)$. Each factor is linear then we have to find 3 reals A, B, C that

$$\frac{2x^2 + 3x + 6}{x^3 + x^2 - 2x} = \frac{2x^2 + 3x + 6}{x(x+2)(x-1)} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{x-1} = \frac{A(x+2)(x-1) + Bx(x-1) + Cx(x-2)}{x(x+2)(x-1)}$$

Then $2x^2 + 3x + 6 = (A + B + C)x^2 + (A - B + 2C)x - 2A$. The identification of the coefficients give the system

$$\begin{cases} A + B + C = 2 \\ A - B + 2C = 3 \\ -2A = 6 \end{cases} \quad \text{and after resolution we find } A = -3; B = \frac{4}{3}; C = \frac{11}{3}$$

$$I = -3 \int \frac{dx}{x} + \frac{4}{3} \int \frac{dx}{x+2} + \frac{11}{3} \int \frac{dx}{x-1} = -3 \ln|x| + \frac{4}{3} \ln|x+2| + \frac{11}{3} \ln|x-1| + c.$$

Example 17. Calculate

$$J = \int \frac{x^2 + 8x + 19}{(x+6)(x-1)^2} dx.$$

Solution:

$$\frac{x^2 + 8x + 19}{(x+6)(x-1)^2} = \frac{A}{x+6} + \frac{B}{x-1} + \frac{C}{(x-1)^2};$$

after calculation we find

$$A = \frac{1}{7}; \quad B = \frac{6}{7}; \quad C = 4.$$

If we replace the values in the partial fractions we have

$$J = \frac{1}{7} \int \frac{dx}{x+6} + \frac{6}{7} \int \frac{dx}{x-1} + 4 \int \frac{dx}{(x-1)^2} = \frac{1}{7} \ln|x+6| + \frac{6}{7} \ln|x-1| - \frac{4}{x-1} + c.$$

Exercises: Verify that

$$M = \int \frac{x+8}{4x+x^3} dx = \ln \frac{x^2}{x^2+4} + \frac{1}{2} \arctan \frac{x}{2} + c; \quad \text{And that}$$

$$N = \int \frac{x^3 + x^2 + 2}{(x^2 + 2)^2} dx = \frac{1}{2} \ln(x^2 + 2) + \frac{1}{\sqrt{2}} \arctan \frac{x}{\sqrt{2}} + \frac{1}{x^2 + 2} + c$$

e) Integration of some form of trigonometric functions.

Form 1: $\int \sin^m x \cdot \cos^n x dx$

• **If m or n is odd:**

If m is odd we suppose $u = \cos x$ and we arrive after simple transformation on an elementary integrals;

If n is odd we suppose $u = \sin x$ and we arrive after simple transformation on an elementary integrals;

• **If m and n are together even:** We use the formulas of Carnot :

$$\sin^2 x = \frac{1 - \cos 2x}{2}; \quad \text{and} \quad \cos^2 x = \frac{1 + \cos 2x}{2} \quad \text{and we return at the first case.}$$

Example 18. Calculate

$$(1) \quad A = \int \sin^2 x \cdot \cos^3 x dx; \quad (2) \quad B = \int \sin^2 x \cdot \cos^4 x dx.$$

Solutions:

$$(1) \quad \int \sin^2 x \cdot \cos^2 x \cdot \cos x dx; \quad \text{suppose} \quad u = \sin x \Rightarrow du = \cos x dx; \quad \text{then}$$

$$A = \int u^2 \cdot (1 - u^2) du = \frac{u^3}{3} - \frac{u^5}{5} + c \quad \text{and} \quad A = \frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + c.$$

$$(2) \quad B = \int \sin^2 x \cdot \cos^4 x dx, \quad \text{We suppose} \quad \sin^2 x = \frac{1 - \cos 2x}{2} \quad \text{and} \quad \cos^2 x = \frac{1 + \cos 2x}{2}; \quad \text{then}$$

$$B = \frac{1}{8} \int (1 + \cos 2x - \cos^2 2x - \cos^3 2x) dx = \frac{1}{8} \int dx + \frac{1}{8} \int \cos 2x dx - \frac{1}{8} \int \cos^2 2x dx - \frac{1}{8} \int \cos^3 2x dx$$

$$\int \cos^2 2x dx = \frac{1}{2} \int (1 + \cos 4x) dx = \frac{1}{2} \left(x + \frac{\sin 4x}{4} \right) \quad \text{and}$$

$$\int \cos^3 2x dx = \int \cos^2 2x \cdot \cos 2x dx = \frac{1}{2} \int (1 - \sin^2 2x) d(\sin 2x) = \frac{1}{2} \left(\sin 2x - \frac{\sin^3 2x}{3} \right) + c.$$

Then

$$B = \frac{x}{8} + \frac{\sin 2x}{16} - \frac{x}{16} - \frac{\sin 4x}{64} - \frac{\sin 2x}{16} + \frac{\sin^3 2x}{48} + c = \frac{x}{16} - \frac{\sin 4x}{64} + \frac{\sin^3 2x}{48} + c$$

Form 2: $\int R(\sin x, \cos x) dx$ where R is a rational function of $\sin x$ and $\cos x$.

$$\text{To this case we take} \quad t = \tan \frac{x}{2} \Rightarrow \begin{cases} dx = \frac{2dt}{1+t^2} \\ \sin x = \frac{2t}{1+t^2} \\ \cos x = \frac{1-t^2}{1+t^2} \\ \tan x = \frac{2t}{1-t^2} \end{cases};$$

after the introduction of t , the function R becomes a rational function of t ; and after its integration we return to the variable x .

Example 19. Calculate

$$(1) \quad A = \int \frac{dx}{1 + \sin x} \quad \text{and} \quad (2) \quad B = \int \frac{dx}{4 \sin x + 3 \cos x + 5}$$

$$(1) \quad t = \tan \frac{x}{2}; \quad \text{then} \quad A = \int \frac{\frac{2}{1+t^2} dt}{1 + \frac{2t}{1+t^2}} = 2 \int \frac{dt}{(1+t)^2} = \frac{-2}{1+t} + c = \frac{-2}{1 + \tan \frac{x}{2}} + c.$$

$$(2) \quad \text{with } t = \tan \frac{x}{2} \quad \text{we use the same way and find} \quad B = \frac{-1}{2 + \tan \frac{x}{2}} + c.$$

EXERCISES 1.4

1. Evaluate the following integrals:

$$a) \int (a + bx^3) dx \quad b) \int \frac{dx}{x^2 - 10} \quad c) \int 3^x e^x dx \quad d) \int \frac{dx}{\sqrt{8 - x^2}} \quad e) \int \frac{(x^m - x^n)^2}{\sqrt{x}} dx$$

2. Evaluate the following integrals:

$$a) \int \left(\frac{2x+3}{2x+1} \right) dx \quad b) \int \frac{dx}{3x^2+5} \quad c) \int \frac{\tan^{-1} \left(\frac{x}{2} \right)}{4+x^2} dx$$

$$d) \int \frac{3x+1}{\sqrt{5x^2+1}} dx \quad e) \int \frac{x^3}{x^8+5} dx.$$

3. Evaluate the following integrals:

$$a) \int \left(\frac{e^x}{\sqrt{e^{2x}-2}} \right) dx \quad b) \int \frac{xdx}{\sqrt{x^4+1}} \quad c) \int \frac{dx}{1+\cos^2 x} \quad d) \int \frac{1}{\sqrt{5x-2}} \quad \int \frac{\sqrt{x^2+1}}{x} dx$$

4. Evaluate the following integrals:

$$a) \int \left(\frac{dx}{x^2+2x+5} \right) \quad b) \int \frac{dx}{\sqrt{x^2+px+q}} \quad c) \frac{e^x dx}{\sqrt{e^{2x}+e^x+1}}$$

$$d) \int \frac{dx}{(x+a)(x+b)} \quad e) \int \frac{\sqrt{x}+1}{\sqrt[3]{x}+1} dx$$

5. Evaluate the following integrals:

$$a) \int \sin 3x \cos 5x dx \quad b) \int \frac{dx}{\sin^3 x} \quad c) \int \frac{dx}{\cos^3 x} \quad d) \int \sin \omega t \sin(\omega t + \varphi) dt \quad e) \int \frac{1 + \tan x}{1 - \tan x} dx$$

6. Evaluate the following integrals:

$$a) \int \sinh^3 x dx \quad b) \int \frac{dx}{\sinh x \cosh^2 x} \quad c) \frac{\cos 2x}{\cos^4 x + \sin^4 x} dx \quad d) \int \sqrt{3-2x-x^2} dx \quad e) \int |x| dx.$$

1.7 DEFINITE INTEGRAL.

1.7.1 Definition and Newton-Leibniz's formula.

Let $y = f(x)$ a positive and continuous function on the interval $[a, b]$. Let us subdivide this interval in n arbitrary parts using the points $a = x_0 < x_1 < x_2 \dots < x_{n-1} < x_n = b$ (*).

If we take in each partial interval $[x_{i-1}, x_i]$ a point c_i and if the length of the interval is Δx_i ; the quantity $\sigma = \sum_{i=1}^n f(c_i) \Delta x_i$ (**) is called the sum of the function f on the segment $[a, b]$ attached to the subdivision (*). Geometrically σ is the area of rectangles of basis Δx_i and heights $f(c_i)$.

The finite limits if it exists of the sum (**) when $n \rightarrow \infty$ is called the definite integral of $f(x)$ on $[a, b]$ and is denoted

$$I = \int_a^b f(x)dx. \quad \text{Then} \quad \int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x_i.$$

The real a and b are called respectively the lower and the upper bounds of the integration. $f(x)$ is the integrand and x is the variable.

Theorem 1.10. *If $f(x)$ is the continuous function on the interval $[a, b]$. If $F(x)$ is the anti-derivative of function $f(x)$ on the interval $[a, b]$. Then*

$$\int_a^b f(x)dx = F(b) - F(a).$$

*This formula is called **the Newton - Leibniz's formula**. The number $F(b) - F(a)$ doesn't depend on the choice of the anti-derivative F . It is denoted by $F(b) - F(a) = [F(x)]_a^b$.*

Example 20. Calculate

$$I = \int_{-1}^1 (x^2 + x + 2)dx; \quad \text{and} \quad J = \int_{\frac{\pi}{12}}^{\frac{\pi}{4}} (\cos^2 x)dx$$

Solution:

$$I = \int_{-1}^1 (x^2 + x + 2)dx = \left[\frac{x^3}{3} + \frac{x^2}{2} + 2x \right]_{-1}^1 = \left(\frac{1}{3} + \frac{1}{2} + 2 - \left(-\frac{1}{3} + \frac{1}{2} - 2 \right) \right) = \frac{14}{3}.$$

$$J = \int_{\frac{\pi}{12}}^{\frac{\pi}{4}} \left(\frac{1 + \cos 2x}{2} \right) dx = \frac{1}{2} \cdot \left[x + \frac{\sin 2x}{2} \right]_{\frac{\pi}{12}}^{\frac{\pi}{4}} = \frac{1}{2} \left[\left(\frac{\pi}{4} + \frac{1}{2} \right) - \left(\frac{\pi}{12} + \frac{1}{4} \right) \right] = \frac{1}{2} \left(\frac{\pi}{6} + \frac{1}{4} \right).$$

1.7.2 Fundamental properties of the definite integral

1) $\int_a^a f(x)dx = 0$.

2) $\int_a^b f(x)dx = -\int_b^a f(x)dx$.

3) $\int_a^b [rf(x) + sg(x)] dx = r \int_a^b f(x)dx + s \int_a^b g(x)dx$.

4) $\forall c \in [a, b]; \quad \int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$ (Chasle's rule).

5) Let $f(x)$ a continuous function on $\mathbb{R}$ then

$$\begin{cases} \int_{-a}^a f(x)dx = 0; & \text{if } f(x) \text{ is an odd function} \\ \int_{-a}^a f(x)dx = 2 \int_0^a f(x)dx; & \text{if } f(x) \text{ is an even function} \end{cases}$$

6) $\forall x \in [a, b];$ if $f(x) \leq g(x)$ then $\int_a^b f(x)dx \leq \int_a^b g(x)dx$.

7) If m and M are respectively the smallest and the largest values of $f(x)$ on $[a, b]$ then

$$m(b-a) \leq \int_a^b f(x)dx \leq M(b-a).$$

Theorem 1.11. The mean value theorem: If $f(x)$ is a continuous function on the segment $[a, b];$ then

$$\exists c \in [a, b] : \int_a^b f(x)dx = (b-a)f(c).$$

The number $\mu = \frac{2}{b-a} \int_a^b f(x)dx$ is called the mean value of the function $f(x)$ on the interval $[a, b]$.

Example 21. Calculate:

$$(1) \quad I = \int_0^2 (|x-1| + x) dx \quad (2) \quad J = \int_{-1}^2 (|x^2 - x| - x^2) dx$$

Solutions:

$$(1) \quad |x-1| = \begin{cases} x-1 & \text{if } x > 1 \\ -x+1 & \text{if } x < 1 \end{cases}; \quad \text{then} \quad I = \int_0^1 1dx + \int_1^2 (2x-1)dx = [x]_0^1 + [x^2 - x]_1^2 = 1 + 2 = 3.$$

(2) The sign table of $x^2 - x$ is

x	$-\infty$	0	1	$+\infty$	
x^2-x	$+++++$	0	$-----$	0	$+++++$

$$|x^2 - x| = \begin{cases} x^2 - x & \text{if } x < 0 \wedge x > 1 \\ -x^2 + x & \text{if } 0 \leq x \leq 1 \end{cases} \Rightarrow J = \int_{-1}^0 -x dx + \int_0^1 (-2x^2 + x) dx + \int_1^2 -x dx$$

$$J = \frac{-1}{2} [x^2]_{-1}^0 - \frac{2}{3} [x^3]_0^1 + \frac{1}{2} [x^2]_0^1 - \frac{1}{2} [x^2]_1^2 = \frac{1}{2} - \frac{2}{3} + \frac{1}{2} - \frac{3}{2} = 1 - \frac{13}{6} = -\frac{7}{6}.$$

Example 22. Calculate the mean value of $f(x) = \cos^2 x$ at the interval $\left[\frac{\pi}{12}, \frac{\pi}{4}\right]$.

Solution: We know that

$$\int_{\frac{\pi}{12}}^{\frac{\pi}{4}} (\cos^2 x) dx = \frac{1}{2} \left(\frac{\pi}{6} + \frac{1}{4} \right) \quad \text{see example 16 J.}$$

$$\text{Then } \mu = \frac{1}{\frac{\pi}{4} - \frac{\pi}{12}} \cdot \frac{1}{2} \left(\frac{\pi}{6} + \frac{1}{4} \right) = \frac{2\pi + 3}{4\pi}$$

1.7.3 Methods of integrations

a) Integration with change of the variable

When we change the variable we have to find the new bound of integration adapted at the new variable like the following exercise:

Example 23. Calculate

$$A = \int_0^1 \frac{(2x+3)}{(x^2+3x+1)} dx$$

Solution: Let us $u = x^2+3x+1$ then $du = (2x+3)dx$. We change the bound: $\begin{cases} x = 1 \Rightarrow u = 5 \\ x = 0 \Rightarrow u = 1 \end{cases}$; then we find

$$A = \int_1^5 \frac{du}{u^2} = - \left[\frac{1}{u} \right]_1^5 = 1 - \frac{1}{5} = \frac{4}{5}.$$

b) Integration by parts

We know that $\int u dv = uv - \int v du$; then if $u(x)$ and $v(x)$ are two continuous functions at $[a, b]$, we have

$$\int_a^b u dv = [u.v]_a^b - \int_a^b v . du$$

Example 24. Find

$$B = \int_0^1 x e^x dx$$

Solution:

$$\begin{cases} u = x \Rightarrow du = dx \\ dv = e^x dx \Rightarrow v = e^x \end{cases} \Rightarrow B = [x.e^x]_0^1 - \int_0^1 e^x dx = e - [e^x]_0^1 = e - e + 1 = 1$$

1.7.4 Applications

a) Calculation of surface areas

i) Surface areas between the graph and the x-axis.

We know that if $f(x)$ is a continuous function on an interval $[a, b]$; then the surface areas between the x – axis and the representative curve of $f(x)$ is given by:

$$\begin{cases} \int_a^b f(x)dx & \text{if } f(x) > 0; \forall x \in [a, b] \\ \left| \int_a^b f(x)dx \right| & \text{if } f(x) < 0; \forall x \in [a, b] \end{cases}$$

Example 25. Find the surface areas between the x-axis and the graph of the function $y = x^2 - 2x$; on the interval $[-1, 2]$.

Solution:

The parabola $P \equiv y = x^2 - 2x$ pass through the x-axis at the points $(0, 0)$ and $(2, 0)$. It has a minimum at $(1, -1)$. Then

$$\begin{cases} f(x) < 0 & \text{if } 0 < x < 2 \\ f(x) > 0 & \text{if } x < 0 \text{ and } x > 2. \end{cases}$$

$$S = \int_{-1}^0 (x^2 - 2x) dx + \left| \int_0^2 (x^2 - 2x) dx \right| = \left[\frac{x^3}{3} - x^2 \right]_{-1}^0 + \left| \left[\frac{x^3}{3} - x^2 \right]_0^2 \right| = \frac{8}{3}$$

ii) Surface areas between two graphs of two functions.

Let $f(x)$ and $g(x)$ two continuous functions on the interval $[a, b]$. The Surface areas between the graphs of the functions $f(x)$ and $g(x)$ is given by:

$$\begin{cases} \int_a^b [g(x) - f(x)] dx & \text{if } g(x) > f(x); \forall x \in [a, b] (*) \\ \left| \int_a^b [g(x) - f(x)] dx \right| & \text{if } g(x) < f(x); \forall x \in [a, b] (**). \end{cases}$$

- (*) Means that the graph of $f(x)$ is under the graph of $g(x)$.
- (**) Means that the graph of $g(x)$ is under the graph of $f(x)$.

Example 26. Calculate the area of surface S ranging between the parabola $P \equiv y = x^2 + 1$ and with the line $D \equiv x + y = 3$ on the interval $[0, 2]$.

Solution: The intersection points of D and P are given by the solution of the system

$$\begin{cases} y = x^2 + 1 \\ x + y = 3 \end{cases}$$

which are $(-2, 5)$ and $(1, 2)$. The graph of P and D show that the curve of P is under the line on $[-2, 1]$ and the line D is under the parabola for $x < -2$ and $x > 1$.

Let $f(x) = x^2 + 1$ and $g(x) = 3 - x$; then
$$S = \int_0^1 (g(x) - f(x)) dx + \left| \int_1^2 (g(x) - f(x)) dx \right|.$$

$$\text{Or } g(x) - f(x) = -x^2 - x + 2$$

$$\begin{aligned} S &= \left[-\frac{x^3}{3} - \frac{x^2}{2} + 2x \right]_0^1 + \left| \left[-\frac{x^3}{3} - \frac{x^2}{2} + 2x \right] \right| \\ &= \left(-\frac{1}{3} - \frac{1}{2} + 2 \right) + \left| \left(-\frac{8}{3} - 2 + 4 \right) - \left(-\frac{1}{3} - \frac{1}{2} + 2 \right) \right| \\ &= -\frac{5}{6} + 2 + \frac{11}{6} \\ S &= 3 \text{ unit of Area} \end{aligned}$$

b) Arc length of a curve

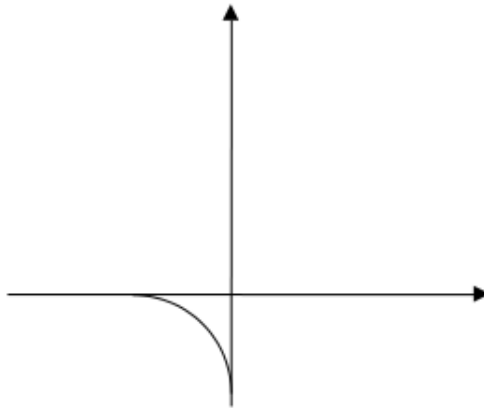
i) Arc length of a curve on rectangular coordinates

The arc length S of a curve of the equation $y = f(x)$ included between two points of abscise $x = a$ and $x = b$ is equal to

$$S = \int_a^b \sqrt{1 + (y')^2} dx$$

Example 27. Calculate the Arc length of an astroid

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$$



Solution:

$$y' = \frac{a^{\frac{1}{3}}}{x^{\frac{1}{3}}} \longrightarrow (y')^2 = \frac{a^{\frac{2}{3}}}{x^{\frac{2}{3}}}$$

The length of quarter of an astroid is

$$\frac{1}{4}S = \int_0^a \sqrt{1 + \frac{a^{\frac{2}{3}}}{x^{\frac{2}{3}}}} dx = \frac{3}{2}a$$

Hence $S = 6a$ Length unit

ii) Arc length of curve given in parametric form

If the curve is given by the equation on parametric form as:

$$\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}$$

Then the arc length S of curve is equal to:

$$S = \int_{t_1}^{t_2} \sqrt{(x')^2 + (y')^2} dt.$$

Where t_1 and t_2 are the values of parameters correspond to extremity of arc.

Example 28. Calculate the Arc length of a cycloid

$$\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases}$$



Solution:

$$\begin{cases} \frac{dx}{dt} = a(1 - \cos t) \\ \frac{dy}{dt} = a \sin t \end{cases}$$

$$S = \int_0^{2\pi} \sqrt{(a(1 - \cos t))^2 + (a \sin t)^2} dt = 2a \int_0^{2\pi} \sin \frac{t}{2} dt = 8a \quad \text{Length Unit}$$

c) Volume of solids of revolution

The Volume of solids obtained by rotation of trapeze line limited by the curve $y = f(x)$, the axis OX and the two vertical straight lines $x = a$ and $x = b$ around the axis OX or OY is given respectively by the formulas:

$$V_x = \pi \int_a^b y^2 dx, \quad V_y = 2\pi \int_a^b xy dx$$

Example 29. Calculate the volume of solid obtain by rotation of figure limited by a semi-wave of solenoid $y = \sin x$ and the $0 \leq x \leq \pi$ segment of the axis OX around

a) of axis OX b) of axis OY .

Solution:

$$a) \quad V_x = \pi \int_0^\pi \sin^2 x dx = \pi \int_0^\pi \left(\frac{1 - \cos 2x}{2} \right) dx = \frac{\pi}{2} \left[x - \frac{\sin 2x}{2} \right]_0^\pi = \frac{\pi^2}{2}$$

$$b) \quad V_y = 2\pi \int_0^\pi x \sin x dx \quad \text{let } u = x \Rightarrow du = dx \text{ and let } dv = \sin x dx \Rightarrow v = -\cos x.$$

$$\text{Then } V_y = 2\pi \left([-x \cos x]_0^\pi + \int_0^\pi \cos x dx \right) = 2\pi ([-\pi \cos \pi + 0] + [\sin x]_0^\pi) = 2\pi^2$$

d) Area of a surface of revolution

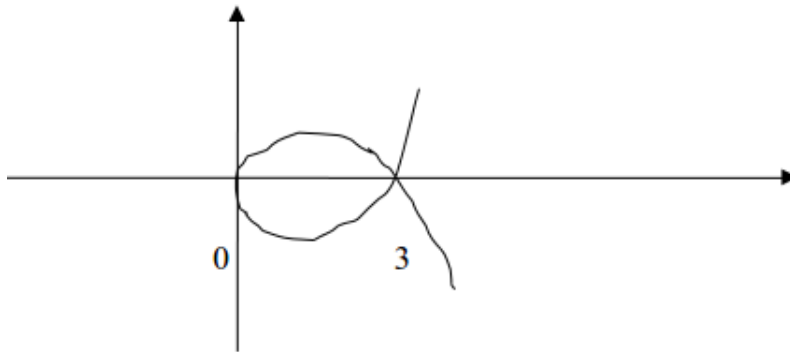
The area of surface generated by the rotation around of the axis OX of an arc of curve $y = f(x)$ included between the points of abscise $x = a$ and $x = b$ is given by the formula:

$$S_x = 2\pi \int_a^b f(x) \frac{ds}{dx} dx = 2\pi \int_a^b f(x) \sqrt{1 + (f'(x))^2} dx$$

$$S_y = 2\pi \int_c^d g(y) \frac{ds}{dy} dy = 2\pi \int_c^d g(y) \sqrt{1 + (g'(y))^2} dy.$$

Where ds is differerntial of the element of arc.

Example 30. Calculate the area of surface generated by rotation around of OX axis of curl of curve $9y^2 = x(3-x)^2$.



Solution: For above part of curve, means that $0 \leq x \leq 3$, we have

$$y = \frac{1}{3}(3-x)\sqrt{x} \longrightarrow \sqrt{1 + (y')^2} = \sqrt{\frac{1+x^2}{2x}}$$

$$S_x = 2\pi \int_0^3 \frac{1}{3}(3-x)\sqrt{x} \sqrt{\frac{1+x^2}{2x}} dx = 2\pi \int_0^3 \frac{\sqrt{2}}{6}(3-x)\sqrt{1+x^2} dx = 3\pi$$

e) Centre of mass

The coordinates of centre of gravity of plane figure (of arc of curve or of plane figure in two dimensions) of mass M is calculated as:

$$\bar{x} = \frac{M_y}{M}, \quad \bar{y} = \frac{M_x}{M}$$

The coordinates of centre of gravity $(\bar{x}, \bar{y})$ of arc plane figure $y = f(x)$, $a \leq x \leq b$:

$$\bar{x} = \frac{\int_a^b x \sqrt{1 + (y')^2} dx}{\int_a^b \sqrt{1 + (y')^2} dx}, \quad \bar{y} = \frac{\int_a^b y \sqrt{1 + (y')^2} dx}{\int_a^b \sqrt{1 + (y')^2} dx}$$

The coordinates of centre of gravity $(\bar{x}, \bar{y})$ of trapeze line $a \leq x \leq b$, $0 \leq y \leq f(x)$

$$\bar{x} = \frac{\int_a^b xy dx}{\int_a^b y dx}, \quad \bar{y} = \frac{\frac{1}{2} \int_a^b y^2 dx}{\int_a^b y dx}$$

Example 31. Determine the centre of gravity for coordinated of semi-circle $x^2 + y^2 = a^2$, ($y \geq 0$).

Solution: We have

$$y = \sqrt{a^2 - x^2} \Rightarrow y' = \frac{-x}{a^2 - x^2}$$

$$\bar{x} = \frac{\int_{-a}^a \frac{ax}{\sqrt{a^2 - x^2}} dx}{\int_{-a}^a \frac{a}{\sqrt{a^2 - x^2}} dx} = 0, \quad \bar{y} = \frac{\int_{-a}^a \sqrt{a^2 - x^2} \frac{a}{\sqrt{a^2 - x^2}} dx}{\int_{-a}^a \frac{a}{\sqrt{a^2 - x^2}} dx} = \frac{2a^2}{\pi a}$$

Exercises 1.5

1. Calculate the total length of curve $r = a \sin^3 \frac{\phi}{3}$, where $0 \leq \phi \leq 3\pi$.
2. Calculate the area of a figure limited by the curve $x = 2 - y - y^2$ and the axis OY?
3. Calculate the area of a figure limited by the parabola $y = \frac{x^2}{2}$, the straight lines $x = 0$ and $x = 1$ and OX axis?
4. Calculate the arc length: a) of the parabola $y = 2\sqrt{x}$ enclosed between abscise $x = 0$ and $x = 1$? b) of the curve $y = e^x$ enclosed between $(0, 1)$ and $(1, e)$?
5. Find the volume of the solid generated when the region between the graphs of $f(x) = \frac{1}{2} + x^2$ and $g(x) = x$ over the interval $[0, 2]$ is revolved about the x-axis?
6. Find the volume of the solid generated when the region enclosed by $y = \sqrt{x}$, $y = 2$ and $x = 0$ is revolved about the y-axis?
7. Find the surface area generated by revolving the curve $y = \sqrt{1 - x^2}$, $0 \leq x \leq \frac{1}{2}$
8. Find the surface area generated by revolving the curve $y = \sqrt[3]{3x}$, $0 \leq y \leq 2$
9. Calculate the centre of mass of asteroide arc $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ which located in first quadrant?
10. Calculate the centre of mass of a figure limited by the curves $y = x^2$, $y = \sqrt{x}$?

1.8 IMPROPER INTEGRALS

In this section we will extend the concept of the definite integral to include:

- a) Integrals with infinite intervals of integration;
- b) Integrals in which the integrand becomes infinite within the interval of integration.

These are called improper integrals.

1st case: If f is continuous on the interval $[a, +\infty]$, then we define the improper integral

$$\int_a^{+\infty} f(x)dx \quad \text{as:} \quad \int_a^{+\infty} f(x)dx = \lim_{l \rightarrow +\infty} \int_a^l f(x)dx$$

If this limit exists, the improper integral is said to **converge**, and the value of the limit is the value assigned to the integral. If the limit does not exist, then the improper integral is said to **diverge**, in which case it is not assigned a value.

Example 32. Find $\int_1^{+\infty} \frac{dx}{x^2}$

Solution:

$$\int_1^{+\infty} \frac{dx}{x^2} = \lim_{l \rightarrow +\infty} \int_1^l \frac{dx}{x^2} = \lim_{l \rightarrow +\infty} \left[\frac{-1}{x} \right]_1^l = \lim_{l \rightarrow +\infty} \left[1 - \frac{1}{l} \right] = 1$$

Thus, $\int_1^{+\infty} \frac{dx}{x^2} = 1$ converges.

Example 33. Find $\int_1^{+\infty} \frac{dx}{x}$

Solution:

$$\int_1^{+\infty} \frac{dx}{x} = \lim_{l \rightarrow +\infty} \int_1^l \frac{dx}{x} = \lim_{l \rightarrow +\infty} [\ln x]_1^l = \lim_{l \rightarrow +\infty} [\ln l] = +\infty$$

Thus, $\int_1^{+\infty} \frac{dx}{x}$ diverges.

2nd case: If f is continuous on the interval $(-\infty, a)$ then we define the improper integral

$$\int_{-\infty}^a f(x)dx \quad \text{as:} \quad \int_{-\infty}^a f(x)dx = \lim_{l \rightarrow -\infty} \int_l^a f(x)dx$$

If this limit exists, the improper integral is said to **converge**, and the value of the limit is the value assigned to the integral. If the limit does not exist, then the improper integral is said to **diverge**, in which case it is not assigned a value.

Example 34. Find $\int_{-\infty}^0 e^x dx$

Solution:

$$\int_{-\infty}^0 e^x dx = \lim_{l \rightarrow -\infty} \int_l^0 e^x dx = \lim_{l \rightarrow -\infty} [e^x]_l^0 = \lim_{l \rightarrow -\infty} [1 - e^l] = 1$$

Thus, $\int_{-\infty}^0 e^x dx = 1$

3rd case: If two improper integrals $\int_{-\infty}^0$ and $\int_0^{+\infty} f(x)dx$ both converge, then we say that $\int_{-\infty}^{+\infty} f(x)dx$ **converges** and we define:

$$\int_{-\infty}^{+\infty} f(x)dx = \int_{-\infty}^0 f(x)dx + \int_0^{+\infty} f(x)dx \quad (-126)$$

Remarks:

- If either integral on the right side of (-126) diverges, then we say that $\int_{-\infty}^{+\infty} f(x)dx$ diverges.
- If f is nonnegative, then the integral $\int_{-\infty}^{+\infty} f(x)dx$ represents the area under $y = f(x)$ over the interval $(-\infty, +\infty)$.

Example 35. Evaluate $\int_{-\infty}^{+\infty} \frac{dx}{1+x^2}$

Solution:

$$\int_0^{+\infty} \frac{dx}{1+x^2} = \lim_{l \rightarrow +\infty} \int_0^l \frac{dx}{1+x^2} = \lim_{l \rightarrow +\infty} [\tan^{-1} x]_0^l = \lim_{l \rightarrow +\infty} [\tan^{-1} l] = \frac{\pi}{2}$$

$$\int_{-\infty}^0 \frac{dx}{1+x^2} = \lim_{l \rightarrow -\infty} \int_l^0 \frac{dx}{1+x^2} = \lim_{l \rightarrow -\infty} [\tan^{-1} x]_l^0 = \lim_{l \rightarrow -\infty} [-\tan^{-1} l] = \frac{\pi}{2}$$

$$\text{Thus, } \int_{-\infty}^{+\infty} \frac{dx}{1+x^2} = \int_0^{+\infty} \frac{dx}{1+x^2} + \int_{-\infty}^0 \frac{dx}{1+x^2} = \frac{\pi}{2} + \frac{\pi}{2} = \pi$$

4th case: If f is continuous on the interval $[a, b]$ but $f(x) \rightarrow +\infty$ or $f(x) \rightarrow -\infty$ as x approaches b from the left, then we define the improper integral

$$\int_a^b f(x)dx \quad \text{as: } \int_a^b f(x)dx = \lim_{l \rightarrow b^-} \int_a^l f(x)dx$$

Example 36. Evaluate

$$\int_0^1 \frac{dx}{\sqrt{1-x}}$$

Solution:

$$\int_0^1 \frac{dx}{\sqrt{1-x}} = \lim_{l \rightarrow 1^-} \int_0^l \frac{dx}{\sqrt{1-x}} = \lim_{l \rightarrow 1^-} [2\sqrt{1-x}]_0^l = \lim_{l \rightarrow 1^-} [2 - 2\sqrt{1-x}] = 2$$

5th case: If f is continuous on the interval $[a, b]$ but $f(x) \rightarrow +\infty$ or $f(x) \rightarrow -\infty$ as x approaches a from the right, then we define the improper integral

$$\int_a^b f(x)dx \quad \text{as: } \int_a^b f(x)dx = \lim_{l \rightarrow a^+} \int_l^b f(x)dx$$

The improper integral **converges or diverges** depending on whether the limit does or does not exist.

Example 37. Evaluate

$$\int_1^2 \frac{dx}{1-x}$$

Solution:

$$\int_1^2 \frac{dx}{1-x} = \lim_{l \rightarrow 1^+} \int_l^2 \frac{dx}{1-x} = \lim_{l \rightarrow 1^+} [-\ln |1-x|_l^2] = \lim_{l \rightarrow 1^+} [\ln |1-l|] = -\infty$$

6th case: If f is continuous on the interval $[a, b]$ with the exception that at some point c satisfying $a \leq c \leq b$, $f(x)$ becomes infinite but $f(x) \rightarrow +\infty$ or $f(x) \rightarrow -\infty$ as x approaches c from the left or right. If the two improper integrals $\int_a^c f(x)dx$ and $\int_c^b f(x)dx$ **converges**, and we define:

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx \quad (-132)$$

If either integral on the right side of (-132) **diverges**, then we say that $\int_a^b f(x)dx$ **diverges**.

Example 38. Evaluate

$$\int_1^4 \frac{dx}{(x-2)^{\frac{2}{3}}}$$

Solution:

$$\int_1^4 \frac{dx}{(x-2)^{\frac{2}{3}}} = \int_1^2 \frac{dx}{(x-2)^{\frac{2}{3}}} + \int_2^4 \frac{dx}{(x-2)^{\frac{2}{3}}}$$

$$\text{But } \int_1^2 \frac{dx}{(x-2)^{\frac{2}{3}}} = \lim_{l \rightarrow 2^-} \int_1^l \frac{dx}{(x-2)^{\frac{2}{3}}} = 3$$

$$\int_2^4 \frac{dx}{(x-2)^{\frac{2}{3}}} = \lim_{l \rightarrow 2^+} \int_l^4 \frac{dx}{(x-2)^{\frac{2}{3}}} = 3\sqrt[3]{2}$$

$$\text{Thus, } \int_1^4 \frac{dx}{(x-2)^{\frac{2}{3}}} = 3 + 3\sqrt[3]{2}$$

EXERCISES 1.6

1) Evaluate the integrals that converge:

$$\begin{array}{llll} a) \int_{-1}^{+\infty} \frac{dx}{1+x^2} & b) \int_{-\inf}^2 \frac{dx}{x^2+4} & c) \int_0^8 \frac{dx}{\sqrt{x}} & d) \int_0^9 \frac{dx}{\sqrt{9-x}} \\ e) \int_{-2}^2 \frac{dx}{x^2} & f) \int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sqrt{1-\sin \theta}} d\theta & g) \int_1^{+\infty} \frac{dx}{\sqrt{x^2-1}} \end{array}$$

2. a) Show that $\int_1^{+\infty} \frac{dx}{x^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

b) Show that $\int_0^1 \frac{dx}{x^p}$ converges if $p < 1$ and diverges if $p \geq 1$.

3) a) Find the area of the region between the x-axis and the curve $y = \frac{8}{x^2-4}$ for $x \geq 3$.

b) Find the area of the region bounded by $y = 0, x = 0, x = 4, \wedge y = (1-x)^{-2}$.

4) a) Show that $\int_{-1}^0 \frac{dx}{x} = -\infty$ and $\int_0^1 \frac{dx}{x} = +\infty$ b) Evaluate $\int_{-1}^1 \frac{dx}{x}$.

5) a) It is possible for improper integral to diverge without becoming infinite. Show that

$$\int_0^{+\infty} \cos x dx \text{ does this . } \quad b) \text{ Evaluate } \int_0^{+\infty} \cos x dx$$

6) For what values of p does $\int_0^{+\infty} e^{px} dx$ converges?

7) In electromagnetic theory, the magnetic potential at a point on the axis of a circular coil is given by

$$u = \frac{2\pi N I r}{k} \int_a^{+\infty} \frac{dx}{(x^2 + r^2)^{\frac{3}{2}}}$$

where N, I, r, k and a are constants. Find u ?

2 Numerical and functional series

2.1 Convergence and divergence of numerical sequences and series

2.1.1 Convergence and divergence of numerical series

a) Definitions

Let $S \subset \mathbb{N}$, we call sequence of real numbers any application from S in $\mathbb{R}$. we denote the range of a natural number n by u_n . Then we get the terms of the sequence which are: $u_1, u_2, \dots, u_n, \dots$ and we denote them by **bracket notation** as $(u_n)_{n \in \mathbb{N}}$ or $\{u_n\}_{n=1}^{+\infty}$. the number u_1 is the first term of the sequence and u_n is the general term of the sequence $(u_n)_{n \in \mathbb{N}}$.

Example 39. In the sequence $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n}, \dots$ the first term is 1, the second term is $\frac{1}{2}$ and $\frac{1}{n}$ is the general term.

In the sequence $1, 4, 8, 16, \dots, n^2, \dots$ the first term is 1; the second term is 4, the general term is n^2 .

- We say that a sequence $(u_n)_{n \in \mathbb{N}}$ is **increasing** $\Leftrightarrow \forall n \in \mathbb{N} : u_n \leq u_{n+1}$; and $(u_n)_{n \in \mathbb{N}}$ is strictly increasing $\Leftrightarrow \forall n \in \mathbb{N} : u_n < u_{n+1}$.
- We say that a sequence $(u_n)_{n \in \mathbb{N}}$ is **decreasing** $\Leftrightarrow \forall n \in \mathbb{N} : u_n \geq u_{n+1}$; and $(u_n)_{n \in \mathbb{N}}$ is strictly decreasing $\Leftrightarrow \forall n \in \mathbb{N} : u_n > u_{n+1}$.
- We say that a sequence $(u_n)_{n \in \mathbb{N}}$ is **stationary (constant)** $\Leftrightarrow \forall n \in \mathbb{N} : u_n = u_{n+1}$.
- A sequence which is increasing (or decreasing) only is monotonous.
- We say that a sequence $(u_n)_{n \in \mathbb{N}}$ is **bounded** if there exists two real numbers m and M such that for any; we have $m \leq u_n \leq M$. The real m is the lower bound of the sequence and M is the upper bound of the sequence.

Exercise 1: We define a sequence $(u_n)_{n \in \mathbb{N}}$ by $u_n = 2 + \frac{1}{n}$. Find the terms of the sequence and show if it is monotonous or bounded.

Solution:

$$u_1 = 3; \quad u_2 = \frac{5}{2}; \quad u_3 = \frac{7}{3}; \quad u_4 = \frac{9}{4}; \dots; u_{n+1} = 2 + \frac{1}{n+1}$$

$(u_n)_{n \in \mathbb{N}}$ is strictly decreasing because

$$u_n - u_{n+1} = \left[2 + \frac{1}{n}\right] - \left[2 + \frac{1}{n+1}\right] = \frac{1}{n} - \frac{1}{n+1} = \frac{n+1-n}{n(n+1)} = \frac{1}{n(n+1)} > 0 \quad u_n > u_{n+1};$$

then it is monotonous strictly. The lower bound is 2 and the upper bound is 3; then the sequence is bounded.

Example 40. Do the same exercise for (i) $v_n = \frac{n}{n+1}$ and (ii) For $w_1 = 2$; and $w_{n+1} = 3 + w_n$

b) Convergent and divergent sequences

- We say that the sequence $(u_n)_{n \in \mathbb{N}}$ tends to l or has a limit l for all $\epsilon > 0$, there is an integer p such that $n > p \Rightarrow |u_n - l| < \epsilon$. We shall denote $\lim_{n \rightarrow \infty} u_n = l$; and we say that $(u_n)_{n \in \mathbb{N}}$ converges to l .

Remark: If the sequence has a limit; this limit is unique.

A sequence which does not converge is divergent. If $(u_n)_{n \in \mathbb{N}}$ is divergent then $\lim_{n \rightarrow \infty} u_n = \infty$.

Theorem 2.1. Suppose that the sequences a_n and b_n converge to limits L_1 and L_2 , respectively, and c is constant. Then:

$$\begin{aligned} a) \quad \lim_{n \rightarrow +\infty} c &= c & b) \quad \lim_{n \rightarrow +\infty} ca_n &= cL_1 & c) \quad \lim_{n \rightarrow +\infty} a_n \pm b_n &= \lim_{n \rightarrow +\infty} a_n \pm \lim_{n \rightarrow +\infty} b_n = L_1 \pm L_2 \\ d) \quad \lim_{n \rightarrow +\infty} a_n \cdot b_n &= \lim_{n \rightarrow +\infty} a_n \cdot \lim_{n \rightarrow +\infty} b_n = L_1 \cdot L_2 & e) \quad \lim_{n \rightarrow +\infty} \frac{a_n}{b_n} &= \frac{\lim_{n \rightarrow +\infty} a_n}{\lim_{n \rightarrow +\infty} b_n} = \frac{L_1}{L_2} \quad \text{if } L_2 \neq 0 \end{aligned}$$

Example 41. Calculate the limit of the following sequences:

$$\begin{aligned} (1) \quad u_n &= \frac{4n^3 - 5n + 7}{2n^3 + 3n^2 + 6}; & (2) \quad u_n &= \frac{3 + 5 + 7 + \dots + (4n + 1)}{2n^2 + n + 6}; \\ (3) \quad u_n &= \frac{\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n}}{1 + \frac{1}{3} + \frac{1}{9} + \dots + \frac{1}{3^n}}; & (4) \quad u_n &= \left(\frac{n^3}{n^2 + 1} - \frac{3n^2}{3n + 1} \right) \end{aligned}$$

Solutions:

$$(1) \quad \lim_{n \rightarrow +\infty} u_n = \lim_{n \rightarrow +\infty} \frac{4n^3 - 5n + 7}{2n^3 + 3n^2 + 6} = \frac{n^3 \left(4 - \frac{5}{n^2} + \frac{7}{n^3} \right)}{n^3 \left(2 + \frac{3}{n} + \frac{6}{n^3} \right)} = \frac{4 - 0 + 0}{2 + 0 + 0} = 2$$

(2) The numerator is the sum S of an arithmetic progression: the reason is 2 and the number p of term is such that $4n + 1 = (p - 1)2 + 3$, then $p = 2n$.

$$S = \frac{(3 + 4n + 1)2n}{2} = n(4n + 4)$$

$$\lim_{n \rightarrow +\infty} u_n = \lim_{n \rightarrow +\infty} \frac{3 + 5 + 7 + \dots + (4n + 1)}{2n^2 + n + 6} = \frac{n^2 \left(4 + \frac{4}{n} \right)}{n^2 \left(2 + \frac{1}{n} + \frac{6}{n^2} \right)} = \frac{4 + 0}{2 + 0 + 0} = 2.$$

(3) The numerator is the sum S of n terms of a geometric progression: the reason is $\frac{1}{2}$. So $S = \frac{1}{2} \cdot \frac{1 - \frac{1}{2^n}}{1 - \frac{1}{2}} = \left(1 - \frac{1}{2^n} \right)$; The denominator is the sum T of $n + 1$ terms of a geometric progression: The reason is $\frac{1}{3}$. So $T = 1 \cdot \frac{1 - \frac{1}{3^{n+1}}}{1 - \frac{1}{3}} = \frac{3}{2} \left(1 - \frac{1}{3^{n+1}} \right)$.

Finally

$$\lim_{n \rightarrow \infty} u_n = \frac{\lim_{n \rightarrow +\infty} \left(1 - \frac{1}{2^n} \right)}{\lim_{n \rightarrow \infty} \frac{3}{2} \left(1 - \frac{1}{3^{n+1}} \right)} = \frac{1}{\frac{3}{2}} = \frac{2}{3}.$$

(4) We put at the same denominator and we follow the method used at (1); finally we find $\frac{1}{3}$.

Example 42. The sequence $(u_n)_{n \in \mathbb{N}}$ is an **alternate sequence** if $\forall n \in \mathbb{N} : u_n$ and u_{n+1} have opposite signs.

Example 43. A sequence which has the general term $u_n = \frac{(-1)^n}{n+3}$ is an **alternate sequence**.

c) Geometric and Arithmetic progressions.

1. Arithmetic progressions

- $(u_n)_{n \in \mathbb{N}}$ is an arithmetic progression if $\forall n \in \mathbb{N}; \exists r \in \mathbb{R} - \{0\} : u_n - u_{n-1} = r$. The real number r is called the **reason of the progression**.
- If $r > 0$ then the progression is strictly increasing.
- If $r < 0$ then the progression is strictly decreasing.
- If $r = 0$ then the progression is constant.

Property: Let $(u_n)_{n \in \mathbb{N}}$ an arithmetic progression with reason r and with the first term u_1

1. $u_n = u_1 + u_{n-1}r$
2. $S_n = \sum_{i=1}^n u_i = u_1 + u_2 + \dots + u_n = \frac{(u_1 + u_n)n}{2}$

Example 44. Given an arithmetic progression $\{b_n\} = (8, 6, 4, 2 \dots)$. Find S_{20} ?

Solution: The reason is -2 and the first term is $b_1 = 8$; then $b_{20} = 8 + 19(-2) = -30$ and $S_{20} = \frac{(8 + (-30)) \cdot 20}{2} = -220$.

2. Geometric progression

- $(u_n)_{n \in \mathbb{N}}$ is a geometric progression if $\forall n \in \mathbb{N}; \exists q \in \mathbb{R} - \{0, 1\} : u_n = u_{n-1} \cdot q$. The real number q is called the **reason of the progression**.
- If $q > 0$ then the term of the progression have the sign of u_1 .
- If $q < 0$ then the progression is an alternate sequence.
- If $q = 1$ then the progression is constant.

Property: Let $\{u_n\}$ a geometric progression with reason q and with the first term u_1 . If u_n is the n^{th} term; and S_n the sum of n first terms (partial sum); then

1. $u_n = u_1 \cdot q^{n-1}$.
2. $S_n = \sum_{i=1}^n u_i = u_1 + u_2 + u_3 + \dots + u_n = u_1 \frac{(1 - q^n)}{1 - q}$.

Example 45. Given an geometric progression $\{b_n\} = (2, 10, 50, 250, \dots)$. Find S_6

Solution: The reason is 5 and the first term is $b_1 = 2$; then $b_6 = 2 \cdot 5^5 = 6250$ and $S_6 = 2 \frac{(1 - 5^6)}{1 - 5} = 7812$.

2.1.2 Convergence and divergence of infinite series

a) Definitions

- An infinite series is an expression of the form

$$u_1 + u_2 + u_3 + \dots + u_k = \sum_{k=1}^{+\infty} u_k$$

The numbers $u_1, u_2, u_3, \dots$ are called the **terms** of the series.

$$s_1 = u_1$$

$$s_2 = u_1 + u_2$$

$$s_3 = u_1 + u_2 + u_3$$

...

$$s_n = u_1 + u_2 + u_3 + \dots + u_n = \sum_{k=1}^n u_k$$

The number s_n is called the n^{th} **partial sum** of the series and the sequence $\{s_n\}_{n=1}^{+\infty}$ is called the **sequence of partial sums**.

Example 46. For the infinite series $\frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \frac{3}{10^4} + \dots$

The partial sums are

$$s_1 = \frac{3}{10}$$

$$s_2 = \frac{3}{10} + \frac{3}{10^2} = \frac{33}{100}$$

$$s_3 = \frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} = \frac{333}{1000}$$

$$s_4 = \frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \frac{3}{10^4} = \frac{3333}{10000}$$

As n increases, the partial sum $s_n = u_1 + u_2 + u_3 + \dots + u_n$ includes more and more terms of the series.

- Let $\{s_n\}$ be the sequence of partial sums of the series $\sum_{k=1}^{+\infty} u_k$.

If the sequence $\{s_n\}$ converges to a limit S , then the series is said to **converge** and S is called $S = \sum_{k=1}^{+\infty} u_k$.

If the sequence of partial sums of a series diverges, then the series is said to **diverge**. A divergent series has no sum.

For above example:

$$s_n = \frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \frac{3}{10^4} + \dots + \frac{3}{10^n} = \frac{3}{10} \left(1 + \frac{1}{10} + \frac{1}{10^2} + \frac{1}{10^3} + \dots + \frac{1}{10^{n-1}} \right)$$

$$s_n = \frac{3}{10} \left[1 \times \frac{1 - \frac{1}{10^n}}{1 - \frac{1}{10}} \right] = \frac{1}{3} \left[1 - \frac{1}{10^n} \right]$$

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} \frac{1}{3} \left[1 - \frac{1}{10^n} \right] = \frac{1}{3}$$

Thus, $\frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \frac{3}{10^4} + \dots = \frac{1}{3}$.

Theorem 2.2. A geometric series $a + ar + ar^2 + ar^3 + \dots + ar^{k-1} + \dots$ converges if $|r| < 1$ and diverges if $|r| \geq 1$. If the series converges the sum is $\frac{a}{1-r} = a + ar + ar^2 + ar^3 + \dots + ar^{k-1} + \dots$

Example 47. The series $5 + \frac{5}{4} + \frac{5}{4^2} + \dots + \frac{5}{4^{k-1}} + \dots$ is a geometric series with $a = 5$ and $r = \frac{1}{4}$. Since $|r| = \frac{1}{4} < 1$ then the series converges and the sum is $\frac{a}{1-r} = \frac{5}{1-\frac{1}{4}} = \frac{20}{3}$

EXERCISES 3.1

1. Express the following sequences in bracket notation:

$$a) \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots \quad b) \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots \quad c) 1, -1, 1, -1, \dots \quad d) \frac{1}{2}, -\frac{2}{3}, \frac{3}{4}, -\frac{4}{5}, \dots \quad e) 1, 3, 5, 7, \dots$$

2. Show that the sequence is strictly monotone:

$$a) \left\{ \frac{n}{2n+1} \right\}_{n=1}^{+\infty} \quad b) \left\{ \frac{1}{n + \ln n} \right\}_{n=1}^{+\infty} \quad c) \{ne^{-2n}\}_{n=1}^{+\infty} \quad d) \left\{ \frac{100^n}{n!} \right\}_{n=1}^{+\infty}$$

3. Show that the sequence is monotone and determine whether it converges:

$$a) \left\{ \frac{n}{5^n} \right\}_{n=1}^{+\infty} \quad b) \left\{ n - \frac{1}{n} \right\}_{n=1}^{+\infty} \quad c) \left\{ 2 + \frac{1}{n} \right\}_{n=1}^{+\infty} \\ d) \left\{ \frac{2^n}{(n+1)!} \right\}_{n=1}^{+\infty} \quad e) \left\{ \cos \frac{\pi}{2n} \right\}_{n=1}^{+\infty} \quad f) \left\{ \frac{4n-1}{5n+2} \right\}_{n=1}^{+\infty}$$

4. Consider $(u_n)_{n \in \mathbb{N}}$ as an arithmetic progression such that $u_8 = 27$ and $u_2 + u_3 + u_4 = 21$. Calculate the common difference, write u_n as function of n ?

5. Three numbers are in arithmetic progression. Find the numbers if their sum is 30 and the sum of their squares is 332?

6. If $a - 4, a + 8$, and 54 are three consecutive terms of a geometric sequence, find the possible values of a , and the numerical value of the next term for each value of a found?

7. Calculate the following limits:

$$(a) \lim_{n \rightarrow \infty} \left(\sqrt{n^2 - 5n + 6} - n \right) \quad (b) \lim_{n \rightarrow \infty} \left(n \sin \frac{\pi}{n} \right) \quad (c) \lim_{n \rightarrow 3} (4n - 20)^{2n-4}$$

8. If $t - 4, t + 8$, and 54 are three consecutive terms of a geometric sequence, find the possible values of a , and the numerical value of the next term for each value of a found?

9. Calculate $1 + 3 + 5 + 7 + 9 + \dots + 129$

2.2 Techniques for testing the convergence series

Theorem 2.3. If the series $\sum u_k$ converges, then $\lim_{k \rightarrow +\infty} u_k = 0$.

Theorem 2.4. The divergence test: If $\lim_{k \rightarrow +\infty} u_k \neq 0$, then $\sum u_k$ diverges.

Example 48.

$$\sum_{k=1}^{+\infty} \frac{k}{k+1} = \frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5} + \dots + \frac{k}{k+1} + \dots \quad \text{diverges since} \quad \lim_{k \rightarrow +\infty} \frac{k}{k+1} = 1 \neq 0$$

Theorem 2.5. (a) If $\sum u_k$ and $\sum v_k$ are convergent series, then $\sum (u_k \pm v_k)$ is convergent series

(b) If $c \neq 0$ then the series $\sum u_k$ and $\sum cu_k$ both converge or both diverge.

(c) Convergence or divergence is unaffected by deleting a finite number of terms from the beginning of a series, that is, for any positive integer K , the series $\sum_{k=1}^{+\infty} u_k$ and $\sum_{k=K}^{+\infty} u_k = u_K + u_{K+1} + \dots$ both converge or both diverge.

Example 49.

$$\sum_{k=1}^{+\infty} \left(\frac{3}{4^k} - \frac{2}{5^{k-1}} \right) \quad \text{converges and} \quad \sum_{k=1}^{+\infty} \left(\frac{5}{k} \right) = 5 \sum_{k=1}^{+\infty} \left(\frac{1}{k} \right) \quad \text{diverges.}$$

Theorem 2.6. The integral test: Let $\sum u_k$ be a positive terms, and let $f(x)$ be the function that results when k is replaced by x by formula for u_k . If f is decreasing and continuous for $x \geq 1$, then $\sum_{k=1}^{+\infty} u_k$ and $\int_1^{+\infty} f(x)dx$ both converge or both diverge.

Example 50. Determine whether $\sum_{k=1}^{+\infty} \frac{1}{k^2}$ converges or diverges?

Answer: $f(x) = \frac{1}{x^2}$ which is decreasing and continuous for all $x \geq 1$.

$$\int_1^{+\infty} \frac{1}{x^2} dx = \lim_{b \rightarrow +\infty} \int_1^b \frac{1}{x^2} dx = \lim_{b \rightarrow +\infty} \left[-\frac{1}{x} \right]_1^b = \lim_{b \rightarrow +\infty} \left[1 - \frac{1}{b} \right] = 1$$

$$\int_1^{+\infty} \frac{1}{x^2} dx \rightarrow 1, \quad \text{then} \quad \sum_{k=1}^{+\infty} \frac{1}{k^2} \rightarrow 1$$

Theorem 2.7. Convergence of p -series

$$\sum_{k=1}^{+\infty} \frac{1}{k^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{3^p} + \dots + \frac{1}{n^p} + \dots, \quad \text{converges if } p \geq 1 \text{ and diverges if } 0 \leq p < 1.$$

Theorem 2.8. Comparison test: Let $\sum a_k$ and $\sum b_k$ be the series with positive terms and suppose $a_i \leq b_i, \forall i$

(a) If $\sum b_k \rightarrow$ (converges), then $\sum a_k \rightarrow$ (converges),

(b) If $\sum a_k \nrightarrow$ (diverges), then $\sum b_k \nrightarrow$ (diverges).

Example 51.

$$\sum_{k=1}^{+\infty} \frac{1}{k^2 + 3k} < \sum_{k=1}^{+\infty} \frac{1}{k^2} \quad (\text{converges}), \text{ then } \sum_{k=1}^{+\infty} \frac{1}{k^2 + 3k} \text{ converges.}$$

Theorem 2.9. The ratio test: Let $\sum u_k$ be a series with positive terms and suppose $\lim_{k \rightarrow +\infty} \frac{u_{k+1}}{u_k} = q$

(a) If $q < 1$, then the series converges.

(b) If $q > 1$, or $q = +\infty$ then the series diverges.

(c) If $q = 1$, then the series may converges or diverges, so that another test must be tried.

Example 52.

$$\text{The series } \sum_{k=1}^{+\infty} \frac{1}{k} \quad \text{and} \quad \sum_{k=1}^{+\infty} \frac{1}{k^2} \quad \text{both have } q = 1.$$

Since the first is **divergent harmonic series** and the second is **a convergent q-series**, the ratio test does not distinguish between convergent and divergent when $q = 1$.

Example 53. The series

$$\text{a) } \sum_{k=1}^{+\infty} \frac{1}{k!} \text{ converges} \quad \text{b) } \sum_{k=1}^{+\infty} \frac{1}{2^k} \text{ converges} \quad \text{c) } \sum_{k=1}^{+\infty} \left(\frac{4k-5}{2k+1} \right)^k \text{ converges}$$

Criterion of comparison

If $\lim_{k \rightarrow +\infty} \frac{a_k}{b_k} = I \in \mathbb{R}$ and $(a_k \approx b_k)$ then $\sum_{k=1}^{+\infty} a_k$ and $\sum_{k=1}^{+\infty} b_k$ convergent or divergent both.

Example 54. 1. The series $1 + \frac{1}{2} + \frac{1}{5} + \frac{1}{2k-1} + \dots$ diverges because

$$\frac{1}{2k-1} \approx \frac{1}{k} \quad \text{and} \quad \lim_{k \rightarrow +\infty} \frac{\frac{1}{2k-1}}{\frac{1}{k}} = \frac{1}{2} \neq 0 \quad \text{and the series } \sum_{k=1}^{+\infty} b_k = \sum_{k=1}^{+\infty} \frac{1}{2k-1} \quad \text{diverges}$$

2. The series

$$\sum_{k=1}^{+\infty} \frac{3k^3 - 2k^2 + 4}{k^5 - k^3 + 2} \leq \sum_{k=1}^{+\infty} \frac{3}{k^2} \quad \text{and} \quad q = \lim_{k \rightarrow +\infty} \frac{\frac{3k^3 - 2k^2 + 4}{k^5 - k^3 + 2}}{\frac{3}{k^2}} = 1 \neq 0$$

$$\text{The series } \sum_{k=1}^{+\infty} \frac{3k^3 - 2k^2 + 4}{k^5 - k^3 + 2} \quad \text{converges because} \quad \sum_{k=1}^{+\infty} \frac{3}{k^2} \quad \text{converges.}$$

Theorem 2.10. Alternating series test (conditionally convergence)

An alternating series

$$\sum_{k=1}^{+\infty} (-1)^k a_k \vee \sum_{k=1}^{+\infty} (-1)^{k+1} a_k.$$

Converges if:

a) $a_1 \geq a_2 \geq a_3 \geq \dots \geq a_k \geq \dots$

b) $\lim_{k \rightarrow +\infty} a_k = 0$.

Example 55.

The series $1 - \frac{1}{2} + \frac{1}{3} - \dots + (-1)^{k+1} \frac{1}{k} + \dots$ is called the **alternating harmonic series**.

$a_k > a_{k+1}$ and $\lim_{k \rightarrow +\infty} a_k = \lim_{k \rightarrow +\infty} \frac{1}{k} = 0$. This series converges by the alternating series test.

Theorem 2.11. The ratio test for absolute convergence

Let $\sum u_k$ be a series with nonzero terms and suppose $\lim_{k \rightarrow +\infty} \frac{|u_{k+1}|}{|u_k|} = q$

a) If $q < 1$, the series converges absolutely.

b) If $q > 1$, or $q = +\infty$ the series diverges.

c) If $q = 1$, no conclusion about convergence can be drawn.

Example 56.

The series $\sum_{k=1}^{+\infty} (-1)^k \frac{2^k}{k!}$ converges absolutely. Because $q = \lim_{k \rightarrow +\infty} \frac{|u_{k+1}|}{|u_k|} = \lim_{k \rightarrow +\infty} \frac{2^{k+1}}{(k+1)!} \div \frac{2^k}{k!} = 0 < 1$.

Theorem 2.12. Cauchy criterion

Let $\sum u_k$ be a series with terms and suppose $\lim_{k \rightarrow +\infty} \sqrt[n]{u_k} = q \in \mathbb{R}$

a) If $q < 1$, then the series converges.

b) If $q > 1$, then the series diverges.

c) If $q = 1$, no conclusion about convergence can be drawn.

EXERCISE 3.2

1. Find the first four partial sum, find a closed form for the nth partial sum, determine whether the series converges, and if so give the sum:

$$a) \sum_{k=1}^{+\infty} \frac{2}{5^{k-1}} \quad b) \sum_{k=1}^{+\infty} \frac{2^{k-1}}{4} \quad c) \sum_{k=1}^{+\infty} \frac{1}{(k+1)(K+2)}$$

2. Determine whether the series converges or diverges. If it converges, find the sum:

$$a) \sum_{k=1}^{+\infty} \frac{1}{5^k} \quad b) \sum_{k=1}^{+\infty} \left(\frac{2}{3}\right)^{k+2} \quad c) \sum_{k=1}^{+\infty} 4^{k-1} \quad d) \sum_{k=1}^{+\infty} \left(\frac{-3}{4}\right)^{k-1} \quad e) \sum_{k=1}^{+\infty} (-1)^{k-1} \frac{7}{6^{k-1}}$$

3. Express the repeating decimal as a fraction:

$$a) 0.4444\dots \quad b) 0.9999\dots \quad c) 0.159159159\dots \quad d) 5.373737\dots \quad e) 0.45114114\dots$$

4. Show that:

$$a) \sum_{k=1}^{+\infty} \ln \left(1 - \frac{1}{k^2}\right) = \ln 2 \quad b) \sum_{k=1}^{+\infty} \frac{\sqrt{k+1} - \sqrt{k}}{\sqrt{k^2 + k}}$$

5. Using geometric series to show:

$$a) \sum_{k=1}^{+\infty} (-1)^k x^k = \frac{1}{1+x} \quad \text{if } -1 < x < 1 \quad b) \sum_{k=1}^{+\infty} (x-3)^k = \frac{1}{4-x} \quad \text{if } 2 < x < 4$$

$$c) \sum_{k=1}^{+\infty} (-1)^k x^{2k} = \frac{1}{1+x^2} \quad \text{if } -1 < x < 1$$

6. Find the sum of series

$$a) \sum_{k=1}^{+\infty} \left(\frac{1}{2^k} + \frac{1}{4^k}\right) \quad b) \sum_{k=1}^{+\infty} \left[\frac{7}{3^k} + \frac{6}{(k+3)(k+4)}\right] \quad c) \sum_{k=1}^{+\infty} \left(\frac{1}{k^2-1} - \frac{7}{10^{k-1}}\right)$$

7. Determine whether the given p-series converges or diverges:

$$a) \sum_{k=1}^{+\infty} \frac{1}{k^3} \quad b) \sum_{k=1}^{+\infty} \frac{1}{\sqrt{k}} \quad c) \sum_{k=1}^{+\infty} \frac{1}{\sqrt[4]{k}} \quad d) \sum_{k=1}^{+\infty} \frac{1}{k^\pi} \quad e) \sum_{k=1}^{+\infty} \frac{1}{\sqrt[3]{k^5}}$$

8. Use the divergence test to show that the series diverge:

$$a) \sum_{k=1}^{+\infty} \frac{K+1}{k+2} \quad b) \sum_{k=1}^{+\infty} \ln k \quad c) \sum_{k=1}^{+\infty} \frac{k^2+k+3}{2k+1} \quad d) \sum_{k=1}^{+\infty} \left(1 + \frac{1}{k}\right)^k$$

9. Apply the ratio test for studying the convergence of the following series:

$$a) \sum_{k=1}^{+\infty} \left(\frac{3K+2}{2k-1}\right)^k \quad b) \sum_{k=1}^{+\infty} \left(\frac{k}{100}\right)^k \quad c) \sum_{k=1}^{+\infty} \frac{k}{5^k} \quad d) \sum_{k=1}^{+\infty} (1 + e^{-k})^k$$

10. Classify the series as: absolutely convergent, conditionally convergent, or divergent

$$a) \sum_{k=1}^{+\infty} \frac{(-1)^{k+1}}{3k} \quad b) \sum_{k=1}^{+\infty} \frac{\cos k\pi}{k} \quad c) \sum_{k=1}^{+\infty} \frac{\sin k}{k^3} \quad d) \sum_{k=1}^{+\infty} \frac{(-1)^k}{\sqrt{k(k+1)}}$$

2.3 POWER SERIES, MACLAURIN SERIES, TAYLOR SERIES

2.3.1 POWER SERIES

In previous sections we studied series with constant terms. In this section we discuss on the series whose terms variables. Such series are of fundamental importance in many branches of mathematics and the physical sciences.

a) Definition

The series $\sum_{k=0}^{+\infty} c_k x^k = c_0 + c_1 x + c_2 x^2 + \dots + c_k x^k + \dots$

Is called a **power series** in x . Where $c_0, c_1, c_2, \dots$ are constants and x is a variable.

Some example of Power series

$$\begin{aligned} i. \sum_{k=0}^{+\infty} x^k &= x + x^2 + \dots & ii. \sum_{k=0}^{+\infty} \frac{x^k}{k!} &= 1 + x + \frac{x^2}{2} + \dots & iii. \sum_{k=0}^{+\infty} (-1)^k \frac{x^{k+1}}{k+1} &= 1 - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \\ iv. \sum_{k=0}^{+\infty} (-1)^k \frac{x^{2k}}{(2k)!} &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots & v. \sum_{k=0}^{+\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} &= 1 - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \end{aligned}$$

c) A fundamental problem

For what values of x does a given series, $\sum c_k x^k$, converges?

The following theorem is the fundamental result on convergence of power series.

Theorem 2.13. *For any power series in x , exactly one of the following is true:*

- a) *The series converges only for $x = 0$,*
- b) *The series converges absolutely for all x ,*
- c) *The series converges absolutely for all x in some finite open interval $(-R, R)$. At the point $x = -R$ and $x = R$ the series may converge absolutely, converge conditionally, or diverge, depending on the particular series.*
 - *In case c), where the power series converges absolutely for $|x| < R$ and diverges for $|x| > R$, we call R the **radius of convergence**.*
 - *In case a), the **radius of convergence** $R = 0$.*
 - *In case b), where the series converges absolutely for all x , we define the radius of convergence to be $R = +\infty$.*
 - *The set of values of x for which a power series converges is called the **interval of convergence**.*

Example 57. 1. Find the interval of convergence and radius of convergence of the power series

$$\sum_{k=0}^{+\infty} x^k$$

Solution:

$$s_k = 1 + x + x^2 + \dots + x^k = \frac{1 - x^{k+1}}{1 - x}$$

$$\lim_{k \rightarrow \infty} s_k = \begin{cases} \lim_{k \rightarrow \infty} \frac{1 - x^{k+1}}{1 - x} & \text{if } |x| < 1 \\ \infty & \text{if } |x| \geq 1 \end{cases}$$

The series converges if $|x| < 1$ and diverges if $|x| \geq 1$.

2. Find the interval of convergence and radius of convergence of the power series

$$\sum_{k=0}^{+\infty} \frac{x^k}{k!}$$

Solution: We apply the ratio test for absolute convergence. For every real number x ,

$$q = \lim_{k \rightarrow \infty} \frac{|u_{k+1}|}{|u_k|} = \lim_{k \rightarrow \infty} \left| \frac{x^{k+1}}{(k+1)!} \cdot \frac{k!}{x^k} \right| = \lim_{k \rightarrow \infty} \left| \frac{x}{k+1} \right| = 0$$

Since $q < 1$, for all x , the series converges absolutely for all x . Thus, the interval of convergence is $(-\infty, \infty)$ and the radius of convergence is $R = +\infty$.

3. Find the interval of convergence and radius of convergence of the power series

$$\sum_{k=0}^{+\infty} \frac{(-1)^k x^k}{3^k(k+1)}$$

Solution:

Since $|(-1)^k| = |(-1)^{k+1}| = 1$, we obtain

$$q = \lim_{k \rightarrow \infty} \frac{|u_{k+1}|}{|u_k|} = \lim_{k \rightarrow \infty} \left| \frac{x^{k+1}}{3^{k+1}(k+2)} \cdot \frac{3^k(k+1)}{x^k} \right| = \lim_{k \rightarrow \infty} \left| \frac{|x|}{3} \cdot \frac{(k+1)}{(k+2)} \right| = \frac{|x|}{3}$$

The series converges absolutely if $q < 1 \iff \frac{|x|}{3} < 1 \iff |x| < 3$, and diverges if $|x| > 3$. The ratio test fails when $|x| = 3$, also the case $x = 3$ and $x = -3$ need separate analysis.

Substituting $x = -3$ in a given series yield:

$$\sum_{k=0}^{+\infty} \frac{(-1)^k (-3)^k}{3^k(k+1)} = \sum_{k=0}^{+\infty} \frac{3^k}{3^k(k+1)} = \sum_{k=0}^{+\infty} \frac{1}{(k+1)}.$$

Which is the divergent harmonic series.

Substituting $x = 3$ in the given series yield:

$$\sum_{k=0}^{+\infty} \frac{(-1)^k (3)^k}{3^k(k+1)} = \sum_{k=0}^{+\infty} \frac{(-1)^k}{(k+1)}.$$

Which is conditionally convergent alternating harmonic series.

Thus the interval of convergence for the given series is $(-3, 3)$ and the radius of convergence is $R = 3$.

d) General case of power series

In addition to power series in x , we shall be interested in series of the form

$$\sum_{k=0}^{+\infty} c_k (x-a)^k = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots + c_k(x-a)^k + \dots$$

Is called a **power series** in $(x-a)$. Where $c_0, c_1, c_2, \dots$ are constants and x is a variable.

e) Some example of power series

$$\begin{aligned} i. \quad & \sum_{k=0}^{+\infty} (-1)^k \frac{(x+3)^k}{k!} = 1 - (x+3) + \frac{(x+3)^2}{2!} - \frac{(x+3)^3}{3!} + \dots \\ ii. \quad & \sum_{k=0}^{+\infty} \frac{(x-1)^k}{k+1} = 1 + \frac{(x-1)}{2} - \frac{(x-1)^2}{3} + \frac{(x-1)^3}{4} + \frac{(x-1)^4}{5} \dots \end{aligned}$$

Theorem 2.14. *For any power series in $x-a$, exactly one of the following is true:*

- a) *The series converges only for $x=a$,*
- b) *The series converges absolutely for all x ,*
- c) *The series converges absolutely for all x in some finite open interval $(-R+a, R+a)$. At the points $x=a-R$ and $x=a+R$ the series may converge absolutely, converge conditionally, or diverge, depending on the particular series.*
 - *In case c), where the power series converges absolutely for $|x-a| < R$ and diverges for $|x-a| > R$, we call R the **radius of convergence**.*
 - *In case a), the **radius of convergence** $R=0$.*
 - *In case b), where the series converges absolutely for all x , we define the **radius of convergence** to be $R=+\infty$.*
 - *The set of values of x for which a power series converges is called the **interval of convergence**.*

Example 58. *Find the interval of convergence and radius of convergence of the power series*

$$\sum_{k=0}^{+\infty} \frac{(x-5)^k}{k^2}$$

Solution: *We apply the ratio test for absolute convergence.*

$$q = \lim_{k \rightarrow \infty} \frac{|u_{k+1}|}{|u_k|} = \lim_{k \rightarrow \infty} \left| \frac{(x-5)^{k+1}}{(k+1)^2} \cdot \frac{k^2}{(x-5)^k} \right| = \lim_{k \rightarrow \infty} \left| x-5 \right| \left(\frac{k}{k+1} \right)^2 = |x-5|$$

The series converges absolutely if $q < 1 \iff |x-5| < 1 \iff -1 < x-5 < 1 \vee 4 < x < 6$, and diverges if $x > 6 \wedge x < 4$. The ratio test fails when $x=4$, and $x=6$ need separate analyse.

Substituting $x = 6$ in a given series yield:

$$\sum_{k=0}^{+\infty} \frac{1^k}{k^2} = \sum_{k=0}^{+\infty} \frac{1}{k^2}$$

Which is the convergent p -series ($p = 2$). Substituting $x = 4$ in a given series yield:

$$\sum_{k=0}^{+\infty} \frac{(-1)^k}{k^2}$$

Which is absolutely convergent series at $x = 4$.

Thus the interval of convergence for the given series is $[4, 6]$, and the radius of convergence is $R = 1$.

2.3.2 TATLOR AND MACLAURIN SERIES

a) Definition of Maclaurin series

If f can be differentiated n times at 0, then we define the n^{th} **Maclaurin polynomial** for f to be

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots + \frac{f^n(0)}{n!}x^n$$

This polynomial has the property that its value and the values of its first n derivatives match the value of $f(x)$ and its first n derivatives when $x = 0$.

Example 59. Find the Maclaurin polynomials P_0, P_1, P_2, P_3 and P_4 for e^x ?

Solution: Let $f(x) = e^x$. Thus $f'(x) = f''(x) = f'''(x) = \dots = f^n(x) = e^x$, and $f'(0) = f''(0) = f'''(0) = \dots = f^n(0) = e^0 = 1$. Therefore,

$$P_0 = f(0) = 1, \quad P_1 = f(0) + f'(0)x = 1 + x, \quad P_2 = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 = 1 + x + \frac{x^2}{2!},$$

$$P_3 = f(0) + f'(0)x + \frac{f'''(0)}{3!}x^3 = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \quad \dots$$

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots + \frac{f^n(0)}{n!}x^n = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$

b) Definition of Taylor series

If f can be differentiated n times at a , then we define the n^{th} **Taylor polynomial** for f about a to be

$$P_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \dots + \frac{f^n(a)}{n!}(x - a)^n$$

This polynomial has the property that its values of its first n derivative match the value of $f(x)$ and its first n derivatives when $x = a$. Observe that the Taylor polynomial include the Maclaurin polynomials as special case (let $x = a$).

Example 60. Find the Taylor polynomials $P_1(x)$, $P_2(x)$, and $P_3(x)$ for $\sin x$ about $x = \frac{\pi}{3}$.

Solution: Let $f(x) = \sin x$. Thus,

$$f'(x) = \cos x, \quad f''(x) = -\sin x, \quad f'''(x) = -\cos x$$

$$f\left(\frac{\pi}{3}\right) = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}, \quad f'\left(\frac{\pi}{3}\right) = \cos \frac{\pi}{3} = \frac{1}{2}, \quad f''\left(\frac{\pi}{3}\right) = -\sin \frac{\pi}{3} = -\frac{\sqrt{3}}{2}, \quad f'''\left(\frac{\pi}{3}\right) = -\cos \frac{\pi}{3} = -\frac{1}{2}$$

Therefore,

$$P_1(x) = f\left(\frac{\pi}{3}\right) + f'\left(\frac{\pi}{3}\right)\left(x - \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2} + \frac{1}{2}\left(x - \frac{\pi}{3}\right)$$

$$P_2(x) = f\left(\frac{\pi}{3}\right) + f'\left(\frac{\pi}{3}\right)\left(x - \frac{\pi}{3}\right) + \frac{f''\left(\frac{\pi}{3}\right)}{2!}\left(x - \frac{\pi}{3}\right)^2 = \frac{\sqrt{3}}{2} + \frac{1}{2}\left(x - \frac{\pi}{3}\right) - \frac{\sqrt{3}}{2 \cdot 2}\left(x - \frac{\pi}{3}\right)^2$$

c) Some other definitions

- If f has derivatives of all orders at a , then we define the **Taylor series** for f about a to be

$$\sum_{k=0}^{+\infty} \frac{f^k(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^k(a)}{k!}(x-a)^k + \dots$$

- If f has derivatives for all orders at 0, then we define the **Taylor series** for f about 0 to be

$$\sum_{k=0}^{+\infty} \frac{f^k(0)}{k!}x^k = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^k(0)}{k!}x^k + \dots$$

Example 61.

$$\sum_{k=1}^n \frac{x^k}{k!} = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \dots + \frac{1}{n!}x^n$$

3 Introduction to Matrix Theory

Let a field $(\mathbb{R}, +, \cdot)$ and $p, q \in \mathbb{N}$;

- The matrix A of the type $p \times q$, defined on a field $\mathbb{R}$ is a table with p lines (rows) and q columns whose entries are elements of the field $\mathbb{R}$ denoted by:

$$A_{p,q} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,q} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,q} \\ \vdots & \vdots & \ddots & \vdots \\ a_{p,1} & a_{p,2} & \cdots & a_{p,q} \end{bmatrix}$$

The element $a_{i,j}$ of the matrix is written at the intersection of the i^{th} row and the j^{th} column. The first index (i.e p) shows the numbering of the row while the second index (i.e. q) indicates the numbering of the column.

Example 62. The matrix $M = \begin{bmatrix} 6 & -5 \\ 4 & 7 \end{bmatrix}$ is a 2×2 matrix and $D = \begin{bmatrix} 4 & 1 & -3 \\ 0 & 2 & 8 \end{bmatrix}$ is a 2×3 matrix.

We denote by $M(p \times q, \mathbb{R})$ the set of the matrices of the type $p \times q$.

- The matrix B is said “**a square matrix of order m**” if B has m rows and m columns. The number of rows and the number of columns are equal.

$$B_{m,m} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,m} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,m} \end{bmatrix} \text{ is a square matrix of order } m.$$

We denote $M(m, \mathbb{R})$ the set of the square matrix of order m .

- A square matrix D is known as a diagonal matrix if its elements are zero except those of the diagonal

$$D = \begin{bmatrix} a_{1,1} & 0 & 0 & \cdots & 0 \\ 0 & a_{2,2} & 0 & \cdots & 0 \\ 0 & 0 & a_{3,3} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{m,m} \end{bmatrix}$$

- The square matrices whose form are

$$L = \begin{bmatrix} a_{1,1} & 0 & 0 & \cdots & 0 \\ a_{2,1} & a_{2,2} & 0 & \cdots & 0 \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & a_{m,3} & \cdots & a_{m,m} \end{bmatrix} \text{ and}$$

$$U = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots & a_{1,m} \\ 0 & a_{2,2} & a_{2,3} & \cdots & a_{2,m} \\ 0 & 0 & a_{3,3} & \cdots & a_{3,m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{m,m} \end{bmatrix} \text{ are called the triangular matrices.}$$

- The matrix B of the type $1 \times n$ is called **row - matrix** and has 1 row and n columns.

Example 63. $B = [a_{1,1} \ a_{1,2} \ a_{1,3} \ \cdots \ a_{1,n}]$

- -The matrix M of the type $n \times 1$ is called a **column - matrix** and it has n rows and 1 column :

Example 64. $M = \begin{bmatrix} a_{1,1} \\ a_{2,1} \\ a_{3,1} \\ \vdots \\ a_{n,1} \end{bmatrix}$

3.1 OPERATIONS ON MATRICES.

a) Equality and transpose

- Two matrices $(a_{i,j})$ and $(b_{i,j})$ are equal if they are at the same type $p \times q$ and if $a_{i,j} = b_{i,j} \ \forall i \in \{1, 2, \dots, p\}; \forall j \in \{1, 2, 3, \dots, q\}$.

Example 65. The matrices $A = \begin{bmatrix} x-4 & 23 & 9 \\ 5 & y+8 & -12 \end{bmatrix}$ and $B = \begin{bmatrix} 3y+x+14 & 23 & 9 \\ 5 & 4y+2x & -12 \end{bmatrix}$

are equal if $x-4 = 3y+x+14$ and if $y+8 = 4y+2x$. It means when $y = -6$ and $x = 13$.

- The transpose of the matrix A is the matrix denoted by tA obtained from the matrix A by interchanging the role of the k^{th} rows and the k^{th} columns where $1 \leq k \leq p$ (or $1 \leq k \leq q$).

Remark:

$${}^t(A_{i,j}) = A_{j,i} \quad \text{and} \quad {}^t({}^tM) = M.$$

b) Addition of Matrices:

- Let $A = (a_{i,j})$ and $B = (b_{i,j})$ two matrices of the same type $p \times q$. The sum of A and B is a matrix C of the type $p \times q$ obtained by adding entry by entry (i.e. $c_{i,j} = a_{i,j} + b_{i,j}$).

Example 66. If we use A and B which are above in (65), we find

$$C = A + B = \begin{bmatrix} 2X+3y+10 & 46 & 18 \\ 10 & 2x+5y=8 & -24 \end{bmatrix}$$

Consequences:

1. The matrix O in which every entry is zero is called a **zero matrix** and it is an **identity element** of the addition of matrices of same types.
2. The addition of matrices of same types is closure, associative, commutative and every matrix $(a_{i,j})$ has its inverse over addition which is $-a_{i,j}$; then the set of matrix of the same type is a commutative group.

c) Multiplication of matrices

- To multiply a matrix by a scalar k , we multiply each entry of the matrix by the scalar k .

Example 67. If we use A and B which are above in (66), we find

$$W = 2C = \begin{bmatrix} 4x + 6y + 20 & 92 & 36 \\ 20 & 4x + 10y + 16 & -28 \end{bmatrix}$$

Exercise

Calculate $M = 2A - 5B$ if $A = \begin{bmatrix} 4 & 5 & 1 \\ 0 & 7 & 2 \\ -3 & 6 & 10 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 5 & 1 \\ 0 & 9 & 2 \\ -1 & 6 & 8 \end{bmatrix}$.

- The multiplication $A.B$ of two matrices A and B is possible if the number of columns in the first matrix A is the same as the number of rows in the second matrix B .

To find $C = (c_{i,j}) = A.B$ where the entry $c_{i,j}$ is at the intersection of the row i and the column j , we fix the i^{th} row of A and the j^{th} column of B and we multiply term by term by storing in the following way:

$$c_{i,j} = a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \dots + a_{i,n}b_{n,j} = \sum_{k=1}^n a_{i,k}b_{k,j}.$$

Example 68. Let $A = \begin{bmatrix} 4 & 2 & 3 \\ 0 & 6 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 3 & -1 \\ 2 & 1 \end{bmatrix}$. Find the matrix $A.B$ and $B.A$.

Solution:

$$A.B = \begin{bmatrix} 4.1 + 2.3 + 3.2 & 4.0 + 2.(-1) + 3.1 \\ 0.1 + 6.3 + (-1).2 & 0.0 + 6.(-1) + (-1).1 \end{bmatrix} = \begin{bmatrix} 16 & 1 \\ 16 & -7 \end{bmatrix}$$

$$B.A = \begin{bmatrix} 1.4 + 0.0 & 1.2 + 0.6 & 1.3 + 0.(-1) \\ 3.4 + (-1).0 & 3.2 + (-1).6 & 3.3 + (-1).(-1) \\ 2.4 + 1.0 & 2.2 + 1.6 & 2.3 + 1.(-1) \end{bmatrix} = \begin{bmatrix} 4 & 2 & 3 \\ 12 & 0 & 10 \\ 8 & 10 & 5 \end{bmatrix}$$

Remark: This example show $A.B \neq B.A$; so the multiplication of matrices is not commutative. The diagonal matrix which have only 1 along the leading diagonal is called unit matrix or identity matrix; it is an **identity element of multiplication of square matrix**.

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{is a unit matrix of order 3.}$$

It means that if M is a 3×3 matrix then $M.I = I.M = M$.

3.2 DETERMINANT OF A SQUARE MATRIX

At a square matrix A , we associate a real number called **determinant of the matrix A** denoted $\det A$ or $|A|$. If $\det A = 0$, the matrix is called “**a singular matrix**”, but if $\det A \neq 0$ then “ **A is a regular matrix**”.

a) Determinant of 2×2 matrix.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ a square matrix of order 2, then $\det A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = a.d - b.c$

b) Determinant of 3×3 matrix by Sarrus' rule.

The Sarrus' rule used to calculate the determinant of a 3×3 matrix only is given by the following way:

If $A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$, then determinant of the matrix A is given by the following formula:

$$|A| = \begin{vmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{vmatrix}$$

$$= (a_{1,1} \cdot a_{2,2} \cdot a_{3,3} + a_{1,2} \cdot a_{2,3} \cdot a_{3,1} + a_{1,3} \cdot a_{2,1} \cdot a_{3,2} - (a_{3,1} \cdot a_{2,2} \cdot a_{1,3} + a_{3,2} \cdot a_{2,3} \cdot a_{1,1} + a_{3,3} \cdot a_{2,1} \cdot a_{1,2}))$$

Example 69. Calculate the determinant of the matrix $A = \begin{bmatrix} 2 & 3 & -2 \\ 4 & 1 & 5 \\ 2 & -1 & -3 \end{bmatrix}$ and $\det B$ if

$B = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$. We show that $\det A = 82$ and we deduce it that A is a regular matrix. $\det B = 4 + 6 = 10$.

c) General method (Minor and Confactor method)

Let A a $n \times n$ matrix below:

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{bmatrix}$$

- The determinant of $M_{i,j}$ which is obtained from A by omitting the i^{th} row and the j^{th} column is called **the minor of the element $a_{i,j}$** .
- The confactor of the entry $a_{i,j}$ is the number $A_{i,j} = (-1)^{i+j} \det M_{i,j}$.
- We define the determinant of A as follow:

$$\det A = \sum_{j=1}^n a_{i,j} \cdot A_{i,j} \quad \text{when the row is fixed.}$$

Example 70. Calculate the determinant of the following matrix:

$$A = \begin{bmatrix} 2 & 0 & 0 & -3 \\ 0 & -1 & 0 & 0 \\ 7 & 4 & 3 & 5 \\ -6 & 2 & 2 & 4 \end{bmatrix}$$

Solution: It is easy to use the second row which has many entries equal zero.

$$\det A = \begin{vmatrix} 2 & 0 & 0 & -3 \\ 0 & -1 & 0 & 0 \\ 7 & 4 & 3 & 5 \\ -6 & 2 & 2 & 4 \end{vmatrix} = (-1) \cdot (-1)^4 A = \begin{bmatrix} 2 & 0 & -3 \\ 7 & 3 & 5 \\ -6 & 2 & 4 \end{bmatrix};$$

and we continue by using the second column:

$$\det A = (-1) \cdot 3 \cdot (-1)^4 \begin{vmatrix} 2 & -3 \\ -6 & 4 \end{vmatrix} + (-1) \cdot 2 \cdot (-1)^5 \begin{vmatrix} 2 & -3 \\ 7 & 5 \end{vmatrix} = -3(8 - 18) + 2(10 + 21) = 92$$

Properties of the determinant

1. $\det A = \det {}^t(A)$.
2. $\det (A.B) = \det A \cdot \det B$.
3. If we interchange two columns (or two rows) of a matrix then its determinant is multiplied by -1 .
4. If two columns (or two rows) of a matrix are identical then its determinant is nul.
5. If we multiply a column (or a row) of a matrix by a number k , then its determinant is multiplied by k .
6. If a column (or a row) is a linear combination of other columns(rows), then its determinant is nul.
7. The determinant of a matrix does not change if a column(or a row) is replaced by a linear combination of others columns(or others rows).

Calculation of the inverse matrix

Let A a $n \times n$ matrix which is regular.

- The inverse matrix of A is a matrix denoted A^{-1} such that $A.A^{-1} = I$ where I is a $n \times n$ unit matrix.
- We call the confactor matrix of A ; the matrix $\tilde{A}$ obtained by replacing each entry $a_{i,j}$ by its confactor $A_{i,j}$.
- We call the adjoint matrix of A denoted $\text{adj}(A)$, the transpose matrix of the cofactor matrix. i.e. $\text{adj}(A) = {}^t\tilde{A}$.
- The inverse matrix of a regular matrix A is given by

$$A^{-1} = \frac{\text{adj}(A)}{\det A}$$

Example 71. Calculate the inverse matrix of $A = \begin{bmatrix} 2 & 3 & -4 \\ 0 & -4 & 2 \\ 2 & -1 & 5 \end{bmatrix}$ and verify the result.

Solution: $\det A = -46$. The matrix A is regular then it is possible to find the inverse matrix of A .

$$\tilde{A} = \begin{vmatrix} \begin{vmatrix} -4 & 2 \\ -1 & 5 \end{vmatrix} & \begin{vmatrix} 2 & 0 \\ 5 & 1 \end{vmatrix} & \begin{vmatrix} 0 & -4 \\ 1 & -1 \end{vmatrix} \\ \begin{vmatrix} -4 & 3 \\ 5 & -1 \end{vmatrix} & \begin{vmatrix} 2 & -4 \\ 1 & 5 \end{vmatrix} & \begin{vmatrix} 3 & 2 \\ -1 & 1 \end{vmatrix} \\ \begin{vmatrix} 3 & -4 \\ -4 & 2 \end{vmatrix} & \begin{vmatrix} -4 & 2 \\ 2 & 0 \end{vmatrix} & \begin{vmatrix} 2 & 3 \\ 0 & -4 \end{vmatrix} \end{vmatrix} = \begin{bmatrix} -18 & 2 & 4 \\ -11 & 14 & 5 \\ -10 & -4 & -8 \end{bmatrix}$$

$$Adj(A) = {}^t \tilde{A} = \begin{bmatrix} -18 & -11 & 7 \\ -2 & 14 & -4 \\ 4 & 5 & -8 \end{bmatrix} \quad and$$

$$A^{-1} = \frac{Adj(A)}{det A} = \frac{1}{-46} \begin{bmatrix} -18 & -11 & 7 \\ -2 & 14 & -4 \\ 4 & 5 & -8 \end{bmatrix} = \begin{bmatrix} \frac{9}{23} & \frac{11}{46} & \frac{5}{23} \\ \frac{1}{23} & \frac{7}{23} & \frac{2}{23} \\ \frac{-2}{23} & \frac{-5}{46} & \frac{4}{24} \end{bmatrix}$$

We verify easily by multiplication that $A.A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$ which is the 3×3 unit matrix.

3.3 SYSTEMS OF LINEAR EQUATIONS

3.3.1 Definitions and examples

- A system of p linear equations with n unknowns is defined by:

[illegible]

The real numbers $a_{i,j}, b_i$ are called the coefficients of the system and the variables x_j are the unknowns of the system.

Example 72.

$$\begin{cases} 5x + 3y = 1 \\ x - 2y = 8 \end{cases}$$

is a system of 2 linear equations with 2 unknowns x and y .

- A solution of the system of linear equations (-227) is the sequence $(x_1, x_2, \dots, x_n)$ of n real numbers which verify the p equations of the system (-227).

- In the example above the solution is $(2, -3)$ because it verify the two equations of the system.

Remark: Every system of equations can be written matricially. For this we consider the matrices

$$A = (a_{i,j}) = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{p,1} & a_{p,2} & \cdots & a_{p,n} \end{bmatrix}; \quad B = (b_j) = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_p \end{bmatrix}; \quad X = (x_i) = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad (-227)$$

The system (-227) is written as $A.X = B$. In (-227) the matrix A is called the **matrix of the system**.

- If B is nul then the system is called an **homogeneous system**.
- In our example above we write matricially the system as follow:

$$\begin{bmatrix} 5 & 3 \\ 1 & -2 \end{bmatrix} \quad \text{and} \quad X = \begin{bmatrix} x \\ y \end{bmatrix}; \quad B = \begin{bmatrix} 1 \\ 8 \end{bmatrix}. \quad \text{The system is not homogeneous.}$$

Resolution of the system by Gauss's method.

The method use L_1 as the pivot of the rows. The by replacing the rows

$$L_i, \quad 2 \leq i \leq p; \quad \text{by} \quad L_i - \frac{a_{i,1}}{a_{1,1}} L_1.$$

We get an equivalent system in which the first column is 0 at the $2^{nd}, 3^{rd}, \dots, p^{th}$ rows. On the next steps we take $L_2; L_3; \dots; L_p$ as a pivot and we continue until the matrix of the system became triangular matrix.

Example: Solve the system bellow by Gauss's method

$$\begin{cases} -2x_1 + 2x_2 - 4x_3 - 6x_4 = -4 \\ -3x_1 + 6x_2 + 3x_3 - 15x_4 = -3 \\ 5x_1 - 8x_2 - x_3 + 17x_4 = 9 \\ x_1 + x_2 + 11x_3 + 7x_4 = 7 \end{cases}$$

Solution: In the beginning we simplify every equation for using the simple number by replacing L_1 by $\frac{-1}{2}LL_1$ and L_2 by $\frac{1}{3}L_2$. The system became

$$\begin{cases} x_1 - x_2 + 2x_3 + 3x_4 = 2 \\ -x_1 + 2x_2 + x_3 - 5x_4 = -1 \\ 5x_1 - 8x_2 - x_3 + 17x_4 = 9 \\ x_1 + x_2 + 11x_3 + 7x_4 = 7 \end{cases}$$

Example 73. Solve the following system of equations:

$$\begin{cases} 2X + 3Y + Z = -1 \\ 4X + 5Y + 2Z = 0 \\ 3X + 2Y - Z = -4 \end{cases}$$

Solution: The system is written matrixly as: $\begin{bmatrix} 2 & 3 & 1 \\ 4 & 5 & 2 \\ 3 & 2 & -1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ -4 \end{bmatrix}$, then the deter-

minant of the system is $\begin{vmatrix} 2 & 3 & 1 \\ 4 & 5 & 2 \\ 3 & 2 & -1 \end{vmatrix} = 5$. The roots are:

$$X = \frac{\begin{vmatrix} -1 & 3 & 1 \\ 0 & 5 & 2 \\ -4 & 2 & -1 \end{vmatrix}}{5}; \quad Y = \frac{\begin{vmatrix} 2 & -1 & 1 \\ 4 & 0 & 2 \\ 3 & -4 & -1 \end{vmatrix}}{5}; \quad Z = \frac{\begin{vmatrix} 2 & 3 & -1 \\ 4 & 5 & 0 \\ 3 & 2 & -4 \end{vmatrix}}{5}.$$

After calculating we find $X = \frac{5}{5} = 1$; $Y = \frac{-10}{5} = -2$; $Z = \frac{15}{5} = 3$. The solution is $S = \{1, -2, 3\}$.

Resolution of the system by using inverse matrix

We know that the matrix notation of a system of equation is $A.X = B$. We use this method when the matrix A is a square matrix and regular. Suppose that A^{-1} is its inverse matrix. Then we have

$$AX = B \Leftrightarrow A^{-1}AX = A^{-1}B \Leftrightarrow I.X = A^{-1}B \Leftrightarrow X = A^{-1}B$$

Example 74. Use the inverse matrix to solve the following system of equations:

$$\begin{cases} 2x + 3y + z = -1 \\ 4x + 5y + 2z = 0 \\ 3x + 2y - z = -4 \end{cases}$$

Solution:

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 5 & 2 \\ 3 & 2 & -1 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} -1 \\ 0 \\ -4 \end{bmatrix} \quad \text{and} \quad \det A = 5 \quad \text{then } A \text{ is regular.}$$

$$\tilde{A} = \begin{vmatrix} \begin{vmatrix} 5 & 2 \\ 2 & -1 \end{vmatrix} & \begin{vmatrix} 2 & 4 \\ -1 & 3 \end{vmatrix} & \begin{vmatrix} 4 & 5 \\ 3 & 2 \end{vmatrix} \\ \begin{vmatrix} 1 & 3 \\ -1 & 2 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 3 & -1 \end{vmatrix} & \begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix} \\ \begin{vmatrix} 3 & 1 \\ 5 & 2 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} & \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} \end{vmatrix} = \begin{bmatrix} -9 & 10 & -7 \\ 5 & -5 & 5 \\ 1 & 0 & -2 \end{bmatrix}$$

$$\text{Adj}(A) = {}^t\tilde{A} = \begin{bmatrix} 9 & 5 & 1 \\ 10 & -5 & 0 \\ -7 & 5 & -2 \end{bmatrix} \quad \text{and} \quad A^{-1} = \frac{\text{Adj}(A)}{\det A} = \begin{bmatrix} -\frac{9}{5} & 1 & \frac{1}{5} \\ 2 & -1 & 0 \\ -\frac{7}{5} & 1 & -\frac{2}{5} \end{bmatrix}$$

Then

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1}B = \begin{bmatrix} -\frac{9}{5} & 1 & \frac{1}{5} \\ 2 & -1 & 0 \\ -\frac{7}{5} & 1 & -\frac{2}{5} \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 0 \\ -4 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$$

EXERCISES

1. If $A = \begin{bmatrix} 3 & 1 & 4 \\ -2 & 0 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 2 \\ -3 & 1 & 1 \\ 2 & -4 & 1 \end{bmatrix}$; then compute:

- (a) $2A$ (b) $A + B$ (c) $2A - 3B$ (d) $(2A)^t - (3B)^t$ (e) AB (f) BA
 (g) $A^t B^t$ (h) $(BA)^t$.

2. For each of the following pairs of matrices, determine whether it is possible to multiply the first matrix times the second. If it is possible, perform the multiplication.

$$(a) \quad \begin{bmatrix} 3 & 5 & 1 \\ -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \\ 4 & 1 \end{bmatrix} \quad (b) \quad \begin{bmatrix} 4 & -2 \\ 6 & -4 \\ 8 & -6 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

$$(c) \quad \begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 1 \\ 4 & 5 \end{bmatrix} \quad (d) \quad \begin{bmatrix} 4 & 6 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 5 \\ 4 & 1 & 6 \end{bmatrix}$$

3. For which of the pairs in exercise 2 is it possible to multiply the second matrix times the first, and what would be the dimension of the product be?

4. Write each of the following systems of equations as a matrix equation.

$$a) \begin{cases} 3x_1 + 3x_2 = 1 \\ 2x_1 + 3x_2 = 5 \end{cases} \quad b) \begin{cases} x_1 + x_2 = 5 \\ 2x_1 + x_2 - x_3 = 6 \\ 3x_1 - x_2 + 2x_3 = 0 \end{cases} \quad c) \begin{cases} 2x_1 + x_2 + x_3 = 4 \\ x_1 - x_2 + 2x_3 = 2 \\ 3x_1 - 2x_2 - x_3 = 0 \end{cases}$$

5. If $A = \begin{bmatrix} 3 & 4 \\ 1 & 1 \\ 2 & 7 \end{bmatrix}$, then verify that: a) $5A = 3A + 2A$ b) $6A = 3(2A)$ c) $(A^t)^t = A$

6. Let $A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$. Compute A^2 and A^3 . What will be A^n turn out to be?

7. Find the inverse of each of the following matrices.

$$a) \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} \quad b) \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \quad c) \begin{bmatrix} 2 & 6 \\ 3 & 8 \end{bmatrix} \quad d) \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad e) \begin{bmatrix} 2 & 0 & 5 \\ 0 & 3 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

8. For each of the following, compute (i) $\det A$ (ii) $\text{adj}(A)$ and A^{-1} .

$$a) \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \quad b) \begin{bmatrix} 1 & 3 & 1 \\ 2 & 1 & 1 \\ -2 & 2 & 1 \end{bmatrix} \quad c) \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

9. Use Cramer's rule to solve each of the following systems.

$$a) \begin{cases} 2x_1 + x_2 + x_3 = 4 \\ x_1 - x_2 + 2x_3 = 2 \\ 3x_1 - 2x_2 - x_3 = 0 \end{cases} \quad b) \begin{cases} x_1 + x_2 = 5 \\ 2x_1 + x_2 - x_3 = 6 \\ 3x_1 - x_2 + 2x_3 = 7 \end{cases} \quad c) \begin{cases} x_1 + x_2 = 0 \\ x_2 + x_3 - 2x_4 = 1 \\ x_1 + 2x_3 + x_4 = 0 \\ x_1 + x_2 + x_4 = 0 \end{cases}$$

10. Given $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$

a) Compute the determinant of A . Is A nonsingular?

b) Compute $\text{adj}(A)$ and the product of $A \cdot \text{adj}(A)$?

4 Element of Vector's Spaces

A vector is a combination of three things:

- a positive number called its magnitude,
- a direction in space,
- a sense making more precise the idea of direction.

Typically a vector is illustrated as a directed straight line. The vector in the above diagram



would be written as $\vec{AB}$ with:

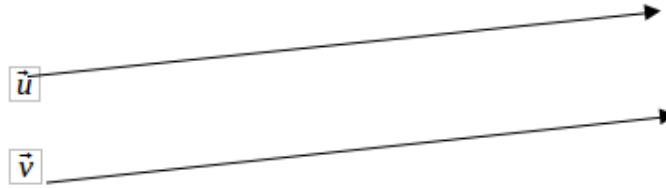
- the direction of the arrow, from the point A to the point B , indicating the sense of the vector,
- the magnitude of given by the length of $\vec{AB}$. The magnitude of $\vec{AB}$ is written $|\vec{AB}|$.

There are very many physical quantities which are best described as vectors; velocity, acceleration and force are all vector quantities. The magnitude of vector $\vec{v}$ is the length of any its representations and is denoted by the symbol $\|\vec{v}\|$ or $|\vec{v}|$. By using the distance formula to compute the length of a segment, we obtain:

- In two-dimensional, $\vec{v} = \langle a_1, a_2 \rangle$, its length is $\|\vec{v}\| = \sqrt{a_1^2 + a_2^2}$.
- In three-dimensional, $\vec{v} = \langle a_1, a_2, a_3 \rangle = \sqrt{a_1^2 + a_2^2 + a_3^2}$

4.1 VECTOR OPERATIONS

Two vectors are equal if they have the same magnitude, the same direction (i.e. they are parallel) and have the same sense. If $\vec{u} = \langle a_1, a_2 \rangle$ and $\vec{v} = \langle b_1, b_2 \rangle$, then:



1. $\vec{u} + \vec{v} = \langle a_1 + b_1, a_2 + b_2 \rangle$
2. $\vec{u} - \vec{v} = \langle a_1 - b_1, a_2 - b_2 \rangle$
3. $\beta \vec{u} = \langle \beta a_1, \beta a_2 \rangle$

Similary, for three-dimensional vectors, if $\vec{u} = \langle a_1, a_2, a_3 \rangle$, and $\vec{v} = \langle b_1, b_2, b_3 \rangle$, then

1. $\vec{u} + \vec{v} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$
2. $\vec{u} - \vec{v} = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$
3. $\beta \vec{u} = \langle \beta a_1, \beta a_2, \beta a_3 \rangle$

4.1.1 Unit vectors

A **unit vector** is a vector whose length is 1. The set of all two-dimensional vectors is denoted by V_2 and the set of all three-dimensional vectors is denoted by V_3 .

More generally, V_n denote the set of all n-dimensional vectors. Three unit vectors in V_3 play a special role:

$$\vec{i} = \langle 1, 0, 0 \rangle \quad \vec{j} = \langle 0, 1, 0 \rangle \quad \vec{k} = \langle 0, 0, 1 \rangle$$

They point in the direction of the positive x , y , and z . Similarly, in two dimensions, we define $\vec{i} = \langle 1, 0 \rangle$ and $\vec{j} = \langle 0, 1 \rangle$

If $\vec{v} = \langle a_1, a_2, a_3 \rangle$, then

$$\vec{v} = a_1 \langle 1, 0, 0 \rangle + a_2 \langle 0, 1, 0 \rangle + a_3 \langle 0, 0, 1 \rangle = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k}$$

Thus, any vector in V_3 can be expressed in terms of the **standard basis vectors** $\vec{i}$, $\vec{j}$ and $\vec{k}$. Similarly, in two-dimensions

$$\vec{v} = \langle a_1, a_2 \rangle = \langle a_1, 0 \rangle + \langle 0, a_2 \rangle = a_1 \langle 1, 0 \rangle + a_2 \langle 0, 1 \rangle = a_1 \vec{i} + a_2 \vec{j}$$

Note: If $\vec{v} \neq 0$, then the unit vector has the same direction as $\vec{v}$ is $\vec{v} = \frac{\vec{v}}{|\vec{v}|}$

Example 75. Find the unit vector in the same direction of the vector $2\vec{i} - \vec{j} - 2\vec{k}$.

solution:

$$|2\vec{i} - \vec{j} - 2\vec{k}| = \sqrt{2^2 + (-1)^2 + (-2)^2} = 3.$$

The required unit vector is $\frac{2}{3}\vec{i} - \frac{1}{3}\vec{j} - \frac{2}{3}\vec{k}$.

4.1.2 Dot product and vector product

Dot product or Scalar products

a) Definition 1:

If $\vec{u} = \langle a_1, a_2, a_3 \rangle$, and $\vec{v} = \langle b_1, b_2, b_3 \rangle$, then the dot product of $\vec{u}$ and $\vec{v}$ is the number $\vec{u} \cdot \vec{v}$ is given by

$$\vec{u} \cdot \vec{v} = a_1.b_1 + a_2.b_2 + a_3.b_3$$

Thus, to find the dot product of two given vectors, we multiply corresponding components and add them. The result is not a vector but real number, that is, a scalar. For this reason, the dot product is sometimes called the **scalar product (or inner product)**.

Example 76. a) $\langle 2, 4 \rangle \cdot \langle 3, -1 \rangle = 2.3 + 4.(-1) = 6 - 4 = 2$

b) $(\vec{i} + 2\vec{j} - 3\vec{k}) \cdot (2\vec{j} - \vec{k}) = 1.0 + 2.2 + (-3).(-1) = 7$

Properties of dot product

If $\vec{u}, \vec{v}$ and $\vec{w}$ are vectors in V_3 and μ is a scalar

1. $\vec{u} \cdot \vec{u} = |\vec{u}|^2$

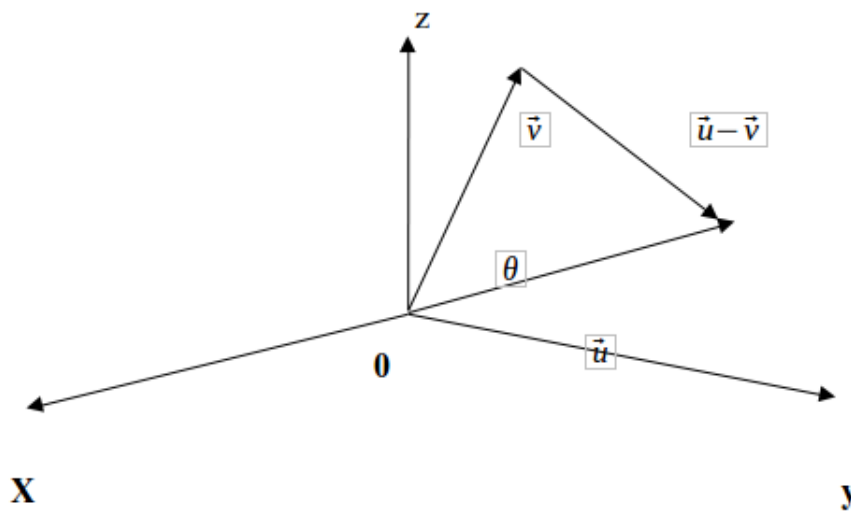
2. $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$

3. $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$
4. $(\mu \vec{u}) \cdot \vec{v} = \mu(\vec{u} \cdot \vec{v}) = \vec{u} \cdot (\mu \vec{v})$
5. $0 \cdot \vec{v} = 0$

c) Definition 2:

The dot product $\vec{u} \cdot \vec{v}$ can be given a geometric interpretation in terms of the angle θ between $\vec{u}$ and $\vec{v}$, which is defined to be the angle between the representations of $\vec{u}$ and $\vec{v}$ that start at the origin, where $0 < \theta < \pi$.

In other words, θ is the angle between the line segments OA and OB . Thus, we may prove



that

$$\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta \Rightarrow \cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|}$$

Example 77. If the vectors $\vec{u}$ and $\vec{v}$ have lengths 4 and 6 and the angle between them is $\frac{\pi}{3}$; find $\vec{u} \cdot \vec{v}$.

Solution: $\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cdot \cos \frac{\pi}{3} = 4 \cdot 6 \cdot \frac{1}{2} = 12$.

Example 78. Find the angle between the vectors $\vec{u} = \langle 2, 2, -1 \rangle$ and $\vec{v} = \langle 5, -3, 2 \rangle$.

Solution:

$$|\vec{u}| = 3, \quad |\vec{v}| = \sqrt{38}, \quad \vec{u} \cdot \vec{v} = 2 \quad \text{so } \cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|} = \frac{2}{3\sqrt{38}} \Rightarrow \theta = \cos^{-1} \left(\frac{2}{3\sqrt{38}} \right) \cong 1.46 \text{ rad} \vee 84^\circ$$

Note: Two nonzero vectors $\vec{u}$ and $\vec{v}$ are called **perpendicular or orthogonal** if the angle between them is $\theta = \frac{\pi}{2}$, that is $\vec{u} \cdot \vec{v} = 0$.

Definition 4.1. The vectors X and Y in $\mathbb{R}^2$ (or $\mathbb{R}^3$) are said to be orthogonal if $X^t Y = 0$.

4.1.3 VECTOR PRODUCTS

The **vector or cross product** takes two vector operands to produce a vector result. The result, like all geometric vectors, has two properties: **length and orientation**. If $\vec{u}$ and $\vec{v}$ are vectors in three dimensional space (only), then $\vec{u} \times \vec{v}$ is a three dimensional vector, where:

length: $|\vec{u} \times \vec{v}| = |\vec{u}||\vec{v}| \sin \theta$, where θ is the angle between $\vec{u}$ and $\vec{v}$.

Orientation: $\vec{u} \times \vec{v}$ is perpendicular to both $\vec{u}$ and $\vec{v}$. The choice (out of two) orientations perpendicular to $\vec{u}$ and $\vec{v}$ is made by the right hand rule.

So, let's start with two vectors $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, then the cross product is given by the formula:

$$\vec{a} \times \vec{b} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle$$

The notation for the determinant is as follows,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \quad (-247)$$

The first row is the standard basis vectors and must appear in the order given here. The second row is the components of $\vec{a}$ and the third row is the components of $\vec{b}$. Now, let's take a look at the different methods for getting the formula.

The first method uses the Method of Cofactors. If you don't know the method of cofactors that is fine, the result is all that we need. Here is the formula.

$$\vec{a} \times \vec{b} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \vec{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \vec{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \vec{k} \quad (-246)$$

Where, $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$.

This is best seen in an example. We'll also use this example to illustrate a fact about cross products.

Example 79. If $\vec{a} = \langle 2, 1, -1 \rangle$ and $\vec{b} = \langle -3, 4, 1 \rangle$. Compute each of the following.

a) $\vec{a} \times \vec{b}$

b) $\vec{b} \times \vec{a}$

Solution: a) Here is the computation of this one.

$$\begin{aligned} \vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & -1 \\ -3 & 4 & 1 \end{vmatrix} \\ &= \vec{i}(1)(1) + \vec{j}(-1)(-3) + \vec{k}(2)(4) - \vec{j}(2)(1) - \vec{i}(-1)(4) - \vec{k}(1)(-2) \\ &= 5\vec{i} + 3\vec{j} + 11\vec{k} \end{aligned}$$

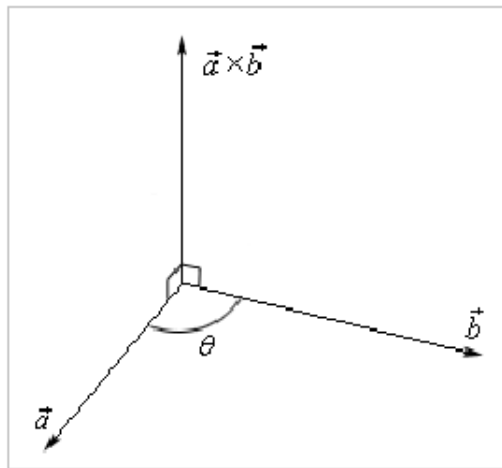
b) And here is the computation of the example b).

$$\begin{aligned}
 \vec{b} \times \vec{a} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -3 & 4 & 1 \\ 2 & 1 & -1 \end{vmatrix} \\
 &= \vec{i}(4)(-1) + \vec{j}(1)(2) + \vec{k}(-3)(1) - \vec{j}(-3)(-1) - \vec{i}(1)(1) - \vec{k}(4)(2) \\
 &= -5\vec{i} - \vec{j} - 11\vec{k}
 \end{aligned}$$

There is also a geometric interpretation of the cross product. First we will let θ be the angle between the two vectors $\vec{a}$ and $\vec{b}$ and assume that $0 < \theta < \pi$, then we have the following fact,

$$\|\vec{a} \times \vec{b}\| = \|\vec{a}\| \|\vec{b}\| \sin \theta \quad (-251)$$

and the following figure There should be a natural question at this point. How did we know



that the cross product pointed in the direction that we've given it here?

First, as this figure, implies the cross product is orthogonal to both of the original vectors. This will always be the case with one exception that we'll get to in a second.

Second, we knew that it pointed in the upward direction (in this case) by the "right hand rule"

This says that if we take our right hand, start at $\vec{a}$ and rotate our fingers towards $\vec{b}$ our thumb will point in the direction of the cross product. Therefore, if we'd sketched in $\vec{b} \times \vec{a}$ above we would have gotten a vector in the downward direction.

Example 80. A plane is defined by any three points that are in the plane. If a plane contains the points $P = (1, 0, 0)$, $Q = (1, 1, 1)$ and $R = (2, -1, 3)$, find a vector that is orthogonal to the plane.

Solution:

The one way that we know to get an orthogonal vector is to take a cross product. So, if we could find two vectors that we knew were in the plane and took the cross product of these two vectors we know that the cross product would be orthogonal to both the vectors. However,

since both the vectors are in the plane the cross product would then also be orthogonal to the plane.

So, we need two vectors that are in the plane. This is where the points come into the problem. Since all three points lie in the plane any vector between them must also be in the plane. There are many ways to get two vectors between these points. We will use the following two,

$$\overrightarrow{PQ} = \langle 1 - 1, 1 - 0, 1 - 0 \rangle = \langle 0, 1, 1 \rangle$$

$$\overrightarrow{PR} = \langle 2 - 1, -1 - 0, 3 - 0 \rangle = \langle 1, -1, 3 \rangle$$

The cross product of these two vectors will be orthogonal to the plane. So, let's find the cross product.

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & 1 \\ 1 & -1 & 3 \end{vmatrix} = 4\vec{i} + \vec{j} - \vec{k}$$

So, the vector $4\vec{i} + \vec{j} - \vec{k}$ will be orthogonal to the plane containing the three points.

Now, let's address the one time where the cross product will not be orthogonal to the original vectors. If the two vectors $\vec{a}$ and $\vec{b}$ are parallel then the angle between them is either 0 or 180 degrees. From (-251) this implies that,

$$\|\vec{a} \times \vec{b}\| = 0$$

From a fact about the magnitude we saw in the first section we know that this implies $\vec{a} \times \vec{b} = \vec{0}$.

In other words, it won't be orthogonal to the original vectors since we have the zero vector. This does give us another test for parallel vectors however.

If $\vec{a} \times \vec{b} = \vec{0}$ then $\vec{a}$ and $\vec{b}$ will be parallel vectors.

Let's also formalize up the fact about the cross product being orthogonal to the original vectors.

Provided $\vec{a} \times \vec{b} \neq \vec{0}$ **then** $\vec{a} \times \vec{b}$ is orthogonal to both $\vec{a}$ and $\vec{b}$.

Here are some nice properties about the cross product.

Properties

If $\vec{u}, \vec{v}$ and $\vec{w}$ are vectors and c is a constant then,

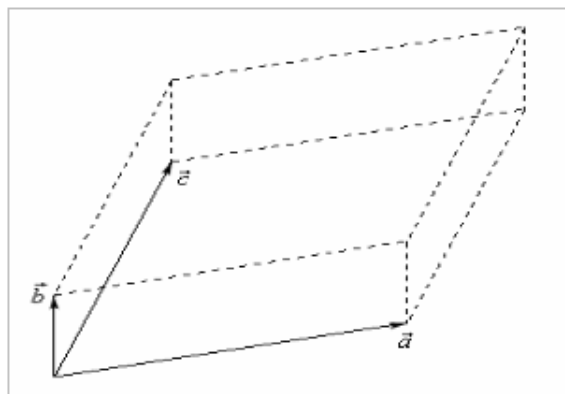
$$1. \vec{u} \times \vec{v} = \vec{v} \times \vec{u}$$

$$2. (c\vec{u}) \times \vec{v} = \vec{u} \times (c\vec{v}) = c(\vec{u} \times \vec{v})$$

$$3. \vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w}$$

$$4. \vec{u} \cdot (\vec{v} + \vec{w}) = (\vec{u} \times \vec{v}) \cdot \vec{w}$$

$$5. \vec{u} \cdot (\vec{v} \times \vec{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$



There are a couple of geometric applications to the cross product as well. Suppose we have three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ and we form the three dimensional figure shown below. The area of the parallelogram (two dimensional front of this object) is given

$$Area = \|\vec{a} \times \vec{b}\| \quad (-251)$$

and the volume of the parallelepiped (the whole three dimensional object) is given by

$$Volume = \left| \vec{a} \cdot (\vec{b} \times \vec{c}) \right| \quad (-250)$$

Note that the absolute value bars are required since the quantity could be negative and volume isn't negative.

We can use this volume fact to determine if three vectors lie in the same plane or not. If three vectors lie in the same plane then the volume of the parallelepiped will be zero.

Example 81. Determine if the three vectors $\vec{a} = \langle 1, 4, -7 \rangle$, $\vec{b} = \langle 2, -1, 4 \rangle$ and $\vec{c} = \langle 0, 9, 18 \rangle$ lie in the same plane or not?

Solution: So, as we noted prior to this example all we need to do is to compute the volume of parallelepiped by these three vectors. If the volume is zero, then they lie in the same plane and if the volume is not zero they don't lie in the same plane.

$$\begin{aligned} \vec{a} \cdot (\vec{b} \times \vec{c}) &= \begin{vmatrix} 1 & 4 & -7 \\ 2 & -1 & 4 \\ 0 & -9 & 18 \end{vmatrix} \\ &= (1)(-1)(18) + (4)(4)(0) + (-7)(2)(-9) - (4)(2)(18) - (1)(4)(-9) - (-7)(-1)(0) \\ &= -18 + 126 - 144 + 36 = 0 \end{aligned}$$

So, the volume is zero and so they lie in the same plane.

4.1.4 TRIPLE SCALAR PRODUCT

The scalar triple product of three vectors A, B , and C is denoted $[A, B, C]$ and defined by

$$\begin{aligned}[A, B, C] &= A \cdot (B \times C) \\ &= B \cdot (C \times A) \\ &= C \cdot (A \times B) \\ &= \det(ABC) \\ &= \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}\end{aligned}$$

The volume of a parallelepiped whose sides are given by the vectors A, B , and C is given by the absolute value of the scalar triple product

$$V_{\text{parallelepiped}} = |A \cdot (B \times C)|$$

4.2 LINEAR INDEPENDENCE, BASES

4.2.1 Linear independence

The vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ in a vector space V are said to be linearly independent if

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_n \vec{v}_n = 0.$$

Implies that all the scalars $c_1, c_2, \dots, c_n$ must equal to zero.

Example 82. The vectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ are linearly independent, since if

$$c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 0$$

Then

$$\begin{cases} c_1 + c_2 = 0 \\ c_1 + 2c_2 = 0 \end{cases} \Rightarrow c_1 = 0 \wedge c_2 = 0$$

The vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ in a vector space V are said to be linearly dependent if there exists scalars $c_1, c_2, \dots, c_n$ not all zero such that

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_n \vec{v}_n = 0.$$

Example 83. The vectors $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 6 \\ -9 \end{bmatrix}$ are linearly dependent, since if

$$c_1 \begin{bmatrix} 2 \\ -3 \end{bmatrix} + c_2 \begin{bmatrix} 6 \\ -9 \end{bmatrix} = 0$$

Then

$$\begin{cases} 2c_1 - 3c_2 = 0 \\ 6c_1 - 9c_2 = 0 \end{cases} \Rightarrow c_1 = \frac{3}{2}c_2 \Rightarrow \begin{bmatrix} 6 \\ -9 \end{bmatrix} = 3 \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

4.2.2 Basis and dimension of a vector space

The vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ in a vector space V form a **basis** for a vector space V if and only if:

1. $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are linearly independent
2. $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ span V means that we can write any vector as linearly combination of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$

Example 84.

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\} \quad \text{form the basis of } \mathbb{R}^3$$

Corollary: If $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ and $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m$ are both bases for a vector space V , then $n = m$.

Definition 4.2. Let V be a vector space. If V has a basis consisting of n vectors, we say that V has **dimension** n . The sub space $\vec{0}$ of V is said to have dimension 0. V is said to be **finite – dimensional** if there is a finite set of vectors that spans V , otherwise, we say that V is **infinite-dimensional**.

Theorem 4.1. If V is a vector space of dimension $n > 0$, then

- i) Any set of n linearly independent vectors spans V ;
- ii) Any n vectors that span V are linearly independent.

Example 85.

$$\vec{u} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix}, \quad \vec{w} = \begin{bmatrix} -6 \\ 4 \\ -8 \end{bmatrix}$$

What are the dimension of $\text{span}(\vec{u}, \vec{v}, \vec{w})$?

Solution: First of all we verify that these are linearly independent as:

$$\begin{vmatrix} 3 & 3 & -6 \\ -2 & 2 & 4 \\ 4 & -4 & -8 \end{vmatrix} = 0$$

This results imply that $\dim \neq 3$, but $\begin{vmatrix} 3 & 3 \\ -2 & 2 \end{vmatrix} \neq 0$. This results imply that $\dim$ of $\text{span}(\vec{u}, \vec{v}, \vec{w}) = 2$.

4.3 Linear operators, Eigenvalues, Eigenvectors, Orthogonal Matrices

4.3.1 LINEAR OPERATORS

Definition 4.3. A mapping L from a vector space V into a vector space W is said to be a **linear transformation or a linear operator** if:

- $L(\vec{u} + \vec{v}) = L(\vec{u}) + L(\vec{v}), \forall \vec{u}, \vec{v} \in V$,

- $L(\alpha\vec{u}) = \alpha L(\vec{u}), \quad \forall \vec{u} \in V, \wedge \forall \alpha \text{ is a scalar, or we can combine both two properties as}$
- $L(\alpha\vec{u} + \beta\vec{v}) = \alpha L(\vec{u}) + \beta L(\vec{v}), \quad \forall \vec{u}, \vec{v} \in V, \quad \wedge \quad \forall \alpha, \beta \quad \text{are scalars.}$

Linear operators on $\mathbb{R}^2$

Example 86. Consider the mapping L defined by $L(\vec{x}) = 3\vec{x}$ for each $\vec{x} \in \mathbb{R}^2$. Show that L is a linear operator?

Solution: L is a linear operator if

- $L(\alpha\vec{x}) = 3(\alpha\vec{x}) = \alpha(3\vec{x}) = \alpha L(\vec{x}).$
- $L(\vec{x} + \vec{y}) = 3(\vec{x} + \vec{y}) = (3\vec{x}) + (3\vec{y}) = L(\vec{x}) + L(\vec{y}), \quad \forall \vec{x}, \vec{y} \in \mathbb{R}^2$

Then, we confirm that L is a linear mapping.

Example 87. Consider the mapping L defined by $L(\vec{x}) = \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix}$ for each $\vec{x} \in \mathbb{R}^2$. Show that L is a linear operator?

Solution: L is a linear operator if

$$L(\alpha\vec{x} + \beta\vec{y}) = \alpha L(\vec{x}) + \beta L(\vec{y}), \forall \vec{x}, \vec{y} \in \mathbb{R}^2 \quad \text{and} \quad \forall \alpha, \beta \quad \text{are scalars.}$$

Verification:

$$\begin{aligned} L(\alpha\vec{x} + \beta\vec{y}) &= L\left(\alpha \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \beta \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) = L\left(\alpha \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \beta \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) \\ &= L\left(\begin{bmatrix} \alpha x_1 + \beta y_1 \\ \alpha x_2 + \beta y_2 \end{bmatrix}\right) = \begin{bmatrix} \alpha x_1 + \beta y_1 \\ -(\alpha x_2 + \beta y_2) \end{bmatrix} \\ &= \alpha \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix} + \beta \begin{bmatrix} y_1 \\ -y_2 \end{bmatrix} = \alpha L(\vec{x}) + \beta L(\vec{y}) \end{aligned}$$

Then, we confirm that L is a linear mapping.

Linear operator from $\mathbb{R}^n$ into $\mathbb{R}^m$

Example 88. $L : \mathbb{R}^2 \longrightarrow \mathbb{R}^3$ defined as $L(\vec{x}) = \begin{bmatrix} x_2 \\ x_1 \\ x_1 + x_2 \end{bmatrix}$. Show that L is a linear operator?

Solution: L is a linear operator if

$$L(\alpha\vec{x} + \beta\vec{y}) = \alpha L(\vec{x}) + \beta L(\vec{y}), \forall \vec{x}, \vec{y} \in \mathbb{R}^2 \quad \text{and} \quad \forall \alpha, \beta \quad \text{are scalars.}$$

Verification:

$$\begin{aligned} L(\alpha\vec{x} + \beta\vec{y}) &= L\left(\alpha \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \beta \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) = L\left(\alpha \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \beta \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) \\ &= L\left(\begin{bmatrix} \alpha x_1 + \beta y_1 \\ \alpha x_2 + \beta y_2 \end{bmatrix}\right) = \begin{bmatrix} \alpha x_2 + \beta y_2 \\ \alpha x_1 + \beta y_1 \\ (\alpha x_1 + \beta y_1) + (\alpha x_2 + \beta y_2) \end{bmatrix} \\ &= \begin{bmatrix} \alpha x_2 \\ \alpha x_1 \\ \alpha(x_1 + x_2) \end{bmatrix} + \begin{bmatrix} \beta y_2 \\ \beta y_1 \\ \beta(y_1 + y_2) \end{bmatrix} = \alpha \begin{bmatrix} x_2 \\ x_1 \\ x_1 + x_2 \end{bmatrix} + \beta \begin{bmatrix} y_2 \\ y_1 \\ y_1 + y_2 \end{bmatrix} = \alpha L(\vec{x}) + \beta L(\vec{y}) \end{aligned}$$

Then, we confirm that L is a linear mapping.

4.3.2 ORTHOGONAL MATRICES

Orthonormal sets

Definition 1: Let $v_1, v_2, v_3, \dots, v_n$ be vectors in an inner product space V . If $\langle v_i, v_j \rangle = 0$, whenever $i \neq j$ then $\{v_1, v_2, v_3, \dots, v_n\}$ is said to be an **orthogonal set** of vectors.

Example 89. Verify that the set $\{(1, 1, 1)^T, (2, 1, -3)^T, (4, -5, 1)^T\}$ is an orthogonal set in $\mathbb{R}^3$?

Solution: Since $(1, 1, 1)(2, 1, -3)^T = 0$, $(1, 1, 1)(4, -5, 1)^T = 0$, $(2, 1, -3)(4, -5, 1)^T = 0$, then the set $\{(1, 1, 1)^T, (2, 1, -3)^T, (4, -5, 1)^T\}$ is an orthogonal set in $\mathbb{R}^3$.

Theorem 4.2. If $\{v_1, v_2, v_3, \dots, v_n\}$ is an orthogonal set of nonzero vectors in inner product space V , then $v_1, v_2, v_3, \dots, v_m$ are linearly independent.

Definition 2: An **orthonormal set** of vectors is an orthogonal set of unit vectors. The set $\{u_1, u_2, u_3, \dots, u_n\}$ will be orthonormal if and only if $\langle u_i, u_j \rangle = \delta_{ij}$. where

$$\delta_{ij} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$$

Remark: Given any orthogonal set of a nonzero vectors $\{v_1, v_2, v_3, \dots, v_n\}$ it is possible to form an orthonormal set of by defining

$$u_i = \left(\frac{1}{\|v_i\|} \right) v_i, \quad \forall i = 1, 2, \dots, n.$$

Example 90. From the orthogonal set $\{(1, 1, 1)^T, (2, 1, -3)^T, (4, -5, 1)^T\}$ form an orthonormal set in $\mathbb{R}^3$?

Solution:

$$\begin{aligned} u_1 &= \left(\frac{1}{\|v_1\|} \right) v_1 = \frac{1}{\sqrt{3}}(1, 1, 1)^T \\ u_2 &= \left(\frac{1}{\|v_2\|} \right) v_2 = \frac{1}{\sqrt{14}}(2, 1, -3)^T \\ u_3 &= \left(\frac{1}{\|v_3\|} \right) v_3 = \frac{1}{\sqrt{42}}(4, -5, 1)^T \end{aligned}$$

$$\left\{ \frac{1}{\sqrt{3}}(1, 1, 1)^T, \quad \frac{1}{\sqrt{14}}(2, 1, -3)^T, \quad \frac{1}{\sqrt{42}}(4, -5, 1)^T \right\} \quad \text{is an orthonormal set in } \mathbb{R}^3.$$

Theorem 4.3. Let $u_1, u_2, u_3, \dots, u_n$ be an orthonormal basis for an inner product space V . If

$$v = \sum_{i=1}^n c_i u_i, \quad \text{then} \quad c_i = \langle u_i, v \rangle.$$

Corollary 1: Let $u_1, u_2, u_3, \dots, u_n$ be an orthonormal basis for an inner product space V . If

$$u = \sum_{i=1}^n a_i u_i, \quad v = \sum_{i=1}^n b_i u_i, \quad \text{then} \quad \langle u_i, v_i \rangle = \sum_{i=1}^n a_i b_i.$$

Corollary 2 (Parseval's Formula): Let $u_1, u_2, u_3, \dots, u_n$ be an orthonormal basis for an inner product space V and

$$v = \sum_{i=1}^n c_i u_i, \quad \text{then} \quad \|v\|^2 = \sum_{i=1}^n c_i^2.$$

Example 91. The vectors $u_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)^T$ and $u_2 = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)^T$ form an orthonormal basis for $\mathbb{R}^2$?

If $x \in \mathbb{R}^2$, then $x^T u_1 = \frac{x_1 + x_2}{\sqrt{2}}$ and $x^T u_2 = \frac{x_1 - x_2}{\sqrt{2}}$.

It follows from the above theorem that $x = \frac{x_1 + x_2}{\sqrt{2}} u_1 + \frac{x_1 - x_2}{\sqrt{2}} u_2$ and it follows corollary 2 that

$$\|x\|^2 = \left(\frac{x_1 + x_2}{\sqrt{2}}\right)^2 + \left(\frac{x_1 - x_2}{\sqrt{2}}\right)^2 = x_1^2 + x_2^2$$

Orthogonal matrices

Definition:

An $n \times n$ matrix Q is said to be an **orthogonal matrix** if the column vector of Q form an orthogonal set in $\mathbb{R}^n$.

Theorem 4.4. An $n \times n$ matrix Q is an **orthogonal matrix** if and only if $Q^T Q = I$.

It follows from this theorem that if Q is an orthogonal matrix, then Q is an invertible and $Q^{-1} = Q^T$.

Example 92. For any fixed θ , show that $Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

Solution: The inverse is $Q^{-1} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$,

$$Q^T Q = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Hence, $Q = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ is an **orthogonal matrix**.

Properties of orthogonal matrix

If Q is an $n \times n$ matrix is an **orthogonal matrix**, then:

- a) The column vectors of Q form an orthonormal basis for $\mathbb{R}^n$,
- b) $Q^T Q = I$,
- c) $Q^{-1} = Q^T$
- d) $\langle Qx, Qy \rangle = \langle x, y \rangle$
- e) $\|Qx\| = \|x\|$.

4.3.3 EIGENVALUES AND EIGENVECTORS

Definition: Let A be an $n \times n$ matrix. A scalar λ is said to be an **eigenvalue** or a **characteristic value** of a A if there exists a nonzero vector x such that $Ax = \lambda x$. The vector x is said to be an **eigenvector** or **characteristic vector** belonging to λ .

Example 93. Let $A = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$ and $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Determine the eigenvalue of a matrix A ?

Solution: As $x \neq 0$, then λ is an eigenvalue of a matrix A if $Ax = \lambda x$.

$$Ax = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 3x = \lambda x \rightarrow \lambda = 3$$

Therefore, $\lambda = 3$ is an eigenvalue of a matrix A and $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is an **eigenvector** or **characteristic vector** belonging to λ .

Remark:

- Any nonzero multiple of x will be an eigenvector, since $A(\alpha x) = \alpha Ax = \alpha \lambda x = \lambda(\alpha x)$.
Thus, for above example, $\begin{bmatrix} 4 \\ 2 \end{bmatrix}$ is also an eigenvector belonging to $\lambda = 3$.
- The equation $Ax = \lambda x$ can be written in the form

$$(A - \lambda x)x = 0 \quad (-266)$$

Thus, λ is an eigenvalue of A if and only if formula (-266) has a nontrivial solution.

Equation (-266) will be a nontrivial solution if and only if $A - \lambda I$ is a singular matrix, means that

$$\det(A - \lambda I) = 0 \quad (-265)$$

If the determinant in (-265) is expanded, we obtain an n^{th} -degree polynomial in the variable λ ,

$$p(\lambda) = \det(A - \lambda I)$$

This polynomial is called the **characteristic polynomial**, and equation (-265) is called **characteristic equation for the matrix A** . The roots of the characteristic polynomial are the eigenvalues of A .

- These roots may be repeated and some are complex numbers.

Example 94. Let $A = \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{bmatrix}$. Find the eigenvalue and the corresponding eigenvector of matrix A ?

Solution:

$$\det(A - \lambda I) = 0 \Leftrightarrow \begin{vmatrix} 2 - \lambda & -3 & 1 \\ 1 & -2 - \lambda & 1 \\ 1 & -3 & 2 - \lambda \end{vmatrix} = 0 \Leftrightarrow -\lambda(\lambda - 1)^2 = 0 \Leftrightarrow \lambda_1 = 0, \lambda_2 = \lambda_3 = 1$$

$$\text{For } \lambda = 0, \text{ we have } Ax = 0 \Leftrightarrow \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \alpha = \beta = \gamma$$

$$x = \alpha \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \forall \alpha \in \mathbb{R}$$

Therefore $\lambda = 0$ is an eigenvalue of a matrix A and $x = \alpha \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\forall \alpha \in \mathbb{R}$ is an **eigenvector** or **characteristic**.

For $\lambda = 1$, we have

$$(A - I)x = 0 \Leftrightarrow \begin{bmatrix} 1 & -3 & 1 \\ 1 & -3 & 1 \\ 1 & -3 & 1 \end{bmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{pmatrix} \alpha - 3\beta + \gamma \\ \alpha - 3\beta + \gamma \\ \alpha - 3\beta + \gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \alpha - 3\beta + \gamma = 0 \Rightarrow \alpha = 3\beta - \gamma, \quad \forall \beta, \gamma \in \mathbb{R}, \quad \text{not zero together.}$$

$$x = \begin{pmatrix} 3\beta - \gamma \\ \beta \\ \gamma \end{pmatrix} = \beta \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Therefore $\lambda = 1$ is an eigenvalue of a matrix A and $x = \beta \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$, $\forall \beta \in \mathbb{R}$ and $x = \gamma \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$, $\forall \gamma \in \mathbb{R}$ are **eigenvectors** or **characteristic** belonging $\lambda = 1$

EXERCISES

1. show that

i) O is orthogonal to every vector in $\mathbb{R}^2$,

ii) The vector $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ 6 \end{pmatrix}$ are orthogonal in $\mathbb{R}^2$,

iii) The vectors $\begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ are orthogonal in $\mathbb{R}^3$,

iv) In $C(-\pi, \pi)$ with inner product $\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)g(x)dx$ the set $\{\frac{1}{2}, \cos x, \cos 2x, \dots, \cos nx\}$ is orthogonal in $C(-\pi, \pi)$

v) From iv) data form orthonormal set.

2. Find the eigenvalues and the corresponding eigenvectors for each of the following matrices

$$\begin{array}{llllll} a) \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix} & b) \begin{bmatrix} 1 & 1 \\ -2 & 3 \end{bmatrix} & c) \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix} & d) \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} & e) \begin{bmatrix} 4 & -5 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix} & f) \begin{bmatrix} 3 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix} \end{array}$$